

Università
degli Studi
di Padova

LECTURE NOTES
OF

ADVANCED STATISTICS

A Collection of Notes Based on the Lectures of Prof. Falasco

Created By

Gabriele Bellavia

University of Padua
Academic year 2025–2026

October 12, 2026

About these notes

These notes are part of my `PoD-collection-of-notes` repository and were written during my MSc in Physics of Data at the University of Padua between 2024 and 2026.

They are intended as freely accessible study material and may contain errors, inaccuracies, or outdated information. Most of the material was written while attending the courses and was not systematically reviewed afterwards. Course contents and syllabi may also change over time.

License

Except where otherwise noted, this work is licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License (CC BY-NC-SA 4.0).

You are free to share and adapt this material for non-commercial purposes, provided that appropriate credit is given, a reference to the original source is included, and any derivative work is distributed under the same license.

Source files

The complete source files used to compile these notes, including figures and other resources, are available in the corresponding folder of the repository:

`https://github.com/gabriele-bellavia/PoD-collection-of-notes`

Citation

If you use these notes, redistribute them, or use them as a starting point for your own material, please cite the original repository and, where applicable, the authors of any external material referenced in these notes.

Errors and corrections

If you find an error, typo, or unclear explanation, please feel free to open an issue on the GitHub repository or contact me at:

`gabriele.bellavia@studenti.unipd.it`

Physics of Data MSc
University of Padua

Contents

1 Introduction **1**
- 1.1 Our framework . . . 1
- 1.2 Bayesian framework . . . 1
  - 1.2.1 Deduction vs induction . . . 2
  - 1.2.2 What is inference? . . . 2
- 1.3 Two statistical philosophies . . . 2
- 1.4 Main idea: Bayes' theorem . . . 3
- 1.5 Practical machinery . . . 5

2 Recap On Probability Theory **7**
- 2.1 Probability space . . . 7
- 2.2 Events and set operations . . . 8
- 2.3 Random variables . . . 9
- 2.4 Distribution functions . . . 9
- 2.5 Cumulative distribution functions . . . 9
- 2.6 Expectation and momenta . . . 10
- 2.7 Quantiles and percentiles . . . 13
- 2.8 Conditional probability . . . 13
- 2.9 Bayes' theorem . . . 14
- 2.10 Joint distribution and independence . . . 15

3 Review of probability distributions, probability theorems and properties **17**
- 3.1 Permutations, combinations and pairings . . . 17
- 3.2 Statistical zoo . . . 18
  - 3.2.1 Probability distribution in `R` . . . 18
- 3.3 Standard discrete distributions . . . 19
  - 3.3.1 Binomial (and multinomial) distribution . . . 19
  - 3.3.2 Geometric distribution . . . 20
  - 3.3.3 Negative binomial (or Pascal) distribution . . . 21
  - 3.3.4 Hypergeometric distribution . . . 22
  - 3.3.5 Poisson distribution . . . 22
- 3.4 Standard continuous distributions . . . 23
  - 3.4.1 Uniform distribution . . . 23
  - 3.4.2 Normal or Gaussian distribution . . . 25
  - 3.4.3 Exponential distribution . . . 25
  - 3.4.4 Erlang distribution . . . 26
  - 3.4.5 Gamma distribution . . . 26
  - 3.4.6 Beta distribution . . . 28

3.5 Limiting theorems and asymptotic structure . . . 30
3.5.1 Law of large numbers . . . 30
3.5.2 Central limit theorem . . . 31
3.5.3 Large deviation principle . . . 32
3.6 Probabilistic inequalities in Bayesian inference . . . 33
3.6.1 Markov's inequality and tail bounds . . . 33
3.6.2 Chebyshev's inequality and posterior concentration . . . 34
3.6.3 Jensen's inequality and variational structure . . . 34

4 Inference Problems **35**
4.1 Model comparison . . . 35
4.1.1 Marginal likelihood and Occam's razor . . . 37
4.2 Hypothesis testing in frequentist approach . . . 40
4.3 Parameter estimation . . . 42
4.4 Choosing priors . . . 44
4.5 Point estimation . . . 47
4.5.1 MAP estimator . . . 48
4.5.2 LMS estimator: the posterior mean . . . 49
4.5.3 Other common loss functions . . . 50
4.5.4 The risk function . . . 50
4.6 Reliability of estimation . . . 51
4.6.1 Peaked posteriors and Gaussian approximation . . . 52
4.6.2 Limitations of Gaussian approximation and credible intervals . . . 53
4.7 Comparing to frequentist parameter estimation . . . 54
4.7.1 Confidence intervals vs credible intervals . . . 57
4.8 Prediction . . . 58
4.9 Comparing to prediction in the frequentist paradigm . . . 59

5 Some Applications **61**
5.1 Histogramming events: frequentist, Bayesian, and large-deviation perspective 61
5.1.1 Frequentist viewpoint . . . 61
5.1.2 Bayesian viewpoint . . . 62
5.1.3 Large deviation and entropy picture . . . 63
5.2 Bayesian inference for the Poisson rate parameter λ . . . 64
5.3 Bayesian inference for the Gaussian mean and variance . . . 69
5.3.1 Inference of the mean with known variance . . . 70
5.3.2 Inference of the variance with known mean . . . 70
5.3.3 Joint inference of mean and variance . . . 71
5.3.4 Credible intervals . . . 73
5.3.5 One- and two-side hypothesis testing . . . 75
5.4 Fitting as a special case of parametric inference . . . 76

6 Monte Carlo Methods **77**
6.1 Why Monte Carlo? . . . 77
6.2 Monte Carlo estimator . . . 79
6.3 Overview . . . 80
6.4 Inversion of the cumulative distribution function . . . 81
6.4.1 CDF inversion algorithm . . . 81
6.5 Rejection sampling . . . 84

6.5.1 Rejection sampling algorithm . . . 85
6.5.2 Rare event problem revisited . . . 86
6.6 Importance sampling . . . 88
6.7 Markov chains . . . 92
6.7.1 Stochastic processes . . . 92
6.7.2 Markov chain . . . 93
6.7.3 Detailed balance . . . 97
6.8 Monte Carlo implementation of discrete-time Markov chains . . . 98
6.8.1 Metropolis-Hastings algorithm . . . 98
6.8.2 Estimating observables from MC . . . 100
6.8.3 Convergence diagnostics and autocorrelation . . . 100
6.8.4 The independence sampler . . . 103
6.9 Gibbs sampling . . . 108
6.10 Hamiltonian Monte Carlo . . . 113
6.10.1 Multimodal targets problem . . . 119
6.11 Multimodality and energy barriers: tempering and annealing . . . 120
6.11.1 Simulated annealing . . . 122
6.11.2 Optimal cooling protocol . . . 123
6.12 Parallel tempering . . . 124
6.12.1 Temperature ladder and swap efficiency . . . 126
6.12.2 Comparison with simulated annealing . . . 127

7 Bayesian regression **129**
7.1 Where are we in the course . . . 129
7.2 Introduction to regression analysis . . . 129
7.2.1 Linear versus nonlinear regression . . . 130
7.2.2 Bayesian linear regression: prior and likelihood . . . 131
7.2.3 Prediction: integrate over parameter uncertainty . . . 133
7.2.4 Unknown noise level and parameter transformations . . . 134
7.2.5 MCMC view of regression fitting . . . 135
7.3 Generalized linear models . . . 136
7.3.1 Logistic regression for binary data . . . 137
7.3.2 Poisson regression for count data . . . 138
7.4 Connection to supervised learning . . . 139
7.5 Applications of Bayesian regression with MCMC: linear and quadratic fitting 139
7.5.1 Model ingredients . . . 140
7.5.2 Linear model . . . 141
7.5.3 Quadratic model . . . 143
7.5.4 Linear vs quadratic: what changes? . . . 144
7.6 Application . . . 145

Appendices **147**
Appendix A: change of variables . . . 147
Appendix B: σ-Algebras . . . 148
Appendix C: Events as a Boolean algebra . . . 149

Chapter 1

Introduction

The goal of this course is to understand Bayesian reasoning and inference. This framework is helpful when one wants to infer unknown quantities from data.

1.1 Our framework

Statistics is the discipline concerned with the collection, organization, analysis, interpretation, and presentation of data. **Probability theory** is the mathematical foundation of all these tasks. It deals with the study of probability, which quantifies how likely, or plausible, an event is to happen, or a proposition is to be true.

Why, in the first place, do we need statistics and probability? The answer is that there is no real world phenomenon that is completely predictable. Any real-world phenomenon involves systems of which we can have only partial knowledge. Therefore, we need to model and quantify the uncertainty that we find in the data.

We will deal with the Bayesian framework of statistics, which is slightly different from the frequentist approach that we have encountered in the physics courses.

1.2 Bayesian framework

Bayesian probability is an interpretation of probability theory in which probabilities quantify **degree of rational belief** about a certain event. What does this mean?

For example, let's consider a proposition A that can be true or false, or an event that may or not happen. When we assign probabilities, these are always conditional available to us, such as previous knowledge, experimental data, and background assumptions. Therefore, $P(A \mid I)$ represents the **degree of rational belief** that proposition A is true, given the background information I.

In the Bayesian framework, probabilities can be assigned to arbitrary propositions, including hypotheses and models, and not only to repeatable events. It can be shown that this is the appropriate framework to develop a consistent logic for reasoning under uncertainty. In practice, when we want to extract information from data, Bayesian inference allows us to perform induction in a controlled and logically consistent way, moving from observed data to conclusions about unknown quantities.

Cox's theorem states that Bayesian probability theory is mathematically equivalent to logical reasoning performed with numbers.

1.2.1 Deduction vs induction

As in mathematics and theoretical physics, in **deduction** one starts from a set of axioms or fundamental laws and derives their consequences. For example:

- Axioms of Euclidean geometry → Pythagorean theorem;
- Newton's laws → planets motion.

In other words, given a well-defined cause, we can determine its consequences.

Induction, instead, starts from observed data and attempts to infer the underlying model or mechanism behind them. For example, we may want to estimate the probability of obtaining heads in repeated coin tosses. Before collecting data, we do not know whether the coin is fair or biased. From the observed outcomes, we try to infer a more general property of the system, namely the probability of heads or tails.

In this sense, Bayesian statistics provides a systematic framework for inductive reasoning, making inference mathematically rigorous and logically consistent.

1.2.2 What is inference?

Inference consists in observing data and learning something unknown from those observation. Examples include:

- Is a coin fair?
- What is the average temperature in Padova?
- How many emails typically arrive per hour?
- Does a medical treatment work?

These are all questions that can be addressed within the Bayesian framework. To answer them, we combine prior knowledge, probabilistic models, and observed data.

In this course, we will mainly focus on three inference tasks:

- **Parameter estimation**. For example, we may have a probabilistic model for coin tossing characterized by a parameter, such as the Bernoulli p. The goal is to estimate this parameter from a sequence of observed heads and tails.
- **Model comparison**, which is related to the frequentist idea of hypothesis testing. In this case, we compare different models describing the same phenomenon and determine which one is better supported by the data.
- **Prediction**. Once we have learned a model or estimated its parameters, we can use it to predict future observations or new data.

1.3 Two statistical philosophies

As we have said, Bayesian statistics is not only a collection of methods, but also a framework, or a philosophy, for interpreting probability and uncertainty. This is quite different from the approach most of us are initially exposed to during physics or mathematics courses, namely the frequentist approach.

In the **frequentist view**:

- Parameters are fixed but unknown quantities, while only the data are random variables (from a *God given probability, cit*).
- Probability is interpreted as the long-run *relative frequency* of an event over a large number of repeated identical trials
- The main tools are parameter estimators and confidence intervals

In the **Bayesian view**:

- The parameters themselves, which describe the distribution of the data, are treated as random variables, *not only* the data.
- Probability represents a *degree of belief* about an event or proposition.
- As new data are collected, these beliefs can be updated through Bayes' theorem.

It can be shown that, under particular conditions and in the large-sample limit, the Bayesian and frequentist approaches may converge to similar results. Both frameworks are extremely useful and, whenever possible, it is important to compare them.

1.4 Main idea: Bayes' theorem

The main idea of Bayesian approach lies in **Bayes' theorem**, which we will study in more detail in the next chapters. This theorem relates observed data to our prior information about the system.

First of all, we treat the parameters θ describing the model as random variables. Given some observed data D, we want to determine the probability distribution of the parameters conditioned on the data. This distribution is called **posterior** distribution and is obtained though Bayes' theorem:

$$\underbrace{P(\theta \mid D)}_{\text{Posterior}} \propto \underbrace{P(\theta)}_{\text{Prior}} \times \underbrace{P(D \mid \theta)}_{\text{Likelihood}} \tag{1.1}$$

The posterior represents our updated degree of belief about the parameters or models θ after observing the data. The **prior** encodes our knowledge or assumptions before collecting any data, while the **likelihood** quantifies how compatible the observed data are with a given choice of parameters.

In practice, we perform an experiment and collect data distributed according to some probabilistic law. The likelihood measures the probability of observing those data assuming that the model parameters are fixed to a given value θ. Combining this information with the prior through Bayes' theorem allows us to update our beliefs about the unknown parameters.

Therefore, Bayesian inference can be seen as a process of *learning from evidence*.

For example, let's consider a suspicious coin. We toss a coin 10 times and get 8 heads.

- What is the probability p of obtaining heads in a single toss? This is a parameter estimation problem.
- Is the coin fair or biased? This corresponds to model comparison or hypothesis testing.
- What is the probability of obtaining heads in the next two tosses? This is a prediction problem.

- How certain are we about these conclusions?

The important point is that, since parameters are treated as random variables, Bayesian inference does not return only a single estimate. Instead, it provides a full probability distribution p over the possible values of the parameters, together with quantified uncertainty and predictions for future data D_{next}.

From this posterior distribution, we may still decide to extract a single representative value, such as the mean, mode, or median. This procedure is known as point estimation.

Bayesian thinking is not as unnatural as it may initially seem. Indeed, we already think this way in daily life:

- Weather forecast updates each day;
- Doctors update diagnosis with tests;
- Machine learning updates model with data.

Thus, it formalizes common sense reasoning.

To familiarize ourselves with Bayesian thinking, let us consider the following example of a medical test, which leads to an apparent *paradox*.

Example 1.4.1

magine that, in a certain population, a disease has a prevalence of 1%. This means that, based on our previous knowledge, about 1% of the population is affected by the disease.
Now suppose we have a diagnostic test to determine whether a person is sick or healthy. In general, every test has a certain sensitivity, for example 99%. The complementary probability corresponds to the false negative rate, namely the probability that a sick person tests negative.
Moreover, we must also consider the possibility that a healthy person tests positive. This is called the false positive rate. Let us assume it is 1%
Summarizing:

- Disease prevalence = 1%
- Test sensitivity = 99% → False negative rate = 1% (sick people testing negative)
- False positive rate = 1% (healthy people testing positive.

Now we ask the following question: if the test result is positive, what is the probability that the person is actually sick?
At first glance, one may think the answer is 99%, but this is incorrect because there are many more healthy individuals than sick ones, and even a small false positive rate can generate many positive tests among healthy people

---

Let us consider a large sample of 10^4 people. With a disease prevalence of 1%, about 100 individuals are sick, while 9900 are healthy.
Among the 100 sick individuals, 99 test positive due to the 99% sensitivity. Among the 9900 healthy individuals, 99 also test positive because of the 1% false positive

rate. Therefore, the total number of positive tests is $99 + 99 = 198$. Hence,

$$P(s \mid +) \equiv P(\text{sick} \mid \text{positive}) = \frac{99}{198} = 0.5$$

We can obtain the same result directly within the Bayesian framework using Bayes' theorem. Since being sick and healthy are mutually exclusive events:

$$P(s \mid +) = \frac{P(+ \mid s)P(s)}{P(+ \mid s)P(s) + P(+ \mid h)P(h)} = \frac{0.99 \cdot 0.01}{0.99 \cdot 0.01 + 0.01 \cdot 0.99} = 0.5$$

where disease prevalence gives the prior probability $P(s) = 1 - P(h)$.

The key lessons from this example are:

- Rare diseases can produce a surprisingly large number of false positive.
- Prior probabilities matter.
- Data must always be interpreted within their context.
- Bayesian reasoning helps prevent misleading conclusions.

This is why screening tests are followed by confirmatory tests.

1.5 Practical machinery

In practice, we will often need to compute integrals of the form:

$$\mathbb{E}[f] = \int d\theta \, f(\theta) p(\theta \mid D)$$

which are typically high-dimensional and/or analytically intractable.

To deal with these integrals, we introduce approximations such as:

$$\int f(\theta) p(\theta \mid D) \, d\theta \simeq \frac{1}{N} \sum_{i=1} f(\theta_i)$$

where the samples θ_i are drawn from the posterior $p(\theta \mid D)$ using **Monte Carlo methods**.

Essentially, this means that instead of computing the integral exactly, we evaluate the function f on a set of sampled points θ_i extracted from the posterior distribution and then compute their average. The main challenge is therefore learning how to sample efficiently, rapidly, and without bias from complicated probability distributions that depend on the observed data.

To better understand the Bayesian approach, let us consider another example involving weather forecasting

Example 1.5.1

The forecast says: "40% chance of rain tomorrow between 1pm and 2pm". What does this actually mean?
In the frequentist interpretation, there is only one *tomorrow*, so the event cannot

literally be repeated infinitely many times to measure a frequency.
In the Bayesian framework, this becomes an inference and prediction problem. Let R denote the proposition "it rains tomorrow between 1pm and 2pm", while D represents the available data, such as measurements of temperature, pressure, humidity, satellite images, and so on. Then, $P(R \mid D)$ represents our degree of belief that it will rain, given all the available information.

First of all, we introduce the model and parameters:

$$P(R \mid D, M) = \int P(R \mid \theta, M) P(\theta \mid D, M)\, d\theta$$

where M is the weather model, for example a numerical simulation based on fluid dynamics equations), θ represents the uncertain model parameters (e.g., location, season), $P(\theta \mid D, M)$ is the posterior over parameters, and $P(R \mid \theta, M)$ is the probability of rain for a given choice of parameters.
However, solving this integral exactly is extremely difficult because of the large number of parameters involved. Instead, we can approximate it using Monte Carlo methods. In practice, we run many simulations with slightly different sampled parameters θ_i, count how many predict rain, and estimate the predictive probability through an average

$$P(R \mid D, M) \simeq \frac{1}{N} \sum_{i=1}^{N} P(R \mid \theta_i, M)$$

Chapter 2

Recap On Probability Theory

This chapter provides a brief recap on probability theory. We will see:

- Axioms framework;
- Random variables and distributions;
- Expectation, variance, moments, and cumulants;
- Conditional probability and Bayes theorem;
- Independence;

2.1 Probability space

A **probability space** is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where:

- Ω is a sample space containing all possible events.
- $\mathcal{F}$ is a σ-algebra describing which events are measurable. In simple terms, $\mathcal{F}$ consists of subsets of Ω closed under complements, countable unions, and countable intersections.
- $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a probability measure.

Probability theory is based on the **Kolmogorov axioms**. Given events $A_i \in \mathcal{F}$, we have:

1. **Non-negativity**. $\mathbb{P}(A_i) \geq 0$ (if interpreted as degree of belief, it cannot be negative);
2. **Normalization**. $\mathbb{P}(\Omega) = 1$;
3. **Countable additivity**. For any countable collection of disjoint events $\{ A_i \}_{i=1}^{\infty}$

$$\mathbb{P}(\cup_i A_i) = \sum_i \mathbb{P}(A_i)$$

for disjoint A_i.

2.2 Events and set operations

In this framework, we need a set of operations corresponding to logical (Boolean) operations. These are:

- **Union** (logical `OR`) $A \cup B$: the event that at least one of the events A or B occurs:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

- **Intersection** (logical `AND`) $A \cap B$: the event that both A and B occur:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$$

  where $\mathbb{P}(\cdot \mid A)$ denotes the **conditional probability**, namely the probability computed under the assumption that event A has occurred;

- **Complement** (negation `NOT`) $\overline{A} = A^c = \Omega \setminus A$: the event that A does not occur:

$$\mathbb{P}(\overline{A}) = 1 - \mathbb{P}(A)$$

- **Set difference** $A \setminus B = A \cap \overline{B}$: the event that A occurs but B does not:

$$\mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$$

When working with complements, we usually use **De Morgan's laws**:

$$\overline{(A \cup B)} = \overline{A} \cap \overline{B}, \qquad \overline{(A \cap B)} = \overline{A} \cup \overline{B}$$

Figure 2.1 illustrates some notable relations between sets. Two sets are said to be **mutually exclusive** if $A \cap B = \emptyset$. A collection of sets A_i is **collectively exhaustive** if $A_1 \cup \ldots \cup A_N = \Omega$. Finally, if the sets A_i are both mutually exclusive and collectively exhaustive, they constitute a **partition** of Ω.

These definitions are particularly useful in hypothesis testing and model comparison. In these contexts, we often consider a space of possible events or hypotheses and compare different partitions, that is, different sets of mutually exclusive and collectively exhaustive possibilities.



Figure 2.1: Notable relation between sets

2.3 Random variables

When dealing with data, we work with numerical quantities rather than sets. For this reason, we introduce random variables. A **random variable** is a measurable function:

$$X : \Omega \to \mathbb{R}$$

that associates a numerical value with each outcome in the sample space, allowing us to work with numbers instead of abstract events. For instance, the number of heads in repeated coin tosses, the decay time in a counting experiment, or the measured voltage across a conductor.

2.4 Distribution functions

How do we describe random variables? We introduce distribution functions, whose form depends on whether the random variable is discrete or continuous.

For discrete random variables, we use the **probability mass function** (PMF):

$$\mathbb{P}(X = x_i) = p_X(x_i)$$

which gives the probability that the random variable X takes the value x_i.

For continuous variables, we use the **probability density function** (PDF):

$$\mathbb{P}(a \leq X \leq b) = \int_a^b p_X(x)\, dx$$

which is defined over the real line. Unlike the PMF, the probability density function is not itself a probability, but a density. Therefore, the quantity $p_X(x)\, dx$ represents the probability that the random variable takes a value inside an infinitesimal interval of width dx around x.

As a consequence, for continuous random variables the probability of observing exactly one value vanishes:

$$\mathbb{P}(X = x) = 0$$

The normalization conditions for the two cases are:

$$\sum_i p_X(x_i) = 1, \qquad \int p_X(x)\, dx = 1$$

2.5 Cumulative distribution functions

Once we defined the PMF or the PDF, we can introduce the **cumulative distribution function** (CDF) of a random variable X:

$$F_X(x) = \mathbb{P}(X \leq x)$$

This represents the probability that the random variable takes a value smaller than or equal to a given threshold x. The CDF is non-decreasing and right-continuous function, and satisfies:

$$\lim_{x \to -\infty} F_X(x) = 0, \qquad \lim_{x \to +\infty} F_X(x) = 1$$

If p is the PDF of X, then the relation with the CDF is:

$$F_X(x) = \int_{-\infty}^{x} p(x')\,dx', \qquad p_X(x) = \frac{\mathrm{d}}{\mathrm{d}x} F_X(x)$$

That is, the CDF is obtained by integrating the density up to x, while the PDF can be recovered by differentiating the CDF.

If instead p_X is the PMF of X, then:

$$F_X(x) = \sum_{y \leq x} \mathbb{P}(X = y) = \sum_{i : x_i \leq x} p_X(x_i)$$

In this case, the CDF has a staircase shape, with jumps whose magnitude is equal to the probability mass: $p_X(x_i) = F_X(x_i) - F_X(x_{i-1})$

2.6 Expectation and momenta

Let's introduce the concept of **expectation value** of a random variable X:

$$\mathbb{E}[X] = \sum_i x_i p(x_i), \qquad \mathbb{E}[X] = \int x p(x)\,dx$$

for discrete and continuous random variables, respectively.

The expectation of the n-th power of X defines the **n-th raw moment**:

$$\mu'_n \equiv \mathbb{E}[X^n]$$

which is the moment computed about zero. Raw moments provide information about the shape of the distribution relative to the origin.

Instead, moments computed with respect to the mean are called **n-th central moments**:

$$\mu_n \equiv \mathbb{E}[(X - \mathbb{E}[X])^n]$$

These describe the shape of the distribution around its average value.

The **variance** corresponds to the $n = 2$ central moment:

$$\mu_2 \equiv \sigma^2 \equiv \mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}\left[X^2\right] - \mathbb{E}[X]^2$$

and the standard deviation is defined as $\sigma \equiv \sqrt{\mu_2}$. This quantity measures the spread of the distribution around the mean.

Higher-order central moments provide information about additional properties of the distribution, such as asymmetry (skewness) and the presence of heavy or long tails.

Some useful properties of expectation values are:

1. **Linearity**:
$$\mathbb{E}[\alpha g(X) + \beta f(Y)] = \alpha \mathbb{E}[g(X)] + \beta \mathbb{E}[f(Y)]$$

2. **Covariance** between random variables X and Y:
$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$
which quantifies how strongly two variables vary together.

3. **Moment generating function** (MGF) of X:

$$M_X(t) = \mathbb{E}\left[e^{tX}\right]$$

whose derivatives generate the n-th raw moments:

$$\frac{\mathrm{d}^n M_X(t)}{\mathrm{d}t^n}\Big|_{t=0} = \mu'_n$$

The MGF completely characterizes the distribution whenever it exists in a neighborhood of $t = 0$, providing an alternative way to represent a probability distribution. Its derivatives allow us to compute moments directly.

4. The **cumulant generating function** (CGF):

$$K_X(t) = \log M_X(t) \tag{2.1}$$

is the n-th cumulant k_n obtained by:

$$\kappa_n = \frac{d^n K_X(t)}{dt^n}\Big|_{t=0} \tag{2.2}$$

The first two cumulants coincide with the mean and the variance:

$$\kappa_1 = \mu'_1, \qquad \kappa_2 = \mu_2$$

Cumulants play a central role in Bayesian inference and statistical mechanics, as characterization of fluctuations and correlations.

Note that MGF and CGF may not exist because the integral (sum) defining the expectation value need not converge absolutely. In this case, one can make the replacement $t \to it$ defining the characteristic function

$$\varphi_X(t) = \mathbb{E}\left[e^{itX}\right],$$

whose derivatives give the central moments,

$$\frac{d^n \varphi_X(t)}{dt^n}\Big|_{t=0} = i^n \mu'_n.$$

Higher standardized moments provide useful dimensionless measures of the shape of a distribution. The *skewness* is defined as the third central moment normalized by the variance,

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{\mathbb{E}\left[(X - \mathbb{E}[X])^3\right]}{\left(\mathbb{E}[(X - \mathbb{E}[X])^2]\right)^{3/2}},$$

and measures the asymmetry of the distribution around its mean. Positive skewness indicates a longer right tail, while negative skewness indicates a longer left tail.

The *kurtosis* is defined as the fourth central moment normalized by the square of the variance,

$$\gamma_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mathbb{E}\left[(X - \mathbb{E}[X])^4\right]}{\left(\mathbb{E}[(X - \mathbb{E}[X])^2]\right)^2}.$$

It quantifies the weight of the tails and the concentration of probability mass around the mean. Often one uses the *excess kurtosis*, $\gamma_2 - 3$, which vanishes for a Gaussian distribution. Skewness and kurtosis are used to compare distributions with the same mean and variance but different shapes, and are particularly useful for detecting asymmetry, heavy tails, and deviations from Gaussian behavior.

Example 2.6.1

As an example, let us consider a continuous random variable X following an exponential distribution. This distribution appears naturally when describing waiting times between independent events, such as the time interval between the arrival of two customers in a shop.
Later we will see that the exponential distribution is related to the Poisson distribution through counting processes, namely exponential waiting times correspond to Poisson-distributed event counts.
Its PDF is:

$$f(x) = \lambda e^{-\lambda x}, \qquad x > 0$$

Its mean is:

$$\mathbb{E}[X] = \lambda \int_0^\infty dx\, x e^{-\lambda x} = \frac{1}{\lambda}$$

The CDF is:

$$F(x) = \int_0^x dx\, f(x) = 1 - e^{-\lambda x}$$

Therefore, the median is obtained from:

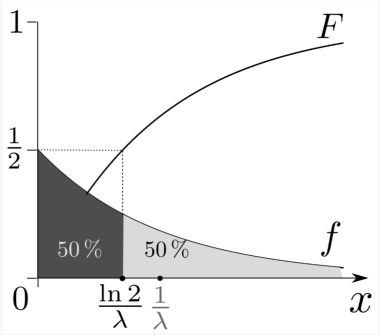
$$F(\tilde{x}) = \frac{1}{2} \quad \Rightarrow \quad \tilde{x} = \frac{\ln 2}{\lambda}$$

The median is therefore the value that divides the probability distribution into two equal halves.
The MGF is:

$$M(t) = \frac{\lambda}{\lambda - t}$$

defined only for $t < \lambda$, otherwise the integral diverges.
One limitation of MGFs is that they are not always defined over the entire real axis. In more advanced settings, alternative generating functions or complex-variable extensions are sometimes introduced.
From the MGF we obtain the raw moments:

$$\mu'_n = \frac{n!}{\lambda^n}$$



2.7 Quantiles and percentiles

A **quantile** describes how probability mass is distributed. The q-quantile x_q of a random variable satisfies:

$$\mathbb{P}\left(X \leq x_q\right)) = F_X(x_q) = q \tag{2.3}$$

Examples:

- Median: $q = 0.5$;
- Lower quartile: $q = 0.25$;
- Upper quartile: $q = 0.75$.

Therefore, the quantile function is the inverse of the CDF: $x_q = F_X^{-1}(q)$.

Percentiles are quantiles expressed in percentages. Thus, the p-th percentile x_p satisfies:

$$\mathbb{P}\left(X \leq x_p\right) = F_X(x_p) = \frac{p}{100} \tag{2.4}$$

Examples:

- 50th percentile = median;
- 25th percentile = first quartile;
- 95th percentile = extreme upper tail.

All these are widely used to summarize empirical data distributions.

2.8 Conditional probability

We want to introduce Bayes' theorem. Before doing that, we first need to define and interpret the concept of **conditional probability**. It is defined as:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \qquad \text{for } \mathbb{P}(B) > 0$$

Therefore, $\mathbb{P}(A \mid B)$ represents the probability that and event A occurs given that event B has already occurred. In practice, we restrict the sample space to the subset where B is true.

More generally, if we have a partition $\{ B_i \}$ of the sample space, the probability of an event A can be written as:

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$$

that is, we sum over all mutually exclusive possibilities weighted by their probabilities. This relation is known as **law of total probability for a partition** $\{ B_i \}$.

In the Bayesian framework, probabilities are *always* conditional on the available information. More generally, we should write probabilities as $\mathbb{P}(A \mid I)$, where I denotes the background information. This emphasizes that probabilities are never completely independent of context or prior knowledge.

For brevity, we will denote joint probability $\mathbb{P}(A \cap B)$ as $\mathbb{P}(AB)$ or $\mathbb{P}(A, B)$.

We can interpret conditional probability visually through Figure 2.2. Suppose we have 16 equally likely outcomes in Ω, with 7 belonging to A, 10 belonging to B, and 3 belonging to the intersection $A \cap B$.

Since all outcomes are equally likely, probabilities can be computed by counting favorable outcomes and dividing by the total number of outcomes. Therefore:

$$\mathbb{P}(\Omega) = 1, \quad \mathbb{P}(A) = \frac{7}{16}, \quad \mathbb{P}(B) = \frac{10}{16}, \quad \mathbb{P}(A \mid B) = \frac{3}{10}, \quad \mathbb{P}(A \cap B) = \frac{3}{16}$$

Thus, we can verify:

$$\frac{3}{16} = \mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)P(B) = \frac{3}{10} \times \frac{10}{16}$$

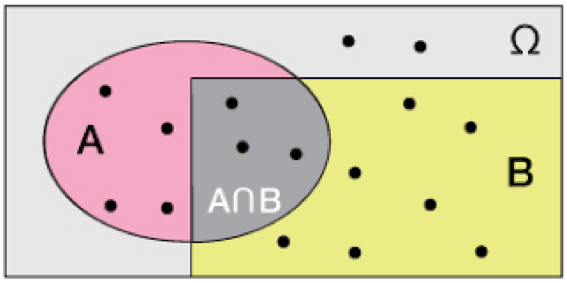


Figure 2.2

For random variables, the conditional distribution of X given $Y = y$ is defined by

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}, \qquad f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)},$$

whenever the denominator is non-zero. The joint distribution factorizes as

$$p_{X,Y}(x,y) = p_{X|Y}(x|y)\, p_Y(y), \qquad f_{X,Y}(x,y) = f_{X|Y}(x|y)\, f_Y(y).$$

2.9 Bayes' theorem

All probabilities are always conditional ones! If I is the background condition, related to information we have, the probability of an event A is of the form $P(A \mid I)$. It makes no sense to talk about the probability of the truth of the statement A without being explicit about the conditions upon which the assignment of probability is based.

Starting from the definition of conditional probability,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$$

Bayes' theorem follows directly:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)} \tag{2.5}$$

Let's interpret each term. In the context of inference, we usually identify $B \to D$ as the observed data, $A \to \theta$ or $A \to H$ as the unknown parameters or the model/hypothesis.

Therefore:

- $\mathbb{P}(\theta)$ is called the **prior**. It encodes our *prior* knowledge or assumptions about the parameters or models before observing any data.

For example, in weather forecasting, if we want to estimate quantities such as temperature, we already know that physically meaningful values may be restricted to a certain range. This information should be encoded in the prior.

Similarly, if a physical process is known to satisfy certain symmetries, this prior knowledge should also be incorporated into the model.

- $\mathbb{P}(D \mid \theta)$ is called the **likelihood**. It represents the probability of observing the data D assuming that the model parameters are fixed to θ, or equivalently assuming hypothesis H.

Importantly, the likelihood is not a probability distribution over the parameters. Instead, it measures how compatible different parameter values are with the observed data.

- $\mathbb{P}(\theta \mid D)$ is called the **posterior**. It represents our updated degree of belief about parameters or hypotheses after observing the data.

- $\mathbb{P}(D)$ is a normalization factor called **evidence** or **marginal likelihood**. It is obtained by averaging the likelihood over all possible parameter values weighted by the prior distribution.

In summary, Bayesian inference consists of defining a probabilistic model for the observed data through the likelihood, encoding previous knowledge through the prior, and then updating these beliefs after observing data using Bayes' theorem to obtain the posterior distribution.

For models that cannot be treated analytically, we will rely on numerical methods such as Markov Chain Monte Carlo (MCMC).

More generally, suppose that A_i is a partition of hypotheses, meaning that the hypotheses are mutually exclusive and collectively exhaustive. Using countable additivity:

$$\mathbb{P}(B) = \sum_{i=1}^{n} \mathbb{P}(B \cap A_i) = \sum_{i=1}^{n} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$$

and Bayes' theorem generalizes to:

$$\mathbb{P}(A_k \mid B) = \frac{\mathbb{P}(B \mid A_k)\mathbb{P}(A_k)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B \mid A_k)\mathbb{P}(A_k)}{\sum_{i=1}^{n} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)}$$

Notice that the normalization factor can be computed entirely from the prior probabilities and the likelihood terms.

2.10 Joint distribution and independence

Another important concept is that of **joint distributions**. Given two random variables X and Y, we can describe them together through a joint distribution $\mathbb{P}(X \cap Y)$. In the discrete case this is written as $p_{X,Y}(x, y)$, while in the continuous case we write $f_{X,Y}(x, y)$.

This distribution describes the probability of simultaneously observing specific values of both variables.

For example, a Gaussian distribution is characterized by two parameters, the mean and the variance. In the Bayesian framework, these parameters are themselves treated as random variables, and therefore we can assign a joint probability distribution to them.

Once we know the *full* joint distribution, we can obtain information about only one variable by integrating or summing over the other variable, that is restricting to *less* knowledge:

$$p_X(x) = \sum_y p_{X,Y}(x, y), \qquad p_X(x) = \int dy\, p_{X,Y}(x, y)$$

for the discrete and continuous case, respectively

This procedure is called **marginalization**, and the resulting distribution is called **marginal distribution** of X. More precisely, we say that we *marginalize* over Y to obtain the distribution of X.

Finally, two random variables are said to be **independent** if their joint distribution factorizes:

$$p_{X,Y}(x, y) = p_{X|Y}(x \mid y)p_Y(y) = p_X(x)p_Y(y)$$

Equivalently,

$$p_{X|Y}(x \mid y) = p_X(x)$$

This means that knowing the value of one variable does not provide any information about the other.

Why is all of this important in Bayesian inference?

Bayesian inference relies heavily on conditional probability, joint distributions, marginalization, and independence assumptions. Marginalization is particularly useful because realistic models often contain many parameters, and in practice we are usually interested only in a subset of them. The remaining parameters can be integrated out and treated as nuisance parameters.

Marginalization also becomes important when dealing with complex or multimodal posterior distributions, where analytical calculations are often impossible.

In practice, essentially every Bayesian computation relies on these concepts.

Typical errors to avoid:

- Confusing $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$;
- Ignoring normalization factors;
- Assuming independence without justification;
- Treating PDF as if it were a PMF.

Chapter 3

Review of probability distributions, probability theorems and properties

Many probability distributions arise from counting the number of ways events can occur (think of entropy). The key combinatorial objects we will see are permutations, combinations, and pairings. These determine the structure of many probability laws such as binomial, multinomial and hypergeometric distributions.

3.1 Permutations, combinations and pairings

A **permutation** is an ordered arrangement of objects. Therefore, we emphasize that *order is important.*

The number of permutations of n distinct objects is:

$$n! = n(n-1)(n-2)\dots 1 \tag{3.1}$$

For instance, arranging three objects A, B, C gives:

$$ABC, \quad ACB, \quad BAC, \quad BCA, \quad CAB, \quad CBA$$

Indeed, the total is $3! = 6$.

Often, we arrange only k objects chosen from n. The number of permutations of k objects among n **without repetition** is:

$$P(n,k) = \frac{n!}{(n-k)!} \tag{3.2}$$

For example, the number of ways to select and order 3 people out of 10 is:

$$P(10,3) = \frac{10!}{7!} = \frac{10 \cdot 9 \cdot 8 \cdot 7!}{7!} = 90 \cdot 8 = 720$$

The number of permutations of k objects among n **with repetition** is:

$$n^k \tag{3.3}$$

Basically, this means that after permuting/drawing an object, it is replaced before the next draw.

A **combination** counts selections where *order does not matter*.

The number of ways to choose k objects among n **without repetition** is:

$$C(n,k) = \frac{P(n,k)}{k!} = \binom{n}{k} = \frac{n!}{k!(n-k)!} \tag{3.4}$$

This is the coefficient of the monomial x^k in the binomial formula:

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

The number of ways to choose k objects among n **with repetition** is:

$$\binom{n+k-1}{k} = \frac{(n+k-1)!}{k!(n-1)!} \tag{3.5}$$

We now list all different ways to pair from n objects:

- **Single pair** (unordered) is the number of ways to choose one pair:

$$\binom{n}{2} = \frac{n(n-1)}{2}$$

- **Selecting k unordered pairs**: choose $2k$ elements and partition them into pairs:

$$\frac{n!}{(n-2k)!2^k k!}$$

- **Selecting k ordered pairs**:

$$\frac{n!}{(n-2k)!k!}$$

- **Partition into pairs (perfect matching)**. If $n = 2m$ and all elements are paired:

$$(2m-1)!! = \frac{(2m)!}{2^m m!}$$

- **Partition in ordered pairs**:

$$\frac{(2m)!}{m!}$$

3.2 Statistical zoo

3.2.1 Probability distribution in `R`

In `R`all standard distributions are available. The naming convention is very easy: a core name is associated with each distribution, and a prefix is appended to indicate the four basic associated functions:

- `d` for the PDF;
- `p` for the CDF;
- `q` for the quantile function;
- `r` for sampling from the distribution.

Notice that `pcore_name()` and `qcore_name()` are one the inverse of one another.

Distribution	Core name	Parameters	Default values
Beta	`beta`	`shape1`, `shape2`	
Binomial	`binom`	`size`, `prob`	
Cauchy	`cauchy`	`location`, `scale`	0, 1
Chi-square	`chisq`	`df`	
Exponential	`exp`	`1/mean`	1
Fisher	`f`	`df1`, `df2`	
Gamma	`gamma`	`shape1`, `1/scale`	`NA`, 1
Geometric	`geom`	`prob`	
Hypergeometric	`hyper`	`m`, `n`, `k`	
Log-Normal	`lnorm`	`mean`, `sd`	0, 1
Logistic	`logis`	`location`, `scale`	0, 1
Normal	`norm`	`mean`, `sd`	0, 1
Poisson	`pois`	`lambda`	
Student	`t`	`df`	
Uniform	`unif`	`min`, `max`	0, 1
Weibull	`weibull`	`shape`	

Table 3.1: List of PDF in `R`.

3.3 Standard discrete distributions

Discrete distributions describe random variables taking values in $\mathbb{N}$ or finite subsets thereof because are based on counting events.

The basic ingredient is the so-called **Bernoulli trial**, which is an idealized event where we can have only two possible outcomes: either is a success with probability p, or a failure with probability $q = 1 - p$. Clearly, they are two independent events, and the probability is normalized: $p + q = 1$. The most common example is coin tossing with a fair coin, i.e. , $p = q = \frac{1}{2}$.

Many distributions arise from repeated Bernoulli trials.

3.3.1 Binomial (and multinomial) distribution

The **binomial distribution** gives the probability of observing x successes in n independent Bernoulli trials, each with success probability p:

$$P(X = x) = \mathrm{B}(n, p) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n \tag{3.6}$$

Therefore, the parameters are n and p. As we saw previously, the binomial coefficient $\binom{n}{x}$ counts the ways to choose an unordered subset of k elements from a fixed set of n distinct elements.

Mean and variance are, respectively,

$$\mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = \sigma_X^2 = np(1-p)$$

The case of a single trial $n = 1$ defines the **Bernoulli distribution**.

A typical simple inference problem is to infer the probability of success p from a sequence of head and tails from coin tossing.

Let's compute compute the binomial mean:

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{x=1}^{n} x \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} = \\
&= \sum_{x=1}^{n} \frac{n!}{(x-1)!(n-x)!} p^x (1-p)^{n-x} = \\
&= np \sum_{x=1}^{n} x \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} (1-p)^{n-x} = \\
&= np \sum_{k=0}^{m} x \frac{m!}{k!(m-k)!} p^k (1-p)^{m-k} = \\
&= np(p + (1-p))^m = np
\end{aligned}$$

where we changed the indices of summation: $x - 1 = k$ and $n - 1 = m$.

Example 3.3.1

In a restaurant 8 entrees of fish, 12 of beef and 10 of poultry are served. What is the probability that 2 of the 4 next customers order fish entrees?

First of all, we can assume that the orders are independent of one another. Then, we can read the probability we're interested in as the probability of having 2 successes for the fish order.
There are $12 + 8 + 10 = 30$ total entrees served. We assume that all dishes can be served an unlimited number of times. So the probability that a random customer orders fish is $8/30 = 4/15$. Then, the number of fish orders among 4 customers is $X \sim \mathrm{B}(n = 4, p = 4/15)$. Therefore,

$$P(X = 2) = \frac{4!}{2!} \left(\frac{4}{15}\right)^2 \left(1 - \frac{4}{15}\right)^{4-2} = 12 \left(\frac{4}{15}\frac{11}{15}\right)^2 \simeq 0.23$$

Binomial can be generalized when we have k different possible outcomes with probabilities $\{ p_i \}_{i=1}^{k}$ and $\sum_i p_i = 1$. For coin tossing, we have $k = 2$, but for a dice with 6 faces, we can have $k = 6$. The joint distribution of counts $(x_1, \ldots, x_k)$ is:

$$p(x_1, \ldots, x_k) = \frac{n!}{x_1! \ldots x_k!} p_1^{x_1} \ldots p_k^{x_k}, \qquad \text{with } \sum_{i=1}^{k} x_i = n \tag{3.7}$$

This generalization is the **multinomial distribution**.

3.3.2 Geometric distribution

The **geometric distribution** can be defined in two slightly different ways. It can be seen as the number of trials N needed in order to obtain the first success:

$$P(N = n) = (1-p)^{n-1} p \tag{3.8}$$

Basically, consider independent Bernoulli trials, for example tossing coin. We ask ourselves: how many time we have to toss a coin before seeing a head or tail? Therefore, it follows easily from the observation that the probability of succeeding at the n-th trial is given by the probability of failing $n-1$ times $(1-p)^{n-1}$ multiplied by the probability of 1 success, p.

The first two moments are:

$$\mathbb{E}[N] = \frac{1}{p}, \qquad \mathrm{Var}(N) = \frac{1-p}{p}$$

It's important to notice that this distribution is **memoryless**:

$$P(N > n+m \mid N > n) = P(N > m)$$

meaning that the probability of waiting an additional m trials for success does not depend on how many trials have already occurred. To show that, we compute the conditional probability:

$$P(N > n+m \mid N > n) = \frac{P(N > n+m)}{P(N > n)}$$

Since

$$P(N > k) = \sum_{j=k+1}^{\infty} (1-p)^{j-1} p = (1-p)^k$$

we obtain directly:

$$P(N > n+m \mid N > n) = \frac{(1-p)^{n+m}}{(1-p)^n} = (1-p)^m = P(N > m)$$

We notice that in the rare-event limit in which $p \to 0$, we have that the variance diverges $\sigma^2 = 1/p^2$. Namely, rare events might happen at any moment, even though they have a negligible probability to happen at any moment.

Notice that `R`counts failures before the first success, namely:

$$P(N_{\text{fail}} = k) = (1-p)^k p \quad \Rightarrow \quad \mathbb{E}[N_{\text{fail}}] = \mathbb{E}[N] - 1 = \frac{1-p}{p}$$

This is the second definition of geometric distribution.

Let's compute the geometric mean:

$$\mathbb{E}[n] = p \sum_{n=0}^{\infty} n(1-p)^{n-1} = -p\partial_p \sum_{n=0}^{\infty} (1-p)^n = \\ = -p\partial_p \frac{1}{p} = \frac{1}{p}$$

where we used the geometric series $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$, with $x < 1$. This is a widely used trick.

3.3.3 Negative binomial (or Pascal) distribution

The geometric distribution generalizes to the so-called **negative binomial distribution**. Basically, instead of the number of failure before first success or the number of trials to get success, it gives the number of failures before the r-th success. Therefore, it gives the probability of observing k failures before the r-th success in independent Bernoulli trials:

$$P(X = k) = \binom{k+r-1}{k} (1-p)^k p^r, \qquad k = 0, 1, 2, \ldots \tag{3.9}$$

The geometric distribution is recovered with $r = 1$. The parameters are then r and p.

3.3.4 Hypergeometric distribution

The **hypergeometric distribution** generalizes the Bernoulli distribution, giving the probability of obtaining exactly k successes in n draws *without replacement* from a finite population of size N containing K successes:

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}} \tag{3.10}$$

This expression follows from counting the number of ways of choosing k successful items among the K available and $n-k$ unsuccessful items among the remaining $N-K$, divided by the total number of possible samples of size n.

The hypergeometric distribution has mean and variance:

$$\mathbb{E}[X] = n\frac{K}{N}, \qquad \sigma^2 = n\frac{K}{N}\left(1 - \frac{K}{N}\right)\frac{N-n}{N-1}$$

In practice, the hypergeometric distribution generalizes Bernoulli trials to sampling without replacement, where the success probability is history-dependent. This is why is expressed combinatorially rather than in terms of a fixed p.

Example 3.3.2

An urn contains 40 white balls and 60 black balls. If you draw 20 balls at random without replacement, what is the probability that at least 50 black balls remain in the urn at the end?

The total number of balls is $N = 100$. Let X be the random number of black balls drawn in the $n = 20$ draws. Since we want at least 50 black balls remaining in the urn, $60 - X \geq 50$, that is $X \leq 10$. Since the draws are without replacement, X follows a hypergeometric distribution:

$$P(X = k) = \frac{\binom{60}{k}\binom{40}{20-k}}{\binom{100}{20}}$$

Therefore, the required probability is:

$$P(X \leq 10) = \sum_{k=0}^{10} \frac{\binom{60}{k}\binom{40}{20-k}}{\binom{100}{20}} \simeq 0.22$$

3.3.5 Poisson distribution

One of the most important discrete distribution is the **Poisson distribution**. It describes the number of events occurring in a fixed interval when events happen independently at constant rate λ:

$$P(X = x) = \frac{\lambda^x}{x!}e^{-\lambda}, \qquad x = 0, 1, 2, \ldots \tag{3.11}$$

For example, the number of customers that arrives in a certain time in a shop are described by a Poisson distribution.

The mean is:

$$\mathbb{E}[X] = e^{-\lambda} \sum_{x=1}^{\infty} x \frac{\lambda^x}{x!} = \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} = \lambda$$

Similarly, one proves all cumulants equal to λ.

Let's see the *Poisson as rare-event limit of Binomial*. Let's start from the binomial distribution:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Assume that we are drawing from this distribution but with a very large sample, that is, $n \to \infty$, with $p \to 0$, meaning that is a rare event. Moreover, $np \to \lambda$. Using the following approximation,

$$\binom{n}{k} \approx \frac{n^k}{k!}, \qquad \binom{n}{k} p^k \approx \frac{\lambda^k}{k!}, \qquad (1-p)^{n-k} \approx e^{-\lambda}$$

we obtain the Poisson distribution:

$$P(X = k) \approx \frac{\lambda^k}{k!} e^{-\lambda}$$

3.4 Standard continuous distributions

Continuous distributions describe random variables taking values in $\mathbb{R}$ or subsets thereof and are characterized by a PDF. The most, widely used in Bayesian inference are:

- Uniform (non-informative prior);
- Normal (CLT + analytical tractability);
- Exponential (waiting times);
- Erlang / Gamma (sums of exponentials);
- Beta.

3.4.1 Uniform distribution

The uniform distribution on an interval $[a, b]$ has constant density:

$$f(x) = \frac{1}{b-a}, \qquad x \in [a, b] \tag{3.12}$$

We denote such a uniformly distribute variables as $X \sim \mathcal{U}(0, 1)$. The expected value and the variance are:

$$\mathbb{E}[X] = \frac{a+b}{2}, \qquad \sigma_X^2 = \frac{(b-a)^2}{12}$$

It models *complete ignorance* with fixed bounds and is often used as a simple prior. Therefore, if we want to estimate a distribution for parameters but we do not have any previous knowledge, this will be updated by data through likelihood but as prior we have to use the uniform, which reflects ignorance.

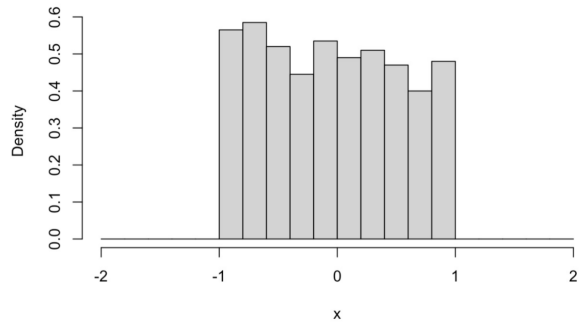


Figure 3.1: Uniform distribution sampled in $[-1, 1]$.

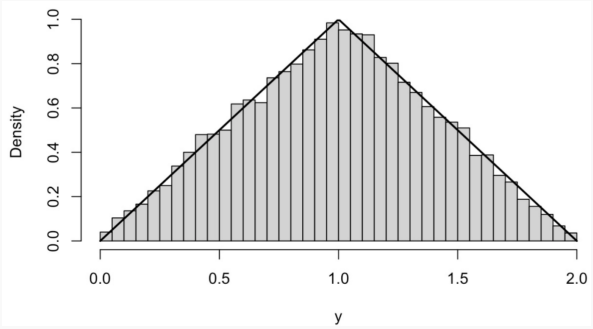
Example 3.4.1

Take X_1 and X_2 independent random variables with uniform distribution in the interval $[0, 1]$. Find the distribution of $Y = X_1 + X_2$.

First, we observed that the support of $f_Y(y)$ will be in the interval $[0, 2]$. By definition, we obtain the PDF of Y as:

$$\begin{aligned} f_Y(y) &= \int_0^1 dx_1 \int_0^1 dx_2 \, f_{X_1}(x_1) f_{X_2}(x_2) \delta(x_1 + x_2 - y) = \\ &= \int_0^1 dx_1 \, \mathbb{1}_{0 \leq y - x_1 \leq 1} = \\ &= \int_{\max(0, y-1)}^{\min(1, y)} dx_1 = \\ &= \min(1, y) - \max(0, y - 1) \end{aligned}$$

where we used $f_{X_i}(x_i) = 1$. This is a *triangular distribution* with expected value $\mathbb{E}[Y] = 1$ and variance $\sigma_Y^2 = \int dy \, (y-1)^2 f_Y(y) = 1/6$. This is the special case $n = 2$ of the Irwin-Hall distribution, which is a sum on n i.i.d. random variables in the interval $[0, 2]$.



3.4.2 Normal or Gaussian distribution

The **normal (Gaussian) distribution** with mean μ and variance σ^2 has PDF:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} \tag{3.13}$$

We denote such a Gaussian distributed variable as $X \sim \mathcal{N}\left(\mu, \sigma^2\right)$. If $X_i \sim \mathcal{N}\left(\mu_i, \sigma_i^2\right)$ are n independent random variables, it holds true that any linear combination $Y = \sum_i a_i X_i$ will be again a Gaussian distributed random variable with mean $\sum_i a_i \mu_i$ and variance $\sum_i a_i^2 \sigma_i^2$. Thus, we say that Gaussian distribution is a **stable** distribution, meaning that the sum is still Gaussian (if independent):

$$\sum_i a_i X_i \sim \mathcal{N}\left(\sum_i a_i \mu_i, \sum_i a_i^2 \sigma_i^2\right)$$

The normal distribution plays a central role due to the central limit theorem (CLT) and its analytical convencienc in Bayesian inference.

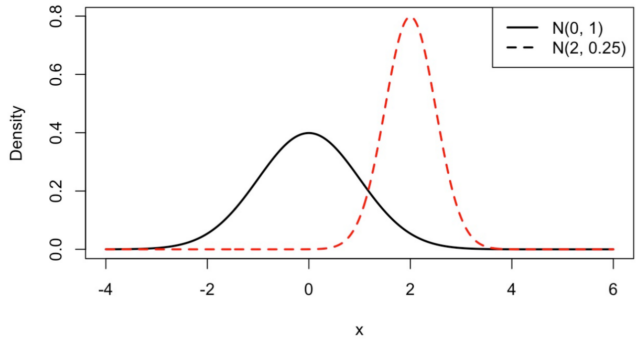


Figure 3.2: Two Gaussian distributions sampled in `R`.

3.4.3 Exponential distribution

The **exponential distribution** with rate $\lambda > 0$ has density:

$$f(t) = \lambda e^{-\lambda t}, \qquad t \geq 0 \tag{3.14}$$

Its mean and variance are, respectively,

$$\mathbb{E}\left[X\right] = \frac{1}{\lambda}, \qquad \mathrm{Var}(X) = \frac{1}{\lambda^2}$$

It describes waiting times between Poissonian events. Consider a Poisson process with constant rate $\lambda > 0$. Let $N(t)$ be the number of events occurring in the time interval $[0, t]$. By definition of a Poisson process,

$$P(N(t) = k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}, \qquad k = 0, 1, 2, \ldots$$

Let T be the waiting time until the first event occurs. The event $\{\, T > t \,\}$ means that no events have occurred up to time t, i.e. $\{\, T > t \,\} \equiv \{\, N(t) = 0 \,\}$. Therefore,

$$P(T > t) = P(N(t) = 0) = e^{-\lambda t}$$

The CDF of T is then:

$$F_T(t) = P(T \leq t) = 1 - e^{-\lambda t}, \qquad t \geq 0$$

Differentiating with respect to t, we obtain the PDF (3.14).

The exponential distribution is memoryless, i.e. ,

$$P(T > t + s \mid T > t) = P(T > s), \qquad \text{with } s, t \geq 0$$

To show it, we compute:

$$P(T > s + t \mid T > s) = \frac{P(T > s + t \cap T > s)}{P(T > s)} = \frac{P(T > s + t)}{P(T > s)}$$

Using the survival function of the exponential distribution:

$$P(T > u) = \int_u^\infty \lambda e^{-\lambda x}\, dx = e^{-\lambda u}$$

we obtain

$$P(T > s + t \mid T > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}$$

3.4.4 Erlang distribution

The **Erland distribution** is the distribution of the sum of n i.i.d. exponential random variables with rate λ:

$$f(t) = \frac{\lambda^n t^{n-1}}{(n-1)!} e^{-\lambda t}, \qquad t \geq 0 \tag{3.15}$$

It describes the waiting time until the n-th Poisson event.

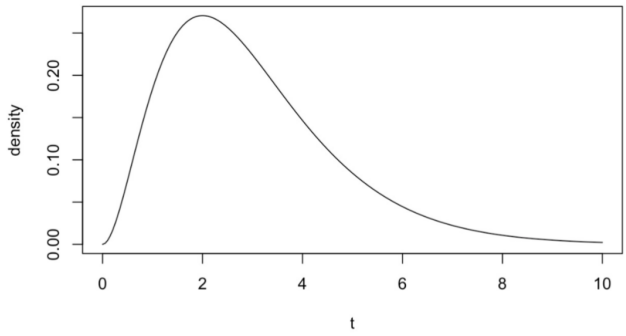


Figure 3.3: Erlang distribution.

3.4.5 Gamma distribution

The **gamma distribution** generalizes the Erland distribution by extending $n \mapsto \alpha$ to a positive real shape parameter $\alpha \in \mathbb{R}^+$. The rate λ is typically denoted with β. Thus, the PDF is:

$$f(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t}, \qquad t \geq 0 \tag{3.16}$$

where $\Gamma(\alpha)$ is the gamma function and it appears as normalization factor as it generalizes the factorial to real values. We denote a gamma distributed variable as $X \sim \text{Gamma}(\alpha, \beta)$ with shape parameter α and scale parameter β. In fact, if $X \sim \text{Gamma}(\alpha, \beta)$, then $cX \sim \text{Gamma}(\alpha, \beta/c)$ for any $x > 0$. This is useful to rescale the rate.

The mean and variance are, respectively,

$$\mu = \frac{\alpha}{\beta}, \qquad \sigma^2 = \frac{\alpha}{\beta^2}$$

The CDF of the gamma distribution does not exist in explicit form.

The gamma distribution is widely used as a conjugate prior for Poisson and exponential likelihoods. A special case is $\alpha = k/2$ and $\gamma = 1/2$, where $\gamma = 1/\beta$, which yields the chi-squared distribution χ^2_k representing the sum of squares of k independent standard normal random variables.

Figure 3.4 shows how the two parameters affect the shape of the Gamma distribution. Increasing the rate parameter β shifts the distribution toward smaller values and reduces both the mean and the variance.

Increasing the shape parameter α shifts the distribution toward larger values. Although the variance also increases with α, the relative dispersion decreases, making the distribution more concentrated around its mean for large values of α. Since α and β affect the mean and variance differently, they can be tuned independently to control both the location and the spread of the distribution.

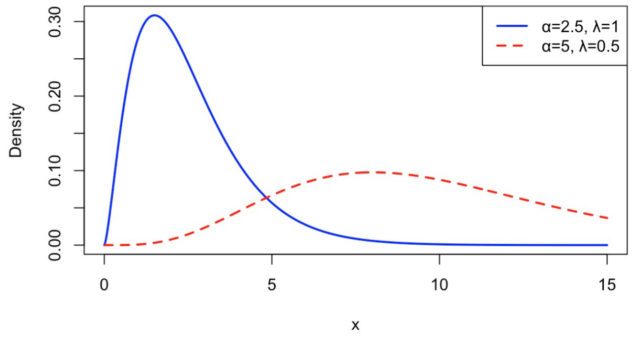


Figure 3.4: Gamma distributions: effect of shape and rate.

An important property is that the *sum of exponential random variables is a gamma distribution*. In fact, Let X_i be i.i.d. exponential random variables with rate $\lambda > 0$. We want to find the distribution of $S_n = \sum_{i=1}^n X_i$. The MGF of a single exponential is:

$$M_{X_i}(t) = \mathbb{E}\left[e^{tX_i}\right] = \frac{\lambda}{\lambda - t}, \qquad t < \lambda$$

Therefore, the MGF of the sum is:

$$M_{S_n}(t) = \mathbb{E}\left[e^{tS_n}\right] = \prod_{i=1}^{n} M_{X_i}(t) = \left(\frac{\lambda}{\lambda - t}\right)^n, \qquad t < \lambda$$

which is the MGF of a $\text{Gamma}(n, \lambda)$.

3.4.6 Beta distribution

The **beta distribution** has support on $[0, 1]$ with parameters $\alpha, \beta > 0$ has density:

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1} \tag{3.17}$$

where the normalization constant is the beta function:

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}\, dt = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

which is a symmetric function.

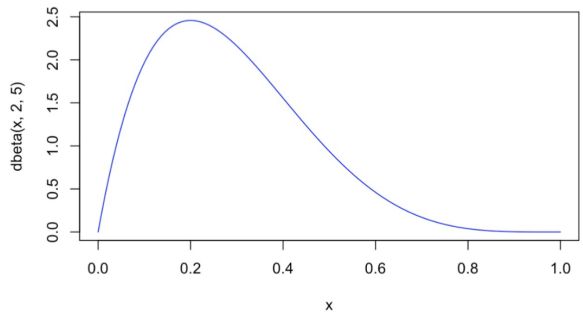


Figure 3.5: Beta distribution.

We denote a beta distributed random variable as $X \sim \text{Beta}(\alpha, \beta)$. Its mean and variance of the beta distribution are, respectively,

$$\mu = \frac{\alpha}{\alpha+\beta}, \qquad \sigma^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

A single beta can be convex or concave. The PDF (3.17) is unimodal when $\alpha, \beta > 1$. Depending on the values of the parameters, the beta can also be uniform, monotonic, or U-shaped with divergences at the boundaries. A common choice to model bimodal distribution in the unit interval is by mixing two different beta distributions, e.g., $X \sim \frac{1}{2}\text{Beta}(\alpha_1, \beta_1) + \frac{1}{2}\text{Beta}(\alpha_2, \beta_2)$. Furthermore, the beta distribution is the natural conjugate

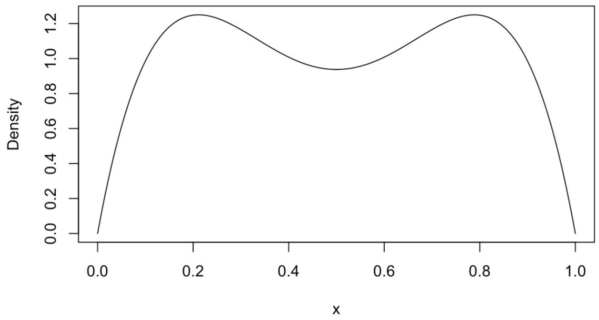


Figure 3.6: $0.5\,\text{Beta}(2, 5) + 0.5\,\text{Beta}(5, 2)$

prior for the Bernoulli and binomial distributions. Therefore, if we use it as a prior for modeling or a parameter in the unit interval, for instance we have a Bernoulli trial with a certain probability p and we want to estimate it. This is a Bayesian inference scheme

of a stochastic object that lies in $[0, 1]$. In this case it becomes handy as prior because it works as conjugate prior for the Bernoulli or binomial. In fact, assuming initially that the p is distributed according to a beta, before looking at any coin tossing, we see that the estimation of p is gonna be itself a beta distribution.

In Figure 3.7 we summarize the probability distributions introduced so far and show how they are related to each other. Starting from Bernoulli trials, we can construct several discrete distributions describing the number of successes or the waiting time in repeated experiments. Their continuous counterparts arise when considering waiting times in continuous-time processes, leading for example to the exponential and gamma distributions.

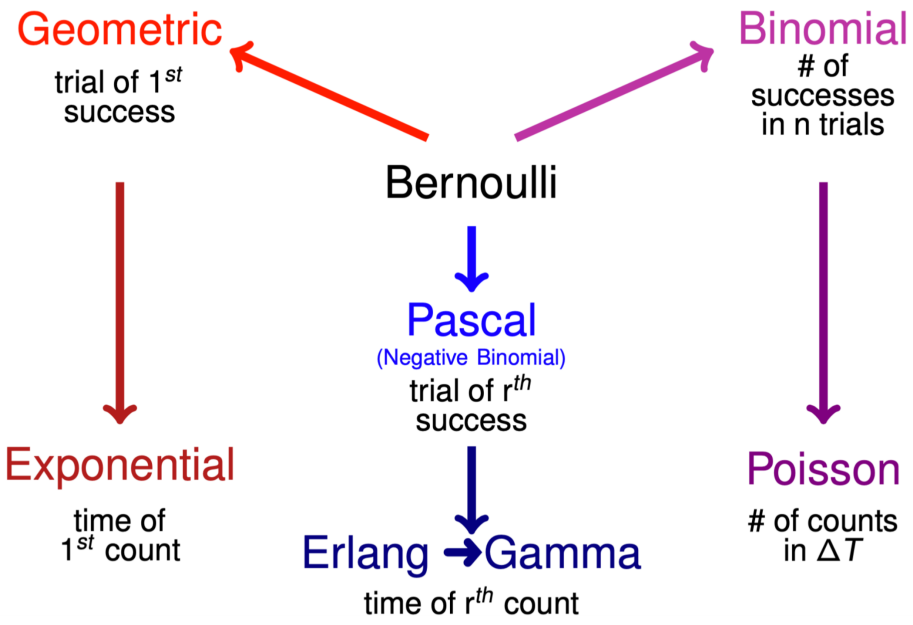


Figure 3.7: Relations between distributions.

3.5 Limiting theorems and asymptotic structure

Limiting theorems describe the behavior of *sums of random variables* in the large-sample limit and form a cornerstone of both classical and Bayesian statistics. They justify the emergence of deterministic behavior, Gaussian fluctuations, and non-Gaussian rare events from microscopic randomness. We review the *Law of Large Numbers* (LLN), the *Central Limit Theorem* (CLT), and the *Large Deviation Principle* (LDP), emphasizing their conceptual unity and their role in Bayesian inference.

Consider n i.i.d. variables $\{ X_i \}$ consider the **sample mean**:

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i \tag{3.18}$$

therefore, they form a hierarchy of asymptotic descriptions:

- the LLN captures deterministic limits $\overline{X}_n \to \mu$;
- the CLT captures typical Gaussian fluctuations $\sim n^{-1/2}$;
- LDP captures rare, non-Gaussian events $\sim e^{-n\mathcal{I}(x)}$.

Together, they provide a multiscale description of uncertainty that is fundamental to modern Bayesian inference and its asymptotic analysis:

$$\text{Randomness: determinism + fluctuations + rare events}$$

3.5.1 Law of large numbers

Let $\{ X_i \}$ be a sequence of n i.i.d. random variables. As previously defined, consider the sample mean:

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$$

There exist two different formulation of the LLN. They can be proved under the assumptions of finite mean $\mathbb{E}[X_i] = \mu$ and variance $\mathrm{Var}(X) = \sigma^2$ (or under milder conditions with some more work):

- **Weak law of large numbers**: the sample mean *converges in probability*[1] to the expected value:

$$\overline{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu$$

- **Strong law of large numbers**: the sample mean *converges almost surely*[2] to the

[1] The sequence X_n is said to converge in probability to X, written $X_n \xrightarrow{\mathbb{P}} X$, if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0$$

[2] The sequence X_n is said to converge almost surely to X, written $X_n \xrightarrow{\text{a.s.}} X$, if,

$$\mathbb{P}\left(\lim_{n\to\infty} X_n = X \right) = 1$$

or equivalently

$$\mathbb{P}(\{ \omega \in \Omega : X_n(\omega) \to X(\omega) \}) = 1$$

expected value:

$$\overline{X}_n \xrightarrow[n\to\infty]{\text{a.s.}} \mu$$

The law of large numbers states that averaging suppresses randomness: extensive quantities become deterministic in the large-sample limit. Fluctuations around the mean vanish relative to the size of the system, meaning that the fluctuations die out.

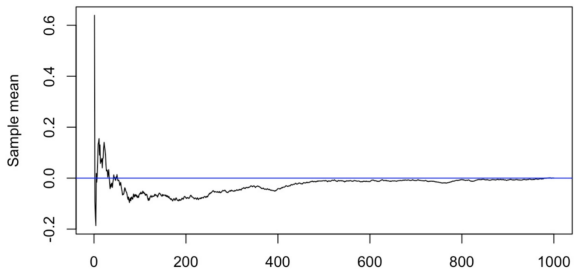


Figure 3.8

Bayesian relevance. In Bayesian inference, the LLN underlies:

- posterior consistency, whereby accumulating data overwhelms the prior. In other words, data accumulation ⇒ prior becomes irrelevant (very nice if our prior is wrong);
- convergence of empirical averages to posterior expectations, that is the posterior concentrates near *true* parameter value;
- the interpretation of likelihoods as empirical averages of log-probability contributions.

In this sense, Bayesian updating can be viewed as a controlled application of the LLN in probability space. Basically, the key idea is that infinite data implies that the inference becomes *deterministic*.

3.5.2 Central limit theorem

Let $\{ X_i \}$ be i.i.d. random variables with mean μ and finite variance σ^2. Then, the PDF of (the scaled and shifted sum):

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} (X_i - \mu) \xrightarrow{n\to\infty} \mathcal{N}\left(0, \sigma^2\right)$$

In other words, we can say that the sample mean $\overline{X}_n \approx \mu \pm \sigma/\sqrt{n}$ as $n \to \infty$.

CLT describes the universal emergence of Gaussian fluctuations around the deterministic limit prescribed by the law of large numbers, whatever the original distribution of data is. Remarkably, the limiting distribution depends only on the first two moments (this property is called **universality**) and is largely independent of the microscopic details of the original distribution.

Bayesian relevance. The CLT explains several ubiquitous Gaussian structures in Bayesian statistics:

- asymptotic normality of posterior distributions (Bernstein-von Mises theorem): Posterior $\approx$ Gaussian for large n;
- Gaussian approximations to likelihoods and posteriors (Laplace's method); Posterior covariance around *true parameter* gives relaiability of estimation;
- the prevalence of quadratic log-likelihoods and Fisher information as a local curvature.

From this viewpoint, Gaussian posteriors arise not as fundamental objects, but as effective descriptions of typical fluctuations in large-data limits.

So here the key idea is: large data $\Rightarrow$ inference is deterministic + Gaussian fluctuations.

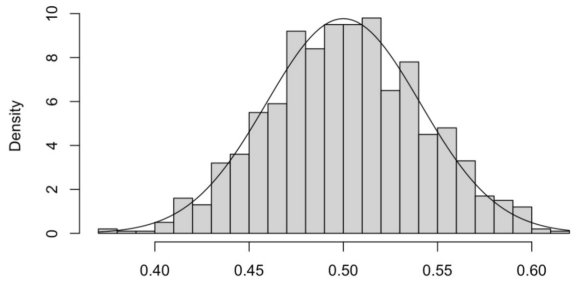


Figure 3.9: Empirical sum of $X \sim \mathcal{U}(0,1)$ with $n = 50$.

3.5.3 Large deviation principle

While the CLT describes typical fluctuations of order $n^{-1/2}$, it does not capture the behavior of rare, large fluctuations. This regime is described by large deviation theory. Let $\{ X_i \}$ be random variables and let $\overline{X}_n$ denote their sample mean. We say that the sample mean satisfies a large deviation principle if[3]:

$$\mathbb{P}\left(\overline{X}_n \approx x\right) \asymp e^{-n\mathcal{I}(x)} \tag{3.19}$$

where $\mathcal{I}(x) \geq 0$ is the **rate function**, vanishing uniquely at $x = \mu$, i.e. $\mathcal{I}(\mu) = 0$. The symbol $\asymp$ means *scaled logarithmic equivalence*, i.e. it is a short-hand for:

$$\mathcal{I}(x) \equiv -\lim_{n\to\infty} \frac{1}{n} \ln \mathbb{P}\left(\overline{X}_n \approx n\right)$$

LDP holds for $\{ X_i \}$ i.i.d. random variables but also under milder conditions (e.g., *short range* correlations between $\{ X_i \}$).

The rate function $\mathcal{I}(x)$ quantifies the exponential cost of atypical outcomes. Expanding $\mathcal{I}(x)$ to second order around its minimum recovers the Gaussian fluctuations of the CLT, while the full function encodes strongly non-Gaussian tails. For $n = \infty$, Equation 3.19 collapses to a Dirac delta functional centered on μ. For this reason, at large but finite n we say that the probability concentrates around the LLN value.

[3]We write $\overline{X}_n \approx x$ to mean that $\overline{X}_n$ takes value in a neighborhood of x since $\overline{X}_n$ is a continuous random variable for $n \to \infty$.

Bayesian relevance. LDPs play a central role in:

- Bayesian consistency and posterior concentration rate;
- the asymptotic form of marginal likelihoods and model evidence;
- PAC-Bayesian bounds and information-theoretic generalization guarantees.

In Bayesian language, the posterior often concentrates as:

$$\pi(\theta \mid \{ y_1, \dots, y_n \}) \asymp e^{-n\mathcal{I}(\theta)}$$

where $\mathcal{I}(\theta)$ acts as an effective rate function combining prior and likelihood contributions.

Here, the key idea is that events far from the mean μ are exponentially rare.

We can summarize everything in a unified picture represented in Table 3.2:

$$\text{LLN} \to \text{CLT} \to \text{LDP} = \text{multiscale structure of uncertainty}$$

Regime	**Scaling**	**Description**
LLN	n	deterministic limit
CLT	$\sqrt{n}$	Gaussian fluctuations
LDP	e^{-n}	tails

Table 3.2: Big picture: probability concentrates as data grows.

3.6 Probabilistic inequalities in Bayesian inference

Several classical inequalities in probability theory play a center role in Bayesian inference. Beyond providing non-asymptotic bounds, they underpin variational approximations, posterior concentration results, and Bayesian optimality principles. We review Markov's, Chebyshev's, and Jensen's inequalities, emphasizing their relevance for Bayesian reasoning.

3.6.1 Markov's inequality and tail bounds

Let X be a non-negative random variable, $X \geq 0$, with finite expectation $\mathbb{E}[X] < \infty$. Then, for any $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a} \tag{3.20}$$

Proof. Since $\frac{X}{a} \geq \mathbb{1}_{X \geq a}$,

$$\mathbb{E}[X] \geq \mathbb{E}[X \mathbb{1}_{X \geq a}] \geq a\mathbb{P}(X \geq a)$$

□

Markov's inequality provides a distribution-free upper bound on tail probabilities using only the first moment. The bound is typically loose, but its generality makes it a fundamental tool.

Bayesian relevance. In Bayesian inference, Markov's inequality is often used to control posterior tails of non-negative quantities. For a posterior distribution $\pi(\theta \mid y)$ and a non-negative loss function $L(\theta)$,

$$\mathbb{P}_{\pi(\theta|y)}(L(\theta) \geq a) \leq \frac{\mathbb{E}_{\pi(\theta|y)}[L(\theta)]}{a} \tag{3.21}$$

Applying Markov's inequality to e^{tX} yields exponential (Chernoff-type) bounds, which form the basis of many Bayesian concentration results.

3.6.2 Chebyshev's inequality and posterior concentration

Let X be a random variable with mean $\mu = \mathbb{E}[X]$ and finite variance $\sigma^2 = \mathrm{Var}(X)$. For any $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2} \tag{3.22}$$

Proof. It follows directly from Markov's inequality applied to the non-negative random variable $(X - \mu)^2$. □

Chebishev's inequality bounds the probability of large deviations from the mean using only second-moment information. It provides a minimal, assumption-free form of concentration.

Bayesian relevance. For a posterior distribution $\pi(\theta \mid y)$,

$$\mathbb{P}(|\theta - \mathbb{E}[\theta \mid y]| \geq k \mid y) \leq \frac{\mathrm{Var}(\theta \mid y)}{k^2} \tag{3.23}$$

As posterior variances shrink (e.g., in large-sample limits), Chebyshev's inequality guarantees posterior concentration in probability, independently of distributional details.

3.6.3 Jensen's inequality and variational structure

Let X be a random variable and let φ be a convex (such as the exp). Then,

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)] \tag{3.24}$$

If φ is concave (such as the log), the inequality is reversed.

Role in Bayesian inference. For observed data y and latent variables θ,

$$\log p(y) = \log \int p(y, \theta)\, d\theta = \log \mathbb{E}_{q(\theta)}\left[\frac{p(y, \theta)}{q(\theta)}\right] \geq \mathbb{E}_{q(\theta)}\left[\log \frac{p(y, \theta)}{q(\theta)}\right] \tag{3.25}$$

This defines the evidence lower bound (ELBO), whose maximization is the basis of variational Bayesian inference.

Finally, notice that the non-negativity of the Kullback-Leibler divergence

$$\mathrm{KL}(q \,\|\, p) \equiv \mathbb{E}_q\left[\log \frac{q}{p}\right] \geq 0 \tag{3.26}$$

is a direct consequence of Jensen's inequality and underlies Bayesian optimality and consistency results.

Chapter 4

Inference Problems

We are now entering into the core of the course. We will explore the different inference tasks, also presenting short comparison between Bayesian and inference viewpoints. We have already seen that in **frequentist inference**, probability refers to long-run frequencies under repeating sampling which converges to *a priori probability* with which data are distributed. Parameters are fixed but unknown, the only randomness is in the data. In **Bayesian inference**, probability quantifies uncertainty or degree of belief: there are no *a priori probability*. Basically, this refers to how much we believe that something is probable. Parameters (or models) are treated as random variables, thus we have to estimate their probability distributions. The objective is to update the knowledge via data, starting from some prior knowledge, we update it.

4.1 Model comparison

In Bayesian inference, a specific hypothesis H is identified with a probabilistic model for the data D. We assign a prior probability $P(H)$ to each competing hypothesis.

Let $\{ H_0, H_1 \}$ be two mutually exclusive and exhaustive hypotheses, so that:

$$P(H_0) + P(H_1) = P(H_0 \cup H_1) = P(\Omega) = 1$$

(and the same holds for the conditionals $P(H_i \mid D)$) each of which specifies a likelihood $P(D \mid H_i)$, meaning that given that H_i is true, we observe data D. Note that, if the model depends on parameters θ, then

$$P(D \mid H) = \int P(D \mid \theta, H) P(\theta \mid H) \, d\theta$$

This quantity is called the **marginal likelihood** or **model evidence**. Hereafter, to illustrate the theory, we stick to the simplest case of two competing models without parameters (like *true* or *false*, or *yes* or *no*).

Applying Bayes' theorem to the hypotheses (treated as random variable that can take only two values) gives:

$$P(H_i \mid D) = \frac{P(D \mid H_i) P(H_i)}{P(D)} \tag{4.1}$$

where the normalization is:

$$P(D) = P(D \mid H_0) P(H_0) + P(D \mid H_1) P(H_1)$$

We now define the **posterior odds ratio** R which reads,

$$R \equiv \frac{P(H_1 \mid D)}{P(H_0 \mid D)} = \frac{P(D \mid H_1)P(H_1)}{P(D \mid H_0)P(H_0)} = \text{BF} \times \frac{P(H_1)}{P(H_0)} \tag{4.2}$$

where we used Bayes' theorem. Basically, this factor tells us the relative probabilities of two hypotheses. The quantity

$$\text{BF} \equiv \frac{P(D \mid H_1)}{P(D \mid H_0)}$$

is called the **Bayes factor** and tells us the proportionality factor between the posterior odds and the prior odds. With this definition, we can rewrite (4.1) for the posterior as:

$$P(H_0 \mid D) = \frac{1}{1+R}, \qquad P(H_1 \mid D) = \frac{1}{1+1/R} \tag{4.3}$$

Basic model selection We say that H_1 is more likely than H_0 conditional on the data D, if the odds ratio $R > 1$. Often, statisticians rather report BF > 1 as a criterion for H_1 to be more likely than H_0. The reason being that BF contains only objective information coming from data, discarding priors "*subjectivity*". The two criteria are identical with uniform priors (check as exercise).

Example 4.1.1

Let's consider the following medical test. A test for a disease is 90% reliable. The probability of testing positive, in absence of the disease, is 0.07. We know that among people aged 40 to 50 with no symptoms 8 in 1000 have the disease.
If a person in their 40s tests positive, what is the probability that they have the disease?

We summarize the above info as:

- $H_0 \equiv H$ sick, $H_1 = \overline{H}$ healthy, D positive test
- $P(D \mid H) = 0.9$
- $P(D \mid \overline{H}) = 0.07$
- $P(H) = 0.008$
- Compute $P(H \mid D)$.

The odds ratio is:

$$R = \frac{P(D \mid H)P(H)}{P(D \mid \overline{H})P(\overline{H})} = \frac{9 \cdot 10^{-1} \times 8 \cdot 10^{-3}}{7 \cdot 10^{-12} \times \underbrace{(1 - 8 \cdot 10^{-3})}_{1-P(H)}} = 0.1035$$

Therefore,

$$P(H \mid D) = \frac{1}{1+R^{-1}} \approx 0.094$$

Interpretation: after seeing a positive test, the odds are about $R \approx 1 : 9.7$ against the disease. Notice that this result is composed on the Bayes factor (just data) and the prior ratio. If we compute separately the Bayes factor, we get BF ≈ 12.86, which is relatively very large, meaning that data (the result of the test) suggest strong evidence of having the disease. However, the disease is very rare (prior odds ≈ 0.008) and this biases the result in the other direction. So strong evidence coming from the Bayes factor is not enough to overcome rarity of the disease which is encoded in the prior odds, resulting in small of probability of being sick given that one results positive.

4.1.1 Marginal likelihood and Occam's razor

Suppose hypothesis H depends on parameters $\theta \in \Theta$ with prior density $\pi(\theta) \equiv P(\theta \mid H)$. Then, the model evidence (marginal likelihood over parameters) is:

$$P(D \mid H) = \int_{\Theta} P(D \mid \theta, H)\pi(\theta)\, d\theta \tag{4.4}$$

We assume that the likelihood $P(D \mid \theta, H)$ is sharply peaked around its maximum $\hat{\theta}$ (small fluctuations) and the prior $\pi(\theta)$ is smooth on the scale of the likelihood peak. Applying Laplace's approximation,

$$P(D \mid H) \approx \underbrace{P(D \mid \hat{\theta}, H)}_{\text{best fit}} \times \underbrace{\pi(\hat{\theta})\Delta\theta_{\text{eff}}}_{\text{Occam factor}}$$

where $\Delta\theta_{\text{eff}}$ is the effective width of the likelihood peak. The Occam factor gives the fraction of parameter space that fits the data. Complex models spread prior mass over large parameter volumes, with only a small region that fits the data well, i.e. $\pi(\hat{\theta})\Delta\theta_{\text{eff}} \ll 1$. The integral automatically penalizes complexity on parameter space, according to **Occam's razor**:

Even though a more flexible model (more parameters) can fit the data better at its optimal parameter value, its overall predictive performance may be worse because most of its parameter space does not explain the data well.

Thus, Bayes structure directly encode that *simple models tend to make precise predictions.*

As a simple illustration for the Occam's razor, consider the two models:

- H_0 with $\theta = \theta_0$, with no free parameters
- H_1 : with $\theta \sim \mathcal{U}(a, b)$

that lead to the posterior distributions, respectively,

$$P(D \mid H_0) = P(D \mid \theta_0, H) \tag{4.5a}$$

$$P(D \mid H_1) = \frac{1}{b-a}\int_a^b P(D \mid \theta, H)\, d\theta \approx P(D \mid \hat{\theta}, H)\frac{\Delta\theta_{\text{eff}}}{b-a} \tag{4.5b}$$

Notice that in the first one, there is no marginalization in the likelihood because the parameter has just one value. In the other case, we marginalize over a flat distribution.

By definition of max, $P(D \mid \theta_0, H) \leq P(D \mid \hat{\theta}, H)$, meaning that H_1 can fit better the data. However, if $\theta_{\text{eff}} \ll b - a$, we end up with $P(D \mid H_1) < P(D \mid H_0)$. Hence, $P(D \mid H_1)$ will be smaller than $P(D \mid H_0)$ despite having a higher maximum likelihood.

Marginal likelihoods balance fit quality and model complexity, with Occam's razor being a mathematical consequence of integration over parameters.

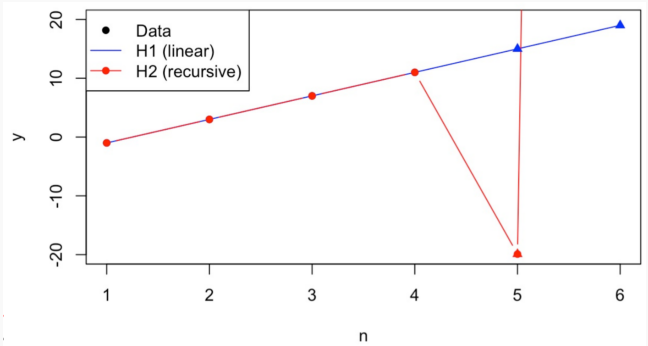
Example 4.1.2

Consider the sequence $D = (-1, 3, 7, 11)$. What are the next two numbers in the sequence?

Let's consider two hypotheses :

- $H_1 : D_{\text{new}} = (15, 19)$ from the arithmetic progression $y_n = y_1 + d(n-1)$
- $H_2 : D_{\text{new}} = (-19.9, 1043.8)$ from cubic interpolation $y_n = -y_{n-1}^3/11 + 9y_{n-1}^2/11 + 23/11$

Both models fit the data perfectly.



We want to estimate the Bayes factor and the odds ratio for the parameters. The simplest model H_1 is an arithmetic progression:

$$y_n - y_{n-1} = d$$

It has one parameter (spacing) and low complexity. The complex one is H_2 (cubic):

$$y_n = \alpha y_{n-1}^3 + \beta y_{n-1}^2 + \gamma y_{n-1}$$

with three parameters (α, β, γ) and high flexibility We assume equal priors to not bias our model selection. In this case, the odds ratio equals the Bayes factor:

$$\frac{P(H_2)}{P(H_1)} = 1 \quad \Rightarrow \quad R = \text{BF} = \frac{P(D \mid H_2)}{P(D \mid H_1)}$$

By construction, both models interpolate the data exactly and the likelihood is:

$$P(D \mid \hat{\theta}_1, H_1) = P(D \mid \hat{\theta}_2, H_2)$$

This means that the best-fit likelihood cancels in the Bayes factor and only the priors over the parameters times the width of the distribution remain

$$\text{BF} \approx \frac{\pi_2(\hat{\theta}_2)\Delta\theta_2}{\pi_1(\hat{\theta}_1)\Delta\theta_1} \equiv \frac{\text{Occam}_2}{\text{Occam}_1}$$

where the **Occam factor** is the fraction of parameter space that fits the data.
We now assume uniform priors over parameter ranges of size L. Thus, the Occam factors are:

$$\text{Occam}_1 \approx \frac{\Delta\theta_d}{L_d}, \qquad \text{Occam}_2 \approx \frac{\Delta\theta_\alpha \Delta\theta_\beta \Delta\theta_\gamma}{L_\alpha L_\beta L_\gamma}$$

Moreover, we assume L of the same magnitude of the values y_n. Roughly, we say $L \approx 10$.
We now want to estimate the width of the likelihood. Thus, in general, we have to assume noisy observations:

$$y_n^{\text{obs}} = y_n(\theta) + \varepsilon_n, \qquad \varepsilon_n \sim \mathcal{N}\left(0, \sigma^2\right)$$

The key idea is that parameter uncertainty is set by how much predictions change under perturbations. Since we consider a polynomial model of the type:

$$y_n = \sum_k \theta_k n^k$$

a small variation $\delta\theta_k$ produces a variation on the predicted value $\delta y_n \sim n^k \delta\theta_k$. In order to remain compatible with the data, the variation must be of the order of the noise, that is $\delta y_n \lesssim \sigma$, so:

$$\Delta\theta_k \sim \frac{\sigma}{n^k} \sim \frac{\sigma}{N^k}$$

where we took $n \sim N$ (largest scale in the data). Therefore, the width of the distribution of the likelihood for each parameter must be of the order of the noise in the observations divided by N^k. This means that the Occam factors scale with the inverse of a power of the number of parameters. In fact, comparing the two parameters:

- **Simple model** H_1 with one parameter d:

$$\Delta d \sim \frac{\sigma}{N} \quad \Rightarrow \quad \text{Occam}_1 \sim \frac{\sigma}{NL}$$

- **Cubic model** H_2 with three parameters (α, β, γ):

$$\Delta\alpha \sim \frac{\sigma}{N}, \quad \Delta\beta \sim \frac{\sigma}{N}, \quad \Delta\gamma \sim \frac{\sigma}{N} \quad \Rightarrow \quad \text{Occam}_2 \sim \frac{\sigma^3}{L^3 N^6}$$

Therefore, we get:

$$R \sim \frac{\text{Occam}_1}{\text{Occam}_2} \sim \frac{\sigma^2}{L^2 N^5} = \frac{\sigma^2}{19^2 \times 4^5} \ll 1$$

The big sensitivity of our parameters go into the denominator, which biases our preference towards a model with low order parameters, meaning that higher-order parameters must be finely tuned and a simpler model is strongly favored:

$$R = \frac{P(H_2 \mid D)}{P(H_1 \mid D)} \ll 1$$

Thus, the cubic model fits the data only in a tiny region of parameter space, while the linear model fits it robustly. The generalization of this is that for $N \gg 1$ independent data, the width of the likelihood distribution goes as $\Delta\theta \sim N^{d/2}$ as a consequence of Laplace and CLT with d parameters.
The physical interpretation is that, for the simple model H_1, many parameter values give similar predictions and thus the model is stable and predictive, while the complex model H_2 requires fine tuning of parameters and fits data as a coincidence.

In the Bayesian framework, complex models are penalized because most of their parameter space does not explain the data.

The final takeaway is the following:

- Fit quality is not enough in choosing a model!!!
- Model complexity matter
- Occam's razor emerges automatically from integration:

$$\text{BF} \approx \frac{\text{good parameter volume}}{\text{total parameter volume}}$$

4.2 Hypothesis testing in frequentist approach

In the frequentist framework, model comparison is formulated as a problem of hypothesis testing based on the sampling properties of observables. Probability is interpreted operationally, as arising from repeated realizations of the same experiment, while model parameters are treated as fixed but unknown quantities. Therefore, we have a probability distribution for data that depends on some parameters which are fixed.

Let $(\Omega, \mathcal{F}, \mathbb{P}_\theta)$ denote a statistical model, where $\theta \in \Theta$ is a vector of unknown parameters on which $\mathbb{P}_\theta$ depends. A hypothesis is defined as a subset of the parameter space. In its simplest form, one considers exclusive hypotheses, i.e.

$$H_0 : \theta \in \Theta_0, \qquad H_1 : \theta \in \Theta_1$$

with

$$\Theta_0 \cap \Theta_1 = \varnothing \quad \text{and} \quad \Theta_0 \cup \Theta_1 \subseteq \Theta$$

In this setting, θ is not a random variable, and no probability measure is assigned to Θ_0 or Θ_1. Instead, the goal is to assess whether the observed data are compatible with the assumption that the true (but unknown) parameter value lies in Θ_0. Different statistical models can be compared by embedding them into a common parametric family and interpreting each model as a hypothesis restricting the admissible parameter values.

Given an observation $X \in \Omega$, inference is based on a measurable test statistic $T : \Omega \to \mathbb{R}$ chosen so that large (or small) values $T(X)$ indicate disagreement with the null hypothesis

H_0. This answers to the question: *how well does parameter θ explain the observed data X?* The distribution of $T(X)$ under H_0,

$$F_{T,0}(t) = \mathbb{P}_\theta(T(X) \leq t), \qquad \theta \in \Theta_0$$

is called the **sampling distribution** of the test statistic.

In many cases, T is constructed from the likelihood function $\mathbb{P}_\theta(X)$, leading to the **likelihood ratio** statistic:

$$\Lambda(X) = \frac{\text{best fit under } H_0}{\text{best fit overall}} = \frac{\sup_{\theta\in\Theta_0} \mathbb{P}_\theta(X)}{\sup_{\theta\in\Theta} \mathbb{P}_\theta(X)}$$

- $\Lambda \simeq 1$: restricted model (null hypothesis) explains as well as full one
- $\Lambda \ll 1$: full model explains much better $\Rightarrow$ evidence against H_0.

By taking the logarithm of the likelihood ratio, we get the **log-likelihood difference**:

$$T(X) = -2\log\Lambda(X) = 2(\log\Lambda_{\text{full}} - \log\Lambda_{H_0})$$

which has good properties. Under suitable regularity conditions and for nested hypotheses, $T(X)$ converges in distribution to a χ^2_k random variable with k degrees of freedom equal to $k = \dim\Theta - \dim\Theta_0$ (Wilk's theorem).

p-values and rejection regions. A test is defined by a *rejection region* $\mathcal{R} \in \Omega$, with the null hypothesis rejected if $X \in \mathcal{R}$. The size of the test is:

$$\sup_{\theta\in\Theta_0} \mathbb{P}_\theta(X \in \mathcal{R}) = \alpha$$

where $\alpha \in (0, 1)$ is the prescribed significance level.

Equivalently, one can define and compute the p**-value** associated with the observed data X_{obs}, which tells us what is the probability that at best the likelihood difference on the data is gonna be larger than the one on the observed data. That is,

$$p(X_{\text{obs}}) = \sup_{\theta\in\Theta_0} \mathbb{P}_\theta(T(X) \geq T(X_{\text{obs}}))$$

for a one-sided test (with obvious modifications for two-sided tests). The decision rule is:

$$\text{reject } H_0 \text{ if } p(X_{\text{obs}}) < \alpha$$

Unlike Bayesian model comparison, this procedure does not assign relative weights or posterior probabilities to competing models. Instead, it provides a binary decision based on whether the observed data fall into a region that is unlikely under H_0.

Error control and power. Frequentist testing is characterized by explicit control of error probabilities under repeated sampling. A **type I error** corresponds to rejecting H_0 when $\theta \in \Theta_0$, with probability bounded by α. A **type II error** corresponds to failing to reject H_0 when $\theta \in \Theta_1$, with probability:

$$\beta(\theta) = \mathbb{P}_\theta(X \notin \mathcal{R}), \qquad \theta \in \Theta_1$$

The **power function** of the test is defined as:

$$\pi(\theta) = \mathbb{P}_\theta(X \in \mathcal{R}) = 1 - \beta(\theta)$$

and quantifies the sensitivity of the test to deviations from the null hypothesis. Optimal tests are often constructed by maximizing power subject to a fixed type I error constraint, as formalized by the Neyman-Pearson lemma.

From this perspective, frequentist model comparison emphasizes long-run performance guarantees and controlled error frequencies, rather than degrees of belief or probabilistic statements about model validity.

Bayesian vs Frequentist

- Bayesian:
  - Assign probabilities to hypotheses
  - Use Bayes factor
  - Penalizes complexity (Occam)
- Frequentist:
  - No probability for hypotheses, only for data
  - Use p-values and tests
  - Focus on error control

4.3 Parameter estimation

Up to now, we have seen that we have to deal with parameters estimation which enter in the definition of the hypotheses. To do this, we have to introduce the concept of parameter estimation. To do it, we need a so-called generative model. A **generative model** is a theoretical model that predicts observable data from a set of parameters. The simplest example of a generative model is a straight line:

$$f(x; a, b) = a + bx$$

where a and b are unknown parameters. Measurements do not exactly follow the model due to noise. We therefore write:

$$y = f(x; \theta_\mathrm{f}) + \varepsilon$$

where ε represents measurement noise and θ_f the model parameters. The statistical properties of ε are described by a *measurement model*, which may itself depend on additional parameters. Given a dataset $D = \{ y_j, x_j \}$, our goal is to infer the values of the parameters of the generative model.

We denote by $\theta = (\theta_\mathrm{f}, \theta_\varepsilon)$ the full set of parameters, including both generative (or deterministic model) and measurement parameters (or noise parameters). Bayesian inference aims to determine the *posterior probability density function* $P(\theta \mid D, H)$, where H represents the background hypotheses, including the choice of the generative model and the choice of the measurement (noise) model.

Again, Bayes' theorem adapted to this setup states:

$$P(\theta \mid D, H) = \frac{P(D \mid \theta, H)P(\theta \mid H)}{P(D \mid H)}$$

The likelihood $P(D \mid \theta, H)$ is the probability of obtaining the measured data assuming specific values of the parameters. For example, assume that each measurement of y at a fixed x is affected by Gaussian noise with variance σ^2, then the parameter is $\theta = (a, b, \sigma^2)$. Then, the probability of a single measurement is:

$$P(y \mid \theta, H) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y - f(x; a, b))^2}{2\sigma^2}\right]$$

This expresses the fact that measurements are normally distributed around the true model value and fluctuate around model prediction with width controlled by σ. In this case, $\theta = (a, b, \sigma)$ combining both generative and measurement parameters.

Assuming independent measurements, the likelihood for the dataset is:

$$P(D \mid \theta, H) = \prod_j P(y_j \mid x_j, \theta, H)$$

and for Gaussian noise it becomes

$$P(D \mid \theta, H) \propto \exp\left[-\frac{1}{2\sigma^2}\sum_j (y_j - f(x_j; a, b))^2\right]$$

which is connected to least squares:

$$\chi^2 = \sum_j (y_j - f(x_j; a, b))^2$$

The *posterior* $P(\theta \mid D, H)$ is the probability density of the parameters given the observed data and background information. Conceptually, Bayesian inference is straightforward: we update our prior knowledge using the information contained in the data. The *prior* encapsulates all information available before observing the data. It reflects the reality that scientific reasoning is never based on data alone.

The quantity:

$$P(D \mid H) = \int P(D \mid \theta, H)P(\theta \mid H)\, d\theta$$

is called the *evidence*, which is obtained by marginalizing over the parameters. The evidence plays a crucial role in model comparison, normalization of the posterior and computation of expectation values. Analytical solutions are possible only in simple cases. In general, we have high-dimensional integrals and very complex likelihoods. For most realistic problems, however, numerical techniques such as Markov Chain Monte Carlo (MCMC) are required

Summarizing everything, we have seen:

- Generative model + noise defines likelihood
- Bayesian inference updates prior using data
- Posterior contains full parameter information
- Evidence enables normalization and model comparison
- Numerical methods often required

4.4 Choosing priors

We now see the guiding principles for the choice of priors and how to extract *best guess* from posterior and uncertainty, that is point estimation (since we want a *number* $\hat{\theta}$ from the distribution we obtain) and assign errors with Hessian likelihood and credible intervals. We will get acquainted by introducing simple analytical examples of estimation of parameters for Bernoulli, Poisson and Gaussian.

Let's consider an experiment of coin tossing. We toss a coin n times and register the sequence of heads and tails:

$$\text{T H H T H T H H H H T T H T} \ldots \text{T}$$

In total, we see k times head. We then ask: is the coin biased or not?

For this question, we consider a Bernoulli model. Assuming that the properties of the coin do not change during the trials, the data model is assumed to be n independent Bernoulli trials distributed with a certain unknown parameter $p \in [0, 1]$ representing the probability of success:

$$X_1, \ldots, X_n \sim \text{Bernoulli}(p)$$

We denote the observed dataset $D = \{ x_1, \ldots, x_n \}$ where $x_i \in \{ 0, 1 \} \longleftrightarrow \{ \text{T}, \text{H} \}$ is a numerical representation of the abstract outcomes heads and tails. The number of successes is then:

$$k = \sum_{i=1}^{n} x_i$$

The parameter n is fixed by the experiment.

Given these hypotheses, the correct likelihood function to consider is:

$$P(k \mid p, n, H) = \binom{n}{k} p^k (1-p)^{n-k} \propto \text{Binomial}(n, p) \tag{4.6}$$

where n is part of the data but is fixed, i.e. not a random variable.

Our goal is then to estimate p knowing k. The coin will be fair if $p = \frac{1}{2}$.

We have seen that within the Bayesian framework, we have to multiply the likelihood times a prior distribution and a normalization constant. Therefore, our first problem is *choosing priors*. We must specify a prior $\pi(p)$ over the parameter space. Different priors encode different levels of prior information.

Choosing priors. We introduce three *principles guiding the choosing of the prior*:

- it must encode prior knowledge, for instance, previous tosses or the guy that tossed the coin is not reliable, etc.
- maintain invariance (when possible), meaning that if a process is independent to change of coordinates or other variables, we would like to maintain this independence.
- prefer analytical tractability (simplifying assumption, mathematical tractability).

To encode agnosticism on p, a common non-informative choice is the uniform or **flat prior**:

$$p \sim \mathcal{U}(0, 1) \quad \text{or} \quad \pi(p) = \mathcal{U}(0, 1)$$

which assign equal weight to all values of p. This corresponds to a particular case of the beta. In fact, it is equivalent to:

$$\text{Beta}\,(\alpha = 1, \beta = 1) \propto p^{\alpha-1}(1-p)^{1-\beta}\big|_{\alpha=1,\beta=1} = \mathcal{U}(0,1)$$

(we emphasize this because we will see that the beta can be taken as a good general prior for this probability). Choosing such a prior is simple and interpretable: only data matter, we skip one passage and the likelihood will coincide with the posterior.

Another possible choice of non-informative prior is **Jeffreys prior**, which is a family of priors. To understand it, let's consider our likelihood (4.6). Taking the log-likelihood and computing how it changes with respect to the parameter p, we get how sensitive the likelihood is to variation of parameter. Taking then the second moment of the likelihood, we obtain an important object:

$$I(p) = \mathbb{E}\left[\left(\frac{\partial}{\partial p}\ln P(k \mid p, n, H)\right)^2\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial p^2}\ln P(k \mid p, n, H)\right] \tag{4.7}$$

which is called **Fisher information** and measures the amount of information data carry about p[1]. For our specific case, the Bernoulli distribution, we have:

$$I(p) = \frac{1}{p(1-p)}$$

Jeffreys prior is defined as:

$$\pi_{\text{J}}(p) \propto \sqrt{I(p)} \tag{4.8}$$

Therefore, we have:

$$\pi_{\text{J}}(p) \propto \frac{1}{\sqrt{p(1-p)}}$$

which is again a particular case of the beta:

$$p \sim \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$$

One may ask why we have to choose such a prior made in this way, that depends on the likelihood. The reason is very useful and it is because the prior results invariant under reparametrization.

In fact, Fisher information kills the Jacobian of the changing of variables. Given a reparametrization of a distribution, we get a new prior:

$$\pi_\varphi(\varphi) = \pi_\theta(\theta)\left|\frac{\mathrm{d}\theta}{\mathrm{d}\varphi}\right|$$

Since the Fisher information transforms as:

$$I_\varphi(\varphi) = I_\theta(\theta)\left(\frac{\mathrm{d}\theta}{\mathrm{d}\varphi}\right)^2 \quad \Leftarrow \quad \partial_\varphi \ln p(x \mid \varphi) = \partial_\theta \ln p(x \mid \varphi(\theta))\frac{\mathrm{d}\theta}{\mathrm{d}\varphi}$$

the choice $\pi_{\theta\text{J}} \propto \sqrt{I_\theta}$ and $\pi_{\varphi\text{J}} \propto \sqrt{I_\varphi}$ gives the desired invariance:

$$\pi_{\theta\text{J}}(\theta) = \pi_{\varphi\text{J}}(\varphi)$$

[1]Notice that we are somewhat deviating from the purely subjective view of priors: Jeffreys prior uses the log-likelihood itself.

preserving in this way the structure of data.

A third family of priors is the one of **conjugate priors**. In general, a prior $\pi(\theta)$ is said to be **conjugate** to a likelihood $p(x \mid \theta)$ if the posterior distribution belongs to the same family as the prior:

$$p(\theta \mid x) \propto p(x \mid \theta)\pi(\theta) \in \text{same family as } \pi(\theta)$$

even with rescaled parameters. This is useful for different reasons. First of all for analytical tractability, in fact by definition we have a closed-form expression for the posterior. In this way, data just do a simple update of the parameters without reshaping the knowledge we have on the parameters. Moreover, we have also a natural interpretation of conjugate priors in terms of pseudo-data, in the sense that it is like we saw some data obtained from a previous experiment and then they are considered as more data the go into the likelihood. Therefore, this turns the Bayesian updating into data-augmentation.

Let's consider the example of beta prior as conjugate in the case of Bernoulli likelihood. The conjugate prior of Bernoulli likelihood is a beta prior:

$$P(p \mid k, n, H) \propto \underbrace{p^k(1-p)^{n-k}}_{\text{Bernoulli likelihood}} \times \underbrace{p^{\alpha-1}(1-p)^{\beta-1}}_{\text{Beta prior}}$$

We can see that the posterior is a beta:

$$p \sim \text{Beta}\,(\alpha, \beta) \quad \Rightarrow \quad p \mid k, n \sim \text{Beta}\,(\alpha + k, \beta + n - k)$$

The interpretation is that choosing the parameters α and β, essentially we can accommodate all previous knowledge that we have about this probability. If $\alpha = \beta$, the prior is symmetric and becomes narrower as α increases. Moreover, concerning the previous pseudo-data interpretation, given that particular form of the prior it is like we already observed $\alpha - 1$ successes and $\beta - 1$ failures and this is equivalent to $\alpha + \beta - 2$ pseudo-observations, then the prior is equivalent to observing these pseudo-observations. As we said, we have a closed-form posterior with easy sequential updates. In fact, doing other experiments, we can repeat everything without changing the type of distributions. Moreover, with particular values of α and β, with beta prior we cover other families of priors (Uniform, Jeffreys).

Posterior summary statistics. Let's now look at some posterior statistics, starting from the posterior mean, which is the expectation value of seeing probability p given k, n.

$$\mathbb{E}\,[p \mid k, n] = \frac{\alpha + k}{\alpha + \beta + n}$$

The mode is:

$$p_{\text{MAP}} = \frac{\alpha + k - 1}{\alpha + \beta + n - 2}$$

which is the value with the most probability. Finally, the variance is:

$$\text{Var}(p \mid k, n) = \frac{(\alpha + k)(\beta + n - k)}{(\alpha + \beta + n)^2(\alpha + \beta + n + 1)}$$

Notice that as $n \to \infty$, posterior concentrates on empirical frequency k/n and mean and mode approaches k/n. Notice also that the variance shrinks with the increasing of n.

So, considering the evolution of posterior with data size, we have that the outcome of only few coin flips tells us little about the fairness of a coin. For example, (H H H) we can't say anything with three trials because fluctuations are too high. By accumulating results, this change. In fact, knowledge after data analysis is strongly dependent on what we assumed a priori for small n. As n grows, we reach the same conclusions irrespective of initial beliefs and the posterior is dominated by the likelihood function, the prior choice becomes largely irrelevant.

In Figure 4.1 is represented the previous example on estimating the probability p of coin tossing. The coin has a true bias $p = 0.7$, then it's not fair. We drawn three different sample sizes of 10, 50, 200 with three different priors: uniform, Jeffreys and a beta concentrated around 0.5. The different colors correspond to different dataset sizes, while the solid, dashed and dotted lines represents respectively our three choices of priors. Notice that in the case of small datasets, the results depend strongly on the prior. For example, for the symmetric beta centered on 0.5, with 50 samples it did not move so much from there. The others, instead, started very broad and became peaked around the true value. Increasing datasets, the distributions move and become more sharper around true value.

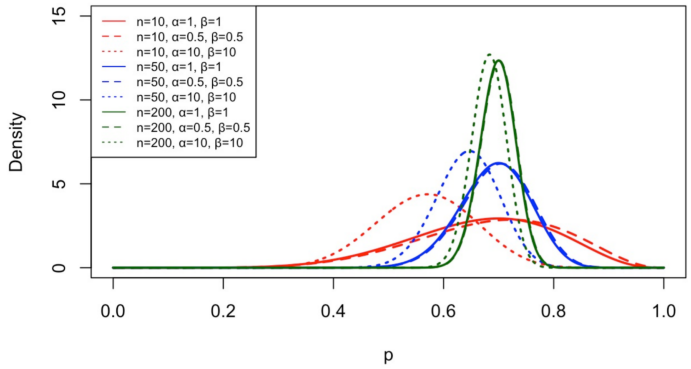


Figure 4.1: Posterior Beta $(\alpha + k, \beta + n - k)$: prior influence vanishes as n grows.

4.5 Point estimation

In the Bayesian paradigm, the posterior distribution is the key object that summarizes our belief about the parameters after we have analyzed the data. However, we may be interested to get a single value $\hat{\theta}$, known as point estimate, which is to serve as a *best guess* of an unknown population parameter.

Let X denote the observed data and θ an unknown parameter. In Bayesian inference, a **point estimator** $\hat{\theta}(X)$ is a function of the observed data X (hence, itself a random variable).

It is defined as the argument minimizing the posterior expected loss:

$$\hat{\theta}(x) = \arg\min_a \mathbb{E}\left[L(a,\theta) \mid X = x\right] = \arg\min_a \int L(a,\theta) p(\theta \mid x)\, d\theta \tag{4.9}$$

Different choices of the **loss function** $L(a, \theta)$ lead to different optimal estimators, then it depends on the chose loss. The loss function quantifies the cost of choosing a when the true parameter is θ. Therefore, while the posterior is the full information, point estimate is a summary and must be accompanied by some uncertainty. Notation: $\hat{\theta}(X)$ is a random variable, $\hat{\theta}(x) \equiv \theta(x)$ is number after data realization.

4.5.1 MAP estimator

The simplest possible estimator is the **MAP estimator**, which selects the single most probable value of the parameter given the observed data, basically the MAP is the posterior mode. It's simple and computationally convenient because one has only to compute the maximum. Even with many parameters, there are quick ways to find the maximum of the distribution. The con is that it uses very little information about the distribution. For instance, considering Figure 4.1, the different color solutions share basically the same maximum but they have completely different widths and this does not appear in the MAP estimator. Another con is that this choice is not invariant under reparametrization. For instance, computing the maximum in θ and then changing variables is not the same of computing the maximum of the changed distribution:

$$\hat{\varphi}_{\text{MAP}} \neq \varphi(\hat{\theta}_{\text{MAP}})$$

To define the MAP estimator, consider the 0-1 loss:

$$L(a,\theta) = \begin{cases} 0 & a = \theta \\ 1 & a \neq \theta \end{cases} \tag{4.10}$$

This loss is strictly appropriate for discrete parameters, or for continuous parameters when interpreted in an infinitesimal neighborhood sense. Namely, the 0-1 loss function is a Kronecker delta in the discrete case, or a *smoothed* Dirac delta in the continuous case.

The posterior expected loss is:

$$\mathbb{E}\left[L(a,\theta) \mid x\right] = \sum_{\theta \neq a} \underbrace{L(a,\theta)}_{=1} p(\theta \mid x) + p(\theta = a \mid x) \underbrace{L(\theta,\theta)}_{=0} = 1 - p(\theta = a \mid x) \tag{4.11}$$

where we used the normalization of $p(\theta \mid x)$.

Minimizing this quantity is equivalent to maximizing the posterior density, yielding the *maximum a posteriori* estimator:

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid x) \tag{4.12}$$

Here we list the caveats with MAP:

- **Ill-defined decision problem in continuous space**. If θ is discontinuously distributed, then:

  $$p(\theta = a \mid x) = 0, \qquad \forall a$$

  so all actions incur the same posterior expected loss. As a result, the 0-1 loss does not define a meaningful optimization problem without further regularization. In continuous spaces, the MAP estimator can only be justified via a limiting procedure, for example by considering losses of the form:

  $$L_\varepsilon(\theta, a) = \mathbb{1}_{|\theta - a| > \varepsilon}$$

  and letting $\varepsilon \to 0$. The resulting estimator depends on how this limit is taken. In this sense, MAP is a limiting, non intrinsic, estimator.

- **Lack of invariance under reparametrization**. The 0-1 loss is not invariant under smooth changes of variables. For example, imagine that we estimated θ via MAP and later re-parametrize to a $\phi = g(\theta)$, for some smooth function g. In general, $g(\hat{\theta}_{\text{MAP}}) \neq \hat{\phi}_{\text{MAP}}$, where $\hat{\phi}_{\text{MAP}}$ is the MAP estimator of ϕ obtained as the maximum of the distribution $p_\phi(\phi \mid x) = p_\theta(\theta \mid x) \left|\det \frac{\mathrm{d}\theta}{\mathrm{d}\phi}\right|$ with $\theta = g^{-1}(\phi)$. The Jacobian determinant of the change of variable shifts the maximum unless it is constant, or varies slowly near the mode.

  Consequently, the MAP estimator depends on the chose parametrization, in contrast with estimators such as the posterior mean that transform covariantly.

- **Insensitivity to posterior geometry**. The 0-1 loss distinguishes only whether the estimate equals the true parameter, ignoring the magnitude of errors. This causes the MAP estimator to depend solely on the local behavior of the posterior density near its maximum, discarding information about spread, skewness, or multimodality.

But if the posterior is unimodal and sharply peaked, MAP is well-behaving since it is often close to mean/median, with the advantage of being computationally convenient.

4.5.2 LMS estimator: the posterior mean

Another choice of point estimator is the posterior mean, which leads to the least mean square.

Given the squared-error loss

$$L(a, \theta) = (a - \theta)^2 \tag{4.13}$$

the posterior expected loss is:

$$\begin{aligned} \mathbb{E}\left[(a-\theta)^2 \mid x\right] &= \int (a-\theta)^2 p(\theta \mid x)\, d\theta = a^2 + \mathbb{E}\left[\theta^2 \mid x\right] - 2a\mathbb{E}\left[\theta \mid x\right] = \\ &= (a - \mathbb{E}\left[\theta \mid x\right])^2 + \mathrm{Var}(\theta \mid x) \end{aligned} \tag{4.14}$$

This expression is minimized when the parameter a coincides with the posterior mean

$$\hat{\theta}_{\text{LMS}}(X) = \mathbb{E}\left[\theta \mid X\right] \tag{4.15}$$

which is called the **least mean square** (LMS) estimator. Hence, the posterior mean minimizes the expected squared estimation error.

This estimator uses the full posterior and it is invariant under reparametrization (the value of the mean does not change with changing variables), but it is sensitive to tails, meaning that having a distribution with a very flat tail, the estimation is biased from the tail.

There are some caveats with LMS are summarized in the following:

- **Non-existence of the posterior mean**. The LMS estimator is well-defined only if the posterior distribution has a finite first moment. Heavy-tailed posteriors or the use of improper priors may lead to divergent posterior means, even when the posterior itself is normalizable.

- **Sensitivity to posterior tails**. Because squared loss penalizes large deviations quadratically, the posterior mean can be strongly influenced by small amounts of posterior mass in the tails. As a consequence, rare but extreme parameter values may dominate the estimate.

- **Poor representativeness for multimodal posteriors**. For multimodal posterior distributions, the posterior mean may lie in regions of low posterior density, for instance between well-separated modes. Ins such cases, the LMS estimator may not correspond to any *typical* parameter value.
- **Violation of natural constraints**. When the parameter space is constrained, the posterior mean may fail to respect these constraints (e.g., positivity, boundedness, or structural conditions), potentially yielding estimates that are not physically or statistically meaningful.
- **Computational sensitivity in high dimensions**. In high-dimensional or complex models, accurate computation of posterior means can be challenging. Moere or Choose Files Start Anglente Carlo estimators of expectations may converge slowly and be sensitive to sampling error, particularly in the presence of heavy tails.

Despite these caveats, the LMS estimator remains a central tool in Bayesian inference due to its invariance properties, admissibility under squared loss, and full use of the posterior distribution.

4.5.3 Other common loss functions

For the loss

$$L(a, \theta) = |a - \theta| \tag{4.16}$$

one can show that the optimal estimator is the **posterior median**. Indeed, for continuous variables we can differentiate the posterior expected loss:

$$\partial_a \mathbb{E}\left[L(a, \theta) \mid x\right] = \operatorname{sgn}(a - \theta) p(\theta \mid x)\, d\theta = \mathbb{P}\left(\theta > a\right) - \mathbb{P}\left(\theta < a\right) \tag{4.17}$$

and set the result to zero to obtain a such that:

$$\mathbb{P}\left(\theta > a \mid x\right) = \mathbb{P}\left(\theta < a \mid x\right)$$

which is satisfied if a is the median of the posterior.

Asymmetric loss functions lead to posterior quantiles as optimal estimators, allowing different penalties for *over* and *under*-estimation.

4.5.4 The risk function

The loss function $L(a, \theta)$ measures how bad an estimate a is when the true parameter value is θ. To assess the overall performance of a point estimator $\hat{\theta}(X)$, one introduces the **risk function**, defined as the *expected loss*[2].

In Bayesian inference, the parameter itself is modeled as a random variable. Conditioning on the observed data $X = x$, one defines the **posterior risk** of an estimate a as:

$$\rho(a \mid x) = \mathbb{E}\left[L(a, \theta) \mid X = x\right] = \int L(a, \theta) p(\theta \mid x)\, d\theta \tag{4.18}$$

[2]In the frequentist setting, the risk is defined as an expectation with respect to the sampling distribution of the data:

$$R(\theta, \hat{\theta}) = \mathbb{E}_{X \mid \theta}[L(\hat{\theta}(X), \theta)]$$

Here the parameter θ is treated as fixed but unknown, and the risk measures the long-run average performance of the estimator under repeated sampling from the model indexed by θ.

which is the expected error given the data (average of the loss).

The Bayesian point estimator is then obtained by minimizing the posterior risk:

$$\hat{\theta}(x) = \arg\min_a \rho(a \mid x) \tag{4.19}$$

This local, data-dependent optimization problem leads directly to the estimators discussed above (MAP, posterior mean, median, etc.), each corresponding to a specific choice of loss function.

We can also evaluate performance on average, ideally over all posssible data sets. One can define the **Bayes risk**, obtained by averaging over all parameters value and data the loss function:

$$\mathcal{R} \equiv \int L(a,\theta)p(\theta,x)\,d\theta\,dx = \int p(x)\,dx \underbrace{\int L(a,\theta)p(\theta \mid x)\,dx}_{\rho(a,x)} \tag{4.20}$$

which measures the overall quality of an estimator and can also be written as the posterior risk averaged over the data. An estimator that minimizes the Bayes risk is called a **Bayes estimator**. Under mild regularity conditions, minimizing the posterior risk pointwise in x yields a Bayes estimator, establishing the equivalence between posterior expected loss minimization and global Bayes optimality. Namely, if we ask which function $\hat{a}(x)$ minimizes the functional $\mathcal{R}$ we obtain:

$$0 = \frac{\delta\mathcal{R}}{\delta\hat{a}(x)} = \int p(x)\frac{\delta\rho(\hat{a},x)}{\delta\hat{a}}\,dx \quad \Rightarrow \quad \frac{\delta\rho(\hat{a},x)}{\delta\hat{a}} = 0 \quad \Longleftrightarrow \quad \text{definition of } \hat{\theta}(X)$$

Since $p(x) > 0$, we have concluded that the posterior risk must be minimized, which is precisely condition (4.19). We therefore conclude that the point estimate $\hat{\theta}(x)$ that minimize the expected loss *pointwise* also minimize the risk *globally*:

$$\text{Local optimum} \equiv \text{Global optimum}$$

Wrapping everything up, we saw that Bayesian estimation is based on *minimize expected loss given the data*. This local decision rule defining $\hat{\theta}(X)$ is already globally optimal over all data sets.

4.6 Reliability of estimation

Once the posterior distribution $P(\theta \mid D, H)$ has been obtained, an important question arises: *how reliable are the inferred parameter values?* The answer depends strongly on the shape of the posterior distribution. Basically, we have two scenarios: if posterior is unimodal and sharply peaked, then we have that:

$$\text{mean} \approx \text{mode} \approx \text{median}$$

In the other case, that is if the posterior is multimodal/asymmetric/broadly peaked, the situation is different.

4.6.1 Peaked posteriors and Gaussian approximation

In many favorable situations, the posterior distribution is unimodal and is sharply peaked around the MAP estimate $\hat{\theta}_{\text{MAP}}$, such that the mode does not differ much from the mean and the median. It is often convenient to work with the logarithm of the posterior,

$$\mathcal{L}(\theta) = \log P(\theta \mid D, H) \tag{4.21}$$

which reaches its maximum at the same point $\hat{\theta}_{\text{MAP}}$. Given that the posterior is sufficiently narrow, we can perform Taylor expansion of $\mathcal{L}(\theta)$ around $\hat{\theta}_{\text{MAP}}$:

$$\mathcal{L}(\theta) \simeq \mathcal{L}\left(\hat{\theta}_{\text{MAP}}\right) - \frac{1}{2}(\theta - \hat{\theta}_{\text{MAP}})^T \mathbf{H} (\theta - \hat{\theta}_{\text{MAP}})$$

where $\mathbf{H}$ is the negative Hessian matrix of second derivatives,

$$H_{ij} = -\frac{\partial^2 \mathcal{L}}{\partial\theta_i \partial\theta_j}\Big|_{\hat{\theta}_{\text{MAP}}} > 0$$

and we also used the fact that first derivative is zero around the maximum. Moreover, we extracted a minus sign to emphasize that under this approximation, the posterior becomes Gaussian $P(\theta \mid D, H) \approx \mathcal{N}\left(\hat{\theta}_{\text{MAP}}, \mathbf{C}\right)$ with covariance matrix $\mathbf{C} = \mathbf{H}^{-1}$. The diagonal elements of $\mathbf{C}$ provide estimates of the parameter variances

$$\sigma_i^2 = C_{ii}$$

while the off-diagonal elements quantify correlations between parameters. One can then provide the final estimate for the vector if parameter $\theta = \hat{\theta}_{\text{MAP}} \pm \text{diag}(\mathbf{C})$, which is in the form:

$$\text{point estimation} + \text{symmetric error}$$

In particular, in the Bernoulli case the log-posterior is:

$$\mathcal{L}(p) = (a-1)\log p + (b-1)\log(1-p)$$

with $a = \alpha + k$ and $b = \beta + n - k$. Then, the second derivative is:

$$-\mathcal{L}''(p_{\text{MAP}}) = (a+b-2)^2 \left(\frac{1}{a-1} + \frac{1}{b-1}\right)$$

Therefore, we find that the uncertainty is:

$$\sigma \sim [-\mathcal{L}''(p_{\text{MAP}})]^{-1/2} \sim n^{-1/2}$$

which basically is a re-stating of the CLT.

Let's now consider the Bernoulli MAP under change of variable. One typical change of variable is the so-called **logit** or log-odds, which is the logarithm of the odds and is defined as:

$$\varphi = \log \frac{\text{prob of success}}{\text{prob of fails}} = \log \frac{p}{1-p} \quad \Longleftrightarrow \quad p = \frac{e^{\varphi}}{1+e^{\varphi}}$$

The Jacobian is:

$$\frac{\mathrm{d}p}{\mathrm{d}\varphi} = p(1-p)$$

Using change of variables and expressing posterior of φ using p, we get:

$$P(\varphi \mid D) = P(p \mid D) \left| \frac{\mathrm{d}p}{\mathrm{d}\varphi} \right| \propto p^{k+\alpha}(1-p)^{n-k+\beta}$$

Therefore, the Jacobian modifies the exponents, and computing the maximum in φ, we get:

$$p^{\varphi}_{\mathrm{MAP}} = \frac{k+\alpha}{n+\alpha+\beta}$$

while the maximum in p is given by:

$$p_{\mathrm{MAP}} = \frac{\alpha+k-1}{n+\alpha+\beta-2}$$

resulting in the shifting of the parameter. We have then

$$\hat{\varphi}_{\mathrm{MAP}} \neq \log \frac{\hat{p}_{\mathrm{MAP}}}{1-\hat{p}_{\mathrm{MAP}}}$$

4.6.2 Limitations of Gaussian approximation and credible intervals

In many realistic inference problems the posterior distribution is not well approximated by a Gaussian since it may be broad, very asymmetric, multimodal, or truncated by parameter boundaries, e.g., $p \in [0,1]$. This is particularly true with small data sets (far from CLT). In such cases, quoting a single best-fit value and a symmetric error bar can be misleading.

A more appropriate characterization of parameter uncertainty is given by **credible intervals**. For a parameter θ_i, a credible interval $[\theta_i^-, \theta_i^+]$ at credibility level γ satisfies:

$$P(\theta_i \int [\theta_i^-, \theta_i^+] \mid D, H) = \gamma \tag{4.22}$$

while for continuous variables, we have to compute the CDF, that is finding values of $\theta_i^{\pm}$ such that:

$$\gamma \equiv \int_{\theta_i^-}^{\theta_i^+} P(\theta_i \mid D, H)\, d\theta_i \tag{4.23}$$

where $P(\theta_i \mid D, H)$ is the marginal posterior for θ_i. Unlike frequentist confidence intervals (see later), credible intervals have a direct probabilistic interpretation: the parameter lies within the interval with probability γ, given the data and the model. In general, inverting the CDF is needed.

Common choices include central credible intervals and highest posterior density (HPD) intervals. A **central credible interval** at credibility level γ is defined by excluding equal probability mass from both tails of the posterior distribution. Hence, the interval $[\theta_i^-, \theta_i^+]$ satisfies:

$$\underbrace{\int_{-\infty}^{\theta_i^-} p(\theta_i \mid D, H)\, d\theta_i}_{\text{left tail}} = \underbrace{\int_{\theta_i^+}^{\infty} p(\theta_i \mid D, H)\, d\theta_i}_{\text{right tail}} = \frac{1-\gamma}{2} \tag{4.24}$$

which is symmetric only for symmetric distributions. For a Gaussian posterior, the central credible interval coincides with the familiar $\hat{\theta} \pm n\sigma$ interval, with $\gamma \simeq 68\%$ at $n = 1$, $\gamma \simeq 95\%$ at $n = 1.96$, and $\gamma \simeq 99.7\%$ at $n = 3$. For asymmetric or non-Gaussian posteriors, however, the resulting interval need not be symmetric around any point estimate.

To find the interval endpoints, we have to invert the CDF, that is we have to compute the quantile function:

$$\theta^{\pm} = F^{-1}\left(\frac{1 \pm \gamma}{2}\right)$$

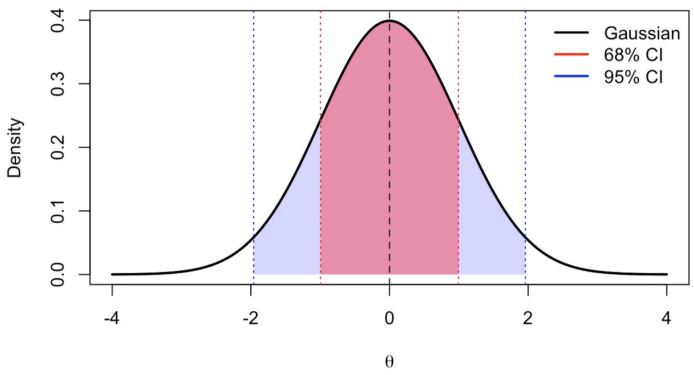


Figure 4.2: Gaussian with central credible intervals.

A **highest posterior density** (HPD) interval at credibility level γ is defined as the region of parameter space containing probability mass γ with the highest posterior density. Formally, the HPD interval is given by:

$$\text{HPD}_\gamma = \{\, \theta : p(\theta_i \mid D, H) \geq k_\gamma \,\}$$

where the threshold k_γ is chosen as the largest constant such that:

$$\int_{\theta_i \in \text{HPD}_\gamma} p(\theta_i \mid D, H)\, d\theta_i = \gamma$$

By construction, every point inside an HPD interval has a higher posterior density than any point outside. Among all regions with probability content γ, the HPD interval has minimum width. For unimodal posterior distributions, the HPD region is a single connected interval, while for multimodal posteriors, it may consist of multiple disjoint intervals, that is it may be disconnected. In the special case of a symmetric unimodal distribution, the HPD and central credible intervals coincide.

The reliability of parameter estimates should always be assessed by inspecting the full posterior distribution, in particular checking for asymmetries, long tails, or verifying the validity of Gaussian approximations. When necessary, credible intervals should be reported rather than bare point estimates.

In Figure 4.3 is represented the HPD interval for a bimodal posterior. This might come from an inference problem such as estimating the probability of heads/tails in coin tossing experiment (Bernoulli trials). We saw that in Bayesian framework this probability is a random variable distributed in $[0, 1]$. For example, starting without any trial, we can start with a belief that the coin is unfair but we don't know which way. Therefore, we would start with a bimodal prior with two peaks, one near zero and the other near one. The bimodal can be built summing two beta distributions.

4.7 Comparing to frequentist parameter estimation

In the frequentist approach, model parameters θ are treated as fixed but unknown quantities. Randomness enters only through the data, which are assumed to be realizations of a random

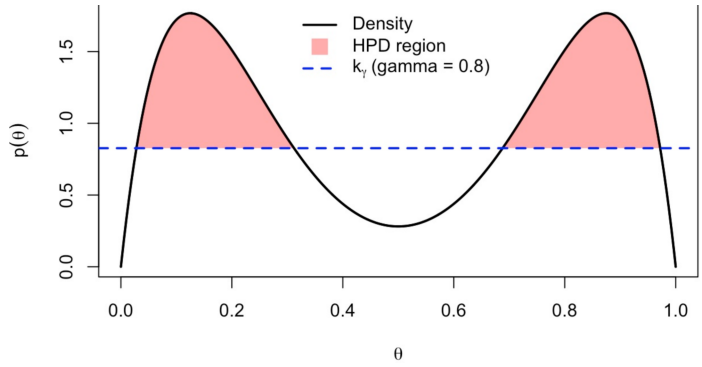


Figure 4.3: HPD interval for a bimodal posterior.

process governed by the true parameter values. The goal of parameter estimation is to construct an *estimator*, that is, a function of the data used to infer the unknown parameters.

Let θ denote an unknown parameter and let $D = \{ x_1, \ldots, x_N \}$ be a set of independent observations drawn from a probability distribution $p(x \mid \theta)$. An **estimator** of θ is a statistic $\hat{\theta} = \hat{\theta}(D)$, which is itself a random variable due to the randomness of the data.

A common construction principle is the **maximum likelihood estimator** (MLE), defined as:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} L(\theta) \tag{4.25}$$

where the likelihood function is:

$$L(\theta) \overset{i.i.d.}{=} \prod_{i=1}^{N} p(x_i \mid \theta)$$

and notice that there is no prior knowledge. Notice also that $\hat{\theta}_{\text{MLE}}$ coincides with Bayesian $\hat{\theta}_{\text{MAP}}$ for uniform priors. Equivalently, one may maximize the log-likelihood:

$$\mathcal{L}(\theta) = \sum_{i=1}^{N} \log p(x_i \mid \theta)$$

Estimators have good properties. First of all, the **bias** of an estimator quantifies systematic deviation from the true parameter value θ:

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta \tag{4.26}$$

where the expectation is taken over repeated realizations of the data drawn from $p(x \mid \theta)$. An estimator is said to be **unbiased** if $\text{Bias}(\hat{\theta}) = 0$, and **biased** otherwise. Bias alone is not sufficient to assess estimator quality, since an unbiased estimator may still have large fluctuations.

The **variance** of an estimator measures its statistical uncertainty:

$$\text{Var}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2 \right]$$

A commonly used global measure of estimator performance is the **mean squared error** (MSE):

$$\text{MSE}(\hat{\theta}) = \mathbb{E}_{\theta}[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

This decomposition highlights the trade-off between bias and variance.

For unbiased estimators, the variance is bounded from below by the **Cramér-Rao lower bound**:

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

where $I(\theta)$ is the Fisher information previously defined:

$$I(\theta) = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\log p(x \mid \theta)\right)^2\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\log p(x \mid \theta)\right]$$

where the second equality can be obtained by integrating by parts. This bound tells us that how the fluctuations of the estimator are bounded by the statistics of the true parameter. Basically, an unbiased estimator that attains the Cramér-Rao bound is called **efficient**. In many regular problems, the maximum likelihood estimator becomes asymptotically efficient as the sample size increases, meaning asymptotically true as $n \to \infty$.

An estimator is said to be **consistent** if it converges in probability to the true parameter value as the number of data point grows:

$$\hat{\theta}_N \xrightarrow[N\to\infty]{P} \theta$$

Consistency ensures that, with enough data, the estimator recovers the true parameter, regardless of its finite-sample bias or variance. Most maximum likelihood estimators are consistent under mild regularity conditions. Therefore, these are two good properties we would like to have for estimators.

Unlike Bayesian inference, all the above properties accessing to the optimality of the estimator are defined with respect to hypothetical repetitions of the experiment, rather than conditioning on the observed data.

Example 4.7.1

Bias of the maximum likelihood estimator of the variance. Let's consider a set of data $x_1, \ldots, x_N \sim \mathcal{N}\left(\mu, \sigma^2\right)$. In this case, the MLEs are:

$$\hat{\mu} = \bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad \hat{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

We want to show the classical result that the MLE $\hat{\sigma}^2$ is biased because it does not converge to the true variance, but to:

$$\mathbb{E}[\hat{\sigma}^2] = \frac{N-1}{N}\sigma^2$$

Starting from the identity:

$$\sum_{i=1}^{N}(x_i - \bar{x})^2 = \sum_{i=1}^{N}(x_i - \mu)^2 - N(\bar{x} - \mu)^2$$

and taking expectations term by term, we get:

$$\mathbb{E}\left[\sum_{i=1}^{N}(x_i - \mu)^2\right] = N\sigma^2$$

since $x_i - \mu \sim \mathcal{N}\left(0, \sigma^2\right)$, and

$$\mathbb{E}\left[N(\bar{x} - \mu)^2\right] = N \operatorname{Var}(\bar{x}) = \sigma^2$$

since $\bar{x} \sim \mathcal{N}\left(\mu, \sigma^2/N\right)$. Considering the two results yields:

$$\mathbb{E}\left[\sum_{i=1}^{N}(x_i - \bar{x})^2\right] = (N-1)\sigma^2$$

Dividing by N conclude the proof:

$$\mathbb{E}[\hat{\sigma}^2] = \frac{1}{N}\mathbb{E}\left[\sum_{i=1}^{N}(x_i - \bar{x})^2\right] = \frac{N-1}{N}\sigma^2$$

The bias originates from estimating the mean μ from the same data used to estimate the variance, effectively removing one degree of freedom. The bias vanishes as $N \to \infty$, so $\hat{\sigma}^2$ is a consistent estimator.

4.7.1 Confidence intervals vs credible intervals

In frequentist language, let X denote the observed data, modeled as random variable with distribution $p_\theta(X)$ depending on an unknown parameter θ living in the parametric space Θ. A **confidence interval** is a *random set* $C(X) \subset \Theta$, since it is a function of the random data X. The parameter θ itself is treated as *fixed but unknown*.

A confidence interval procedure $C(X)$ is said to have *coverage probability* $1 - \alpha$ if:

$$\mathbb{P}\left(\theta \in C(X)\right) = 1 - \alpha \quad \forall\, \theta \in \Theta \tag{4.27}$$

The probability is taken with respect to the sampling distribution of the data X, while the parameter θ is held fixed. Coverage is thus a statement about *repeated sampling*: the experiment is repeated many times, each time new data X are generated from $p_\theta(X)$, a new confidence interval $C(X)$ is constructed. If the procedure has 95% coverage, then in the long run, 95% of the constructed intervals will contain the true value of θ. In general, over repeated experiments, $100(1-\alpha)\%$ of intervals contain the true parameter value θ.

Coverage does not imply the conditional probability statement:

$$\mathbb{P}\left(\theta \in C \mid X = x\right) = 1 - \alpha$$

Such a statement is meaningless in the frequentist framework, because the parameter θ is not a random variable and probability statements are not conditioned on the observed data. Once the data x are observed, the interval $C(x)$ either contains θ or it does not.

As an example, take the normal distribution for the mean with known variance, namely, let $X_1, \ldots, X_n \sim \mathcal{N}\left(\mu, \sigma^2\right)$ with known σ. A $1 - \alpha$ confidence interval for μ is:

$$C(X) = \overline{X} \pm z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}$$

where $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ quantile of the standard normal distribution[3]. This interval

[3] $z_{1-\alpha/2}$ is the number such that a standard normal random variable is less than or equal to it with probability $1 - \alpha/2$. For $\alpha = 0.317$, $z_{1-\alpha/2} = 1$, for $\alpha = 0.05$, $z_{1-\alpha/2} = 1.96$, For $\alpha = 0.003$, $z_{1-\alpha/2} = 3$.

satisfies the coverage property:

$$\mathbb{P}\left(\mu \in C(X)\right) = 1 - \alpha \tag{4.28}$$

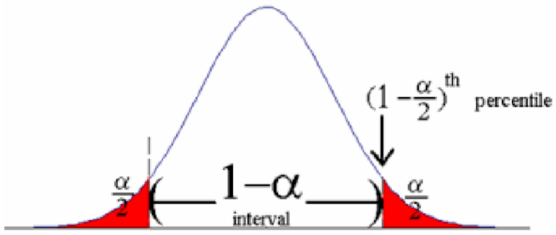


Figure 4.4: In red is represented the area equal to $1 - \alpha/2$ which corresponds to the value $z_{1-\alpha/2}$.

In Bayesian inference, θ is treated as a random variable with posterior distribution $p(\theta \mid X)$. A Bayesian credible interval may or may not satisfy frequentist coverage. In special cases (e.g., conjugate priors, large samples), credible intervals may have approximate frequentist coverage (e.g., Bernstein–von Mises theorem). All in all, coverage is a property of the **interval-producing procedure**, not of a single realized interval.

Summarizing,

- **Bayesian:**
  - probability on parameters
  - conditional on observed data

$$\mathbb{P}\left(\theta \in [\theta^-, \theta^+] \mid D\right) = \gamma$$

- **Frequentist:**
  - probability on data
  - long run frequency

$$\mathbb{P}\left(\theta \in C(X)\right) = 1 - \alpha$$

- **Remark:**
  - Always inspect the full posterior
  - Gaussian approximation may fail
  - Prefer credible intervals in Bayesian inference
  - Uncertainty quantification depends on the *interpretation of probability*

4.8 Prediction

In Bayesian inference, all sources of uncertainty are represented probabilistically. Model parameters are treated as random variables, and inference proceeds by updating prior beliefs in light of observed data. A central object in this framework is the *posterior predictive distribution*, which governs predictions for future observations.

Prediction for a new observation x_{new} is obtained by marginalizing over the posterior distribution of the parameters:

$$p(x_{\text{new}} \mid x) = \int p(x_{\text{new}} \mid \theta) p(\theta \mid x)\, d\theta \tag{4.29}$$

This distribution is referred to as the **posterior predictive distributions**. It is a genuine probability distribution on the space of future data, conditioned only on the observations x, with the parameter θ fully integrated out.

From a probabilistic standpoint, the predictive distribution follows directly from the joint model:

$$p(x_{\text{new}}, \theta \mid x) = \underbrace{p(x_{\text{new}} \mid \theta)}_{\substack{\text{likelihood comp.} \\ \text{on new points}}} \times \underbrace{p(\theta \mid x)}_{\text{posterior}}$$

by marginalization over θ. No point estimate of the parameters is required at any stage.

A key conceptual feature of Bayesian inference is that uncertainty in the model parameters is automatically propagated to predictions. Since $p(\theta \mid x)$ typically has nonzero variance, predictions for x_{new} incorporate both intrinsic data variability and parameter uncertainty.

To make this explicit, consider a model in which the conditional variance, predictions for x_{new} given θ is finite. The total predictive variance satisfies the law of total variance:

$$\text{Var}(x_{\text{new}} \mid x) = \mathbb{E}_{\theta|x}[\text{Var}(x_{\text{new}} \mid \theta)] + \text{Var}_{\theta|x}[\mathbb{E}(x_{\text{new}} \mid \theta)]$$

The first term reflects irreducible noise in the data-generating process, while the second term quantifies uncertainty due to incomplete knowledge of the parameters. Both contributions are present automatically in the Bayesian predictive distribution. By contrast, predictive schemes based on point estimates $\hat{\theta}(x)$ typically replace $p(x_{\text{new}} \mid x)$ with $p(x_{\text{new}} \mid \hat{\theta})$, thereby discarding the second contribution. In the Bayesian framework, no such approximation is made: the predictive distribution remains valid even in regimes of limited data or weak identifiability.

More generally, Bayesian prediction is invariant under reparameterization and naturally extends to hierarchical and latent-variable models, where multiple layers of uncertainty are integrated out through successive marginalizations. This makes the predictive distribution the fundamental object of inference, rather than parameter estimates themselves.

4.9 Comparing to prediction in the frequentist paradigm

In the frequentist framework, prediction is formulated in terms of the sampling distribution of future observations, conditional on unknown but fixed model parameters. Since parameters are not treated as random variables, uncertainty about their values cannot be integrated out. Instead, prediction typically relies on point estimates and procedures designed to achieve correct long-run coverage under repeated sampling.

Let $x = (x_1, \ldots, x_n)$ denote observed data drawn from a model $p(x \mid \theta)$, with $\theta \in \Theta$ fixed but unknown. A future observation x_{new} is assumed to be generated from the same conditional distribution $p(x_{\text{new}} \mid \theta)$. A common frequentist approach to prediction is the plug-in principle[4], whereby θ is replaced by an estimator $\hat{\theta}(x)$, yielding the predictive

[4]The plug-in principle says that a feature of a given distribution can be approximated by the same feature of the empirical distribution of a sample of observations drawn from the given distribution.

distribution:

$$p_{\text{plug}}(x_{\text{new}} \mid x) = p(x_{\text{new}} \mid \hat{\theta}(x))$$

This distribution is interpreted as an approximation to the true sampling distribution of x_{new} under the unknown parameter value. The validity of plug-in prediction is assessed indirectly through the sampling properties of $\hat{\theta}$, such as consistency and asymptotic normality. In large samples, $p(x_{\text{new}} \mid \theta)$ typically converges to the true conditional distribution $p(x_{\text{new}} \mid \theta)$, but for finite samples it neglects the uncertainty associated with estimating θ.

Rather than focusing on full predictive distributions, frequentist prediction often centers on the construction of prediction intervals. A prediction interval $\mathcal{I}(x)$ for x_{new} is a data-dependent set satisfying the coverage condition:

$$\mathbb{P}_\theta(x_{\text{new}} \in \mathcal{I}(X)) \geq 1 - \alpha, \quad \forall\, \theta \in \Theta$$

where the probability is taken with respect to the joint sampling distribution of the observed data X and the future observation x_{new}.

The defining property of a frequentist prediction interval is therefore its long-run coverage: over repeated realizations of the experiment, a proportion at least $1 - \alpha$ of intervals constructed in this way will contain the future observation. This guarantee holds regardless of the realized value of θ, provided the model assumptions are satisfied. In practice, prediction intervals are often derived using asymptotic approximations. For example, if $\hat{\theta}$ is asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2\right)$$

then an approximate prediction interval for x_{new} can be constructed by combining the variability of $p(x_{\text{new}} \mid \theta)$ with the estimated sampling variability of $\hat{\theta}$.

Unlike Bayesian credible intervals for predictions, frequentist prediction intervals are not interpreted as probability statements conditional on the observed data. Instead, their meaning is entirely operational: they control the frequency of correct predictions under repeated sampling.

Chapter 5

Some Applications

5.1 Histogramming events: frequentist, Bayesian, and large-deviation perspective

Let's now see the statistical problem behind a histogram. Consider collection of n i.i.d. data points taking values in an interval $[a, b]$. Fix a bin $B = [u, v] \subset [a, b]$, and denote by X the number of data points falling into B. Our goal is to infer p from a single observation $X = x_0$.

Different statistical paradigms correspond to different ways of treating p with a different estimation of the error associated to a bin count.

5.1.1 Frequentist viewpoint

In the frequentist paradigm, the probability:

$$p = \mathbb{P}\left(\text{a single data point falls in } B\right)$$

is fixed but unknown parameter, wile X is a random variable. Assuming independent sampling:

$$X \sim \text{Binomial}(n, p)$$

with:

$$\mathbb{E}\left[X\right] = np, \qquad \text{Var}(X) = np(1-p)$$

We introduce the estimator of p as the empirical frequency

$$\hat{p} = \frac{X}{n}$$

This estimator is unbiased, i.e. its expected value coincides with p:

$$\mathbb{E}(\hat{p}) = p$$

and it's also consistent, namely, the variance shrinks to zero as n increases:

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} \xrightarrow{n\to\infty} 0$$

Plug-in principle. This principle reads that after we have data from an experiment, we transform our estimator into a number that comes from the experiment. Thus, after observing a realization $X = x_0$, one typically replaces $p \mapsto \hat{p} = x_0/n$ in the binomial formulas, yielding the plug-in estimates:

$$\widehat{\mathbb{E}}[X] = x_0, \qquad \widehat{\mathrm{Var}}(X) = x_0 \left(1 - \frac{x_0}{n}\right)$$

Note that in the Poisson limit $n \to \infty$ ($p \to 0$), the plug-in variance reads $\widehat{\mathrm{Var}}(X) = x_0$, whose square root is the typical uncertainty associated to a bin value, therefore this justify the familiar error bar $\sqrt{x_0}$ in histograms. These quantify sampling fluctuations under repeated experiments, without assigning any probability distribution to p itself.

5.1.2 Bayesian viewpoint

In the Bayesian equivalent, the parameter p is treated as a random variable. Uncertainty is described by a probability distribution.

We assign a prior distribution, typically a beta distribution

$$p \sim \mathrm{Beta}\,(\alpha, \beta)$$

Given an observed count $X = x_0$, the likelihood is binomial and conjugacy yields the posterior:

$$p \mid x_0 \sim \mathrm{Beta}\,(\alpha + x_0, \beta + n - x_0)$$

which reduces to $(x_0 + 1)/(n + 2)$ for a uniform prior ($\alpha = \beta = 1$) and boils down tot the frequency estimation only for $x_0, n \gg 1$. Therefore, the key difference here is that the frequentist approach has the estimator $\hat{p}$, while the Bayesian has the full posterior distribution of p.

From the posterior predictive distribution (4.29), we can make predictions for future data, with the likelihood and posterior being a binomial and a beta distribution, respectively.

From the posterior, we can extract some posterior summaries such as the posterior mean:

$$\mathbb{E}\,[p \mid x_0] = \frac{\alpha + x_0}{\alpha + \beta + n}$$

which becomes

$$\mathbb{E}\,[p \mid x_0]\, 0 \frac{x_0 + 1}{n + 2}$$

assuming uniform prior ($\alpha = \beta = 1$).

Therefore, point estimates appear as derived quantities from the posterior distribution which remain the primary object.

We can quantify uncertainty via credible intervals:

$$P(p \in [p^-, p^+] \mid x_0) = \gamma$$

For instance, central intervals implicitly defined by:

$$\int_0^{p^-} P(p \mid x_0)\, dp = \frac{1 - \gamma}{2} = \int_{p^+}^1 P(p \mid x_0)\, dp$$

Notice that there is no closed-form expression, we have to compute the inverse of the CDF (inversion of Beta function) and it can be computed numerically via quantiles.

Future bin count X_{new} based on n trials,

$$X_{\text{new}} \mid x_0 \sim \text{Beta-Binomial}(n, \alpha + x_0, \beta + n - x_0)$$

The beta-binomial distribution is the binomial distribution in which the probability of success at each of the trials is not fixed but randomly drawn from a beta distribution. Its PDF reads:

$$\begin{aligned} f(x \mid n, \alpha, \beta) &= \int_0^1 \text{Binomial}(x \mid n, p)\, \text{Beta}\,(\alpha, \beta)\; dp = \\ &= \binom{n}{x} \frac{1}{B(\alpha, \beta)} \int_0^1 p^{x+\alpha-1}(1-p)^{n-x+\beta-1}\, dp = \\ &= \binom{n}{x} \frac{B(x+\alpha, n-x+\beta)}{B(\alpha, \beta)} \end{aligned} \tag{5.1}$$

where $B(\cdot)$ is the beta function. Its variance exceeds the frequentist plug-in variance, as it incorporates both binomial sampling noise and uncertainty about p.

5.1.3 Large deviation and entropy picture

The full distribution of estimator, seen as the empirical frequency $\hat{p} = X/n$, satisfies a large deviation principle: *the probability of $\hat{p}$ concentrates exponentially fast in $n \to \infty$ around its mean value.* Indeed, for large n we can derive:

$$P(\hat{p} = y) \asymp \exp[-nD\,(y \,||\, p)]$$

where the rate

$$D\,(y \,||\, p) = y \log \frac{y}{p} + (1-y) \log \frac{1-y}{1-p}$$

is the Kullback-Leibler divergence[1] (also called relative entropy) between Bernoulli distributions. Basically, interpreted as rate function, it quantifies how unlikely a fluctuation y away from true value p is. The symbol $\asymp$ means log equality in the limit $n \to \infty$, namely:

$$-\lim_{n\to\infty} \frac{1}{n} \mathbb{P}\,(\hat{p} = y) = D\,(y \,||\, p)$$

This results descends from the fact that $X = ny$ follows the binomial distribution:

$$\mathbb{P}\,(\hat{p} = y) \propto \mathbb{P}\,(X = ny) = \binom{n}{ny} p^{ny}(1-p)^{n(1-y)} \asymp e^{nH(y) + ny \ln p + n(1-y)\ln(1-p)}$$

where the entropy, obtained using Stirling approximation of the binomial factor:

$$H(y) = -y \log y - (1-y) \log(1-y)$$

[1] The Kullback-Leibler divergence for two probability distribution P and Q is defined as:

$$D\,(P \,||\, Q) = \int dx\, P(x) \log \frac{P(x)}{Q(x)}$$

It can be interpreted as a measure of *dissimilarity* between distributions, but is not a distance since it is not symmetric:

$$D\,(P \,||\, Q) \neq D\,(Q \,||\, P)$$

In general, $D\,(P \,||\, Q) \geq 0$ and the equality holds only iff $P = Q$.

counts the logarithmic number of sequences compatible with a given empirical frequency (it's the entropic factor).

Expanding $D\,(y \,||\, p)$ around its minimum at $y = p$ yields:

$$D\,(y \,||\, p) = \frac{(y-p)^2}{2p(1-p)} + \mathcal{O}\big((y-p)^3\big)$$

This result reproduces the central limit result:

$$\mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

Therefore, the interpretation is that empirical frequencies y concentrate around p and the deviations are exponentially suppressed near $y = p$. In particular, expanding $D\,(y \,||\, p)$, we recover the Gaussian fluctuations and this justifies confidence intervals as local approximations.

5.2 Bayesian inference for the Poisson rate parameter λ

Consider a set of independent observations $D = \{\, x_1, x_2, \ldots, x_n \,\}$, where each $x_i \in \mathbb{N}_0$ is drawn from a Poisson distribution with unknown rate parameter λ:

$$x_i \sim \mathrm{Poisson}(\lambda), \qquad \lambda > 0$$

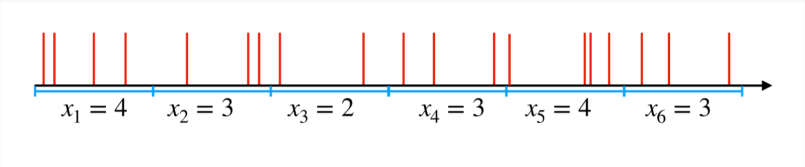


Figure 5.1

Our goal is to estimate λ. The likelihood function is:

$$P(\lambda \mid D) = \prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \left(\prod_{i=1}^{n} \frac{1}{x_i!}\right) \lambda^{\sum_i x_i} e^{-n\lambda} \propto \lambda^{y} e^{-n\lambda}$$

The likelihood depends on the data only through the **sufficient statistic**[2]. This means that if we are able to write the likelihood as the product of two functions, one of the data and the other of the parameters, we have a sufficient statistic. Here:

$$T(x) = y = \sum_{i=1}^{n} k_i, \quad h(x) = \prod_{i=1}^{n} \frac{1}{x_i!}, \quad g(\lambda, y) = \lambda^{y} e^{-n\lambda}$$

[2] A sufficient statistic for a parameter θ is a function of the data X that contains all the information in the sample relevant for inference of θ. More formally, a statistic $T(X)$ is sufficient for θ if the conditional distribution of the data given the statistic does not depend on θ, i.e. $p(x \mid \theta, T(x)) = p(x \mid T(x))$. In practice, one recurs to the Fisher-Neyman factorization criterion: T is sufficient for θ if and only if non-negative functions g and h can be found such that $p(x \mid \theta) = h(x)g(\theta, T(x))$, where one function depends only on the data, and the other on the parameters.

A commonly used but formally improper (because not normalizable, the integral is infinite) choice is the uniform prior on λ:

$$\pi(\lambda) \propto 1, \qquad \lambda \geq 0$$

Combining this with the likelihood yields the posterior:

$$p(\lambda \mid D) \propto \lambda^y e^{-n\lambda}$$

which is a gamma distribution:

$$\lambda \mid D \sim \text{Gamma}\,(y+1, n)$$

Given the pdf of the Gamma (α, β):

$$f(\lambda) = \frac{\lambda^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$$

the posterior mean and variance are:

$$\mathbb{E}\left[\lambda \mid D\right] = \frac{y+1}{n}, \qquad \text{Var}(\lambda \mid D) = \frac{y+1}{n^2}$$

Compared to the MLE $\hat{\lambda}_{\text{MLE}} = y/n$, the posterior mean is slightly shifted due to the prior.

Another choice is Jeffreys prior $\pi_{\text{J}}(\lambda) \propto \sqrt{I(\lambda)}$, which is invariant under reparametrizations. $I(\lambda)$ is the Fisher information of the likelihood:

$$I(\lambda) = -\mathbb{E}[\partial_\lambda^2 \ln p(D \mid \lambda)] = \mathbb{E}[\partial_\lambda^2(-y \ln \lambda + n\lambda)] = \frac{\mathbb{E}\left[y\right]}{\lambda^2} = \frac{n}{\lambda}$$

This leads to:

$$\pi_{\text{J}}(\lambda) \propto \lambda^{-1/2}$$

and the resulting posterior:

$$p(\lambda \mid D) \propto \lambda^{y-1/2} e^{-n\lambda}$$

corresponding to:

$$\lambda \mid D \sim \text{Gamma}\left(y + \frac{1}{2}, n\right)$$

The posterior mean and variance are:

$$\mathbb{E}\left[\lambda \mid D\right] = \frac{y+\frac{1}{2}}{n}, \qquad \text{Var}(\lambda \mid D) = \frac{y+1}{n^2}$$

Finally, one can take the conjugate prior for the Poisson, i.e. the gamma distribution:

$$\pi(\lambda) = \text{Gamma}\,(\alpha, \beta)\,, \propto \lambda^{\alpha-1} e^{-\beta\lambda}$$

This choice leads to a posterior of the same functional form:

$$\lambda \mid D \sim \text{Gamma}\,(\alpha + y, \beta + n)$$

This contains both uniform and Jeffreys prior, respectively with $\alpha = 1$, $\beta = 0$ and $\alpha = 1/2$, $\beta = 0$.

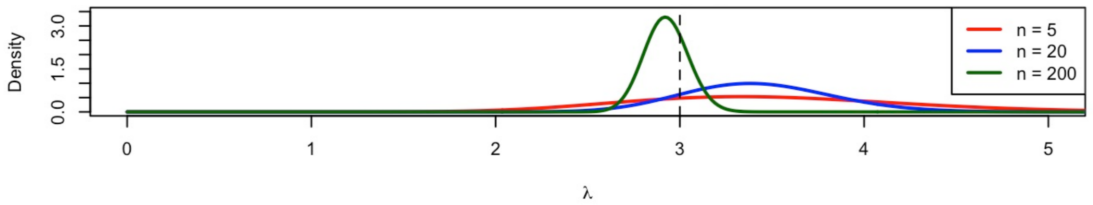
The posterior mean and variance are:

$$\mathbb{E}\left[\lambda \mid D\right] = \frac{\alpha + y}{\beta + n}, \qquad \mathrm{Var}(\lambda \mid D) = \frac{\alpha + y}{(\beta + n)^2}$$

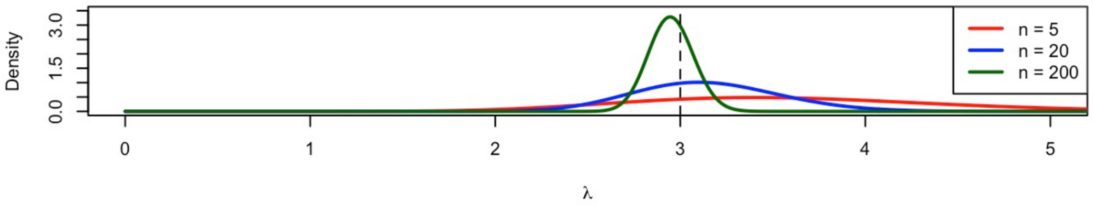
The hyperparameters (α, β) can be interpreted as encoding prior pseudo-counts and pre-exposure time respectively.

For finite n, different priors lead to different posterior distributions and estimators, particularly when the total count y is small. The influence of the prior is most pronounced in the low-data regime, where it can act as regularization for the estimates and prevent unphysical results (see later example).

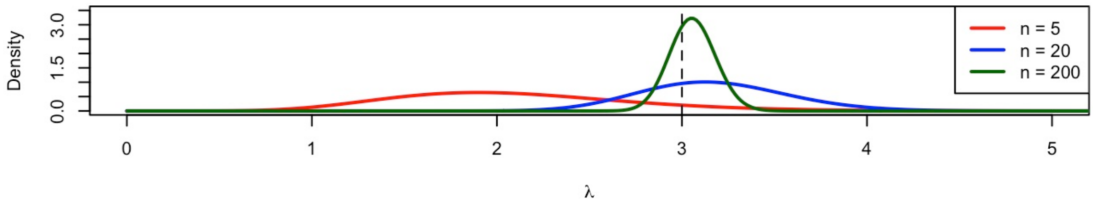
All these results can be visualized in Figure 5.2, where the posterior for different choices of priors is shown.



(a) Posterior with Gamma $(2, 1)$.



(b) Posterior with Uniform.



(c) Posterior with Jeffreys.

Figure 5.2

As the number of observations grow large, the likelihood becomes increasingly sharply peaked. In the limit $n \to \infty$ with $y/n \to \lambda_0$, the likelihood dominates and the posterior distribution concentrates around λ_0 for all reasonable priors:

$$p(\lambda \mid D) \xrightarrow{N \to \infty} \mathcal{N}\left(\lambda_0, \frac{\lambda_0}{n}\right)$$

and the differences between priors become negligible. This illustrates a general manifestation of the:

Theorem 5.2.1 (Bernstein-von Mises theorem)**.** *Under mild conditions, with sufficient data, posterior becomes Gaussian and prior-independent. Bayesian and frequentist inference agree asymptotically*

$$\left\| P(\theta \mid D) - \mathcal{N}\left(\hat{\theta}_{MLE}, n^{-1} I(\theta_0)^{-1}\right) \right\| \xrightarrow{n\to\infty} 0$$

Example 5.2.1

Consider the example of extreme low-data regime in which there are no events:

$$x_1 = \ldots = x_n = 0 \quad \Rightarrow \quad y = 0$$

In this case, the MLE is:

$$\hat{\lambda}_{\text{MLE}} = \frac{y}{n} = 0$$

This implies $P(X > 0) = 0$, which means that there is no probability to see any event. This is unphysical since finite observation does not imply events are impossible.
In this case, Bayesian regularization thanks to priors comes in. With Uniform prior:

$$\lambda \mid D \sim \text{Gamma}\,(1, n) \quad \Rightarrow \quad \mathbb{E}\,[\lambda] = \frac{1}{n}$$

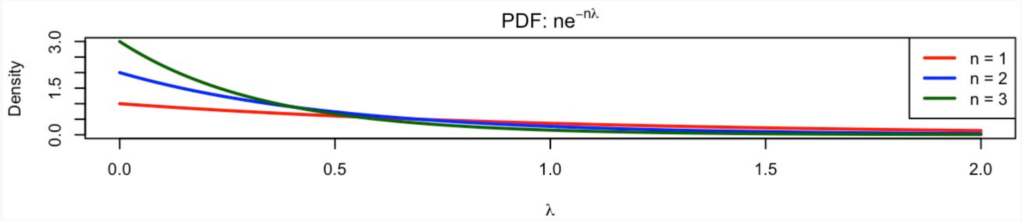
which can be interpreted as our belief that we are not seeing something in any intervals, then it could be that the rate might be of the order of one over the intervals. Then it means that i could not see any event if we just look at one interval. We have to look many more and this makes a lot of sense.
With a Jeffreys prior:

$$\lambda \mid D \sim \text{Gamma}\left(\frac{1}{2}, n\right) \quad \Rightarrow \quad \mathbb{E}\,[\lambda] = \frac{1}{2n}$$

In both cases, $\mathbb{E}\,[\lambda \mid D] > 0$ implies $P(X > 0) \neq 0$ and priors prevent the unphysical boundary estimate $\lambda = 0$.
Moreover, for $y = 0$, uniform prior gives posterior:

$$p(\lambda \mid D) = ne^{-n\lambda}$$

which, for $\lambda \geq 0$, is broad. In the low-data regime, small λ are likely but not forced to zero. Therefore, as n grows large, and we keep seeing no events, we accumulate belief that λ is very small.



However, we are not only interested in λ, but also in future counts. We have seen that the Bayesian prediction for a new observation (4.29) is obtained by marginalizing over all

possible values of the parameter weighted by the likelihood. So

$$p(x_{\text{new}} \mid D) = \int p(x_{\text{new}} \mid \theta) p(\theta \mid D)\, d\theta \tag{5.2}$$

Therefore, no point estimate is required, only law of total probability is used. From Poisson likelihood:

$$p(x_{\text{new}} \mid \lambda) = \frac{\lambda^{x_{\text{new}}} e^{-\lambda}}{x_{\text{new}}!}$$

we get the prediction for the next observation:

$$P(x_{n+1} = 0 \mid D) = \int_0^\infty e^{-\lambda} n e^{-n\lambda}\, d\lambda = \frac{n}{n+1}$$

and next time interval will show the first event with probability:

$$P(x_{n+1} > 0 \mid D) = 1 - P(x_{n+1} = 0 \mid D) = \frac{1}{n+1}$$

For instance, if in $n = 2$ intervals we see no event, there's $1/3$ chance to see it at the next observation.

Equation 5.2 is the Bayesian approach for predicting new data. Another one we have seen is the plug-in approach, in which we substitute our parameters D with the empirical estimation $\hat{\lambda}$:

$$p(x_{\text{new}} \mid D) \approx p(x_{\text{new}} \mid \hat{\lambda})$$

but which $\hat{\lambda}$ to use? The empirical frequency is still unphysical, in fact if it is zero, we would get:

$$\hat{\lambda} = 0 \quad \Rightarrow \quad P(x_{\text{new}} = 1) = 0$$

For instance, using posterior mean (Bayesian plug-in):

$$\hat{\lambda} = \mathbb{E}\left[\lambda \mid D\right] = \frac{1}{n} \quad \Rightarrow \quad P(x_{\text{new}} = 0) = e^{-1/n}$$

Thus, plug-in ignores completely parameter uncertainty, while in the Bayesian approach, integrating over the full parameter distribution takes into account the uncertainty of the parameter. Therefore, Bayesian approach predicts randomness in data and randomness in the estimation of the parameters:

$$\text{data noise} + \text{parameter uncertainty}$$

This can be see for instance by looking at the variance. Uncertainty in λ propagates to predictions:

$$\text{Var}(x_{\text{new}} \mid D) = \mathbb{E}_{\lambda|D}[\text{Var}(x_{\text{new}} \mid \lambda)] + \text{Var}_{\lambda|D}[\mathbb{E}\left[x_{\text{new}} \mid \lambda\right]]$$

This example takes us to another theoretical concept, which is the one sided upper credible interval. So far, we considered central credible intervals and HPD intervals. For a parameter θ a **one sided upper credible interval** $[0, \theta^*]$ at credibility level

$$P(\theta \leq \theta^*) = \gamma$$

It coincides with HPD interval if the posterior is monotonically decreasing. A typical example use is when one wants to bound small rates in low-data regimes.

Considering again a zero-count dataset:

$$x_1 = \ldots = x_n = 0 \quad \Rightarrow \quad y = 0$$

with uniform prior. the posterior is:

$$\lambda \mid D \sim \text{Gamma}\,(1, n)$$

which corresponds to density:

$$p(\lambda \mid D) = ne^{-n\lambda}$$

We impose $P(\lambda \leq \lambda^* \mid D) = 1 - \alpha$ with α tail probability (α is our γ). We compute the CDF:

$$P(\lambda \leq \lambda^*) = \int_0^{\lambda^*} ne^{-n\lambda}\, d\lambda = 1 - e^{-n\lambda^*}$$

We then choose the value of λ^* to satisfy:

$$1 - e^{-n\lambda^*} = 1 - \alpha \quad \Rightarrow \quad \lambda^* = -\frac{1}{n} \ln \alpha$$

For $\alpha = 0.05$, we have:

$$\lambda^* = \frac{\ln(20)}{n} \approx \frac{3}{n}$$

Then, once one specifies the probability level γ, one finds the upper edge of credibility bound.

Finally, a 95% credible upper bound leads to the so-called *rule of three*:

If you observed zero events in n trials, then the true rate is at most $3/n$ *with* 95% *credibility*:

$$\lambda \lesssim \frac{3}{n}$$

5.3 Bayesian inference for the Gaussian mean and variance

Let $D = \{\, x_1, x_2, \ldots, x_n \,\}$ be independent samples drawn from a normal distribution with unknown mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$:

$$p(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

We define the empirical mean and the empirical variance:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad S = \frac{1}{n}\sum_{i=1}^{n} (x_i - \bar{x})^2$$

which will serve as sufficient statistics in what follows.

5.3.1 Inference of the mean with known variance

Assume that σ^2 is a known constant and that only the mean μ is to be inferred. The likelihood is:

$$p(D \mid \mu) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left[-\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}\right] \propto \exp\left[-\frac{n}{2\sigma^2}(\mu-\bar{x})^2\right]$$

where we used the key identity:

$$\sum_{i=1}^{n}(x_i-\mu)^2 = \sum_{i=1}^{n}(x_i-\bar{x})^2 + n(\bar{x}-\mu)^2$$

and isolated the term that depends on μ. This shows that the empirical mean is a sufficient statistic when the variance is known.

A common non-informative choice is the improper uniform prior, $p(\mu) \propto 1$ that yields the Gaussian posterior:

$$\mu \mid D \sim \mathcal{N}\left(\bar{x}, \frac{\sigma^2}{n}\right)$$

The posterior mean coincides with the frequentist's MLE and the variance shrinks as $1/n$. Note that the flat prior also coincides with the Jeffreys prior since the Fisher information of the Gaussian is independent of μ:

$$-\frac{\mathrm{d}^2}{\mathrm{d}\mu^2}\ln p(D \mid \mu) = \frac{n}{\sigma^2}$$

Using the conjugate prior for the mean, which is itself Gaussian:

$$p(\mu) = \mathcal{N}\left(\mu_0, \sigma_0^2\right)$$

returns a Gaussian posterior:

$$\begin{aligned} p(D \mid \mu)p(\mu) &\propto \exp\left[-\frac{n}{2\sigma^2}(\mu-\bar{x})^2\right]\exp\left[-\frac{1}{2\sigma_0^2}(\mu-\mu_0)^2\right] \\ &\propto \exp\left[-\mu^2\left(\frac{n}{2\sigma^2}+\frac{1}{2\sigma_0^2}\right)+\mu\left(\frac{\mu_0}{\sigma_0^2}+\frac{n\bar{x}}{\sigma^2}\right)\right] \end{aligned}$$

Therefore:

$$\mu \mid D \sim \mathcal{N}\left(\frac{\mu_0/\sigma_0^2 + n\bar{x}/\sigma^2}{1/\sigma_0^2 + n/\sigma^2}, \frac{1}{1/\sigma_0^2 + n/\sigma^2}\right)$$

As $n \to \infty$, the influence of the prior vanishes and the posterior concentrates around $\bar{x}$.

5.3.2 Inference of the variance with known mean

Assume now that μ is a known constant and only the variance σ^2 is to be inferred. The likelihood is:

$$\begin{aligned} p(D \mid \sigma^2) &= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right] \\ &\propto \frac{1}{\sigma^n} \exp\left[-\frac{n}{2\sigma^2}(S+(\bar{x}-\mu)^2)\right] \end{aligned} \tag{5.3}$$

$\bar{x}$ together with S are sufficient statistics. For the variance parameter, Jeffreys prior is:

$$p_{\text{J}}(\sigma^2) \propto \frac{1}{\sigma^2}$$

that given an inverse-gamma distribution posterior:

$$\sigma^2 \mid D \sim \text{Inv-Gamma}\left(\frac{n}{2}, \frac{1}{2}\sum_{i=1}^{n}(x_i - \mu)^2\right)$$

The conjugate prior for σ^2 is the inverse-gamma distribution[3]:

$$p(\sigma^2) = \text{Inv-Gamma}(\alpha, \beta)$$

which gives a posterior

$$\sigma^2 \mid D \sim \text{Inv-Gamma}\left(\alpha + \frac{n}{2}, \beta + \frac{1}{2}\sum_{i=1}^{n}(x_i - \mu)^2\right)$$

which contains uniform and Jeffreys priors ($\beta = 0$ and $\alpha = -1, 0$). A last comment is that variance σ^2 is a **scale parameter**. Meaning that, given a conditional PDF, one can always find a function such that we can re-write:

$$p(x \mid \sigma) = \frac{1}{\sigma} g\left(\frac{x}{\sigma}\right)$$

So, it depends on units. For instance, having random data, we can always rescale them by changing the units, leading to a change in the σ with the scaling constant squared:

$$x \mapsto ax \quad \Rightarrow \quad \sigma^2 \mapsto a^2\sigma^2$$

For example $\sigma = 100\,\text{cm}$ and $\sigma = 1\,\text{m}$ are physical identical because the absolute value is not intrinsically meaningful, rather ratios σ_1^2/σ_2^2 are (through with we drop the units). Therefore, considering PDF defined in terms of some scaling parameter, the natural requirement to enforce the meaningfulness of absolute value is invariance under rescaling:

$$\sigma^2 \mapsto a^2\sigma^2$$

Jeffreys prior is exactly like that, because it is uniform in $\log \sigma^2$:

$$p_{\text{J}}(\sigma^2)\, d\sigma^2 \propto \frac{d\sigma^2}{\sigma^2} = d(\log \sigma^2)$$

This ensures that model comparison (BF) is independent of units.

5.3.3 Joint inference of mean and variance

When both μ and σ^2 are unknown, the likelihood is again (5.3). In order to handle the bivariate prior, we use factorization as:

$$p(\mu, \sigma^2) = p(\mu \mid \sigma^2)p(\sigma^2)$$

[3]It is the distribution of the reciprocal of a variable distributed according to the gamma distribution. In other words, if $X \sim \text{Gamma}\,(\alpha, \beta)$, then $\frac{1}{X} \sim \text{Inv-Gamma}(\alpha, \beta)$.

If we want to use reparametrization invariant priors, then we employ the Jeffreys prior previously identified.

The conjugate prior for the joint inference problem is the Normal-Inverse-Gamma distribution:

$$\mu \mid \sigma^2 \sim \mathcal{N}\left(\mu_0, \frac{\sigma^2}{\kappa_0}\right), \qquad \sigma^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0)$$

The posterior is by definition Normal-Inverse-Gamma with updated parameters:

$$\kappa_n = \kappa_0 + n \tag{5.4a}$$

$$\mu_n = \frac{\kappa_0 \mu_0 + n\bar{x}}{\kappa_0 + n} \tag{5.4b}$$

$$\alpha_n = \alpha_0 + \frac{n}{2} \tag{5.4c}$$

$$\beta_n = \beta_0 + \frac{n}{2}S + \frac{\kappa_0 n}{2(\kappa_0 + n)}(\bar{x} - \mu_0)^2 \tag{5.4d}$$

entering the PDF:

$$p(\mu, \sigma^2 \mid D) = \frac{\beta_n^{\alpha_n}}{\Gamma(\alpha_n)}\sqrt{\frac{\kappa_n}{2\pi}}(\sigma^2)^{-(\alpha_n + \frac{3}{2})} \exp\left[-\frac{1}{\sigma^2}\left(\beta_n + \frac{\kappa_n}{2}(\mu - \mu_n)^2\right)\right]$$

Marginalizing over σ^2, we get the posterior of the mean conditioned on the data. The posterior distribution of μ is then a Student-t distribution with $2\alpha_n$ degrees of freedom:

$$p(\mu \mid D) = t_{2\alpha_n}\left(\mu_n, \frac{\beta_n}{\alpha_n \kappa_n}\right)$$

We can write the PDF with ν degrees of freedom and $t = (\mu - \mu_n)/\frac{\beta_n}{\alpha_n \kappa_n}$ as:

$$f(t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu}\Gamma(\frac{\nu}{2})}\left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}$$

This is a heavy-tailed distribution ($\sim |\mu|^{-(2\alpha_n+1)}$) reflects the additional uncertainty associated with estimating the variance.

As $n \to \infty$, the joint posterior distribution concentrates around the MLE:

$$\hat{\mu}_{\text{MLE}} = \bar{x}, \qquad \hat{\sigma}^2_{\text{MLE}} = S$$

Under standard regularity conditions on the prior (in particular, that it is positive and smooth in a neighborhood of the true parameter), for large n, $2\alpha_n \sim n \Rightarrow t_{2\alpha_n} \to \mathcal{N}$ the Bernstein-von Mises theorem implies that the posterior distribution becomes asymptotically Gaussian:

$$p(\theta \mid D) \xrightarrow{n\to\infty} \mathcal{N}\left(\hat{\theta}, I_n^{-1}(\hat{\theta})\right)$$

where $\hat{\theta} = (\hat{\mu}, \hat{\sigma}^2)$ is the MLE (index dropped to lighten notation) and $I_n(\theta)$ is the Fisher information matrix for n observations.

For a single observation $x \sim \mathcal{N}\left(\mu, \sigma^2\right)$, the log-likelihood is:

$$\mathcal{L}\left(\mu, \sigma^2\right) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}$$

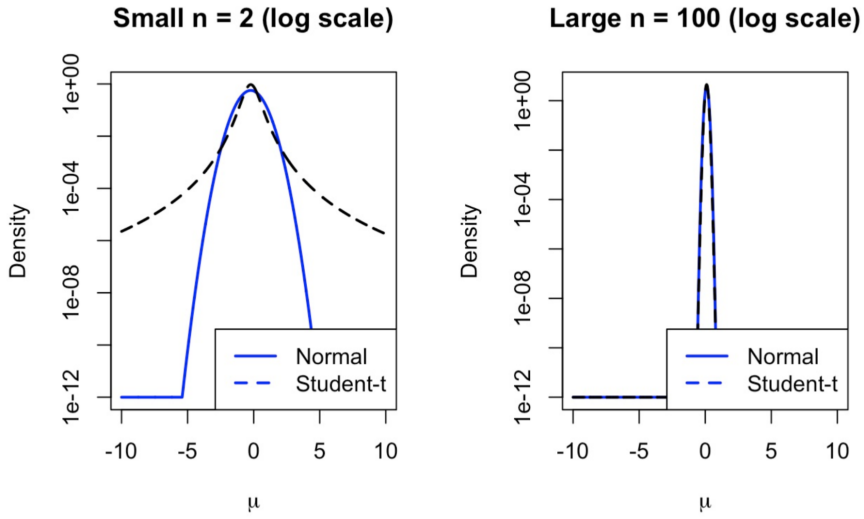


Figure 5.3

A straightforward calculation yields the Fisher information matrix for one observation:

$$I_1(\mu, \sigma^2) = -\mathbb{E}\left[\partial_{\theta_i}\partial_{\theta_j} \ln p(X \mid \theta)\right] = \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2(\sigma^2)^2} \end{pmatrix}$$

Since the observations are independent, the Fisher information for N data points is additive:

$$I_n(\mu, \sigma^2) = nI_1(\mu, \sigma^2) = \begin{pmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{pmatrix}$$

The asymptotic covariance matrix of the posterior (or equivalently of the MLE) is given by the inverse Fisher information calculated at $\hat{\theta}$:

$$I_n^{-1}(\hat{\mu}, \hat{\sigma}^2) = \begin{pmatrix} \frac{\hat{\sigma}^2}{n} & 0 \\ 0 & \frac{2(\hat{\sigma}^2)^2}{n} \end{pmatrix}$$

Thus, in the large-sample limit, the posterior factorizes asymptotically and the parameters μ and σ^2 become uncorrelated. Both uncertainties shrink as $n^{-1/2}$ with:

$$\mathrm{Var}(\mu) \sim \frac{\hat{\sigma}^2}{n}, \qquad \mathrm{Var}(\sigma^2) \sim \frac{2(\hat{\sigma}^2)^2}{n}$$

Importantly, this asymptotic result is independent of the specific choice of prior, provided the prior is regular.

5.3.4 Credible intervals

Credible intervals summarize posterior uncertainty about parameters.

Credible interval for the mean. We have:

$$\mu \mid D \sim t_{2\alpha_n}\left(\mu_n, \frac{\beta_n}{\alpha_n \kappa_n}\right)$$

A *central* $(1-\alpha)\%$ credible interval for μ is therefore:

$$\mu_n \pm t_{2\alpha_n, 1-\alpha/2}\sqrt{\frac{\beta_n}{\alpha_n \kappa_n}}$$

where $t_{\nu,p}$ denotes the p-quantile of a Student-t distribution with ν degree of freedom. In the large-sample limit $n \to \infty$, the Student-t converges to a Gaussian and the interval becomes:

$$\mu_n \pm z_{1-\alpha/2}\sqrt{\frac{\hat{\sigma}^2}{n}}$$

recovering the familiar asymptotic confidence interval (4.28).

Credible interval for the variance. The marginal posterior of the variance is:

$$\sigma^2 \mid D \sim \text{Inv-Gamma}(\alpha_n, \beta_n)$$

Let q_p denote the p-quantile of a gamma distribution with parameters (α_n, β_n). Since $1/\sigma^2$ is gamma-distributed, a *central* $100(1-\alpha)\%$ credible interval for σ^2 is:

$$\left[\frac{1}{q_{1-\alpha/2}}, \frac{1}{q_{\alpha/2}}\right]$$

For large n, using asymptotic normality of the MLE:

$$\sigma^2 \approx \hat{\sigma}^2 \pm z_{1-\alpha/2}\sqrt{\frac{2(\hat{\sigma}^2)^2}{n}}$$

Joint credible regions. In the large-sample limit the posterior is approximately Gaussian with covariance matrix $I_n^{-1}(\hat{\theta})$. A joint $(1-\alpha)\%$ credible region for $\theta = (\mu, \sigma^2)$ is given by the ellipse:

$$(\theta - \hat{\theta})^T I_n(\hat{\theta})(\theta - \hat{\theta}) \leq \chi^2_{2,1-\alpha}$$

where $\chi^2_{2,1-\alpha}$ is the $1-\alpha$ percentile of the χ^2. We keep all parameter values whose *squared distance* is below the cut off.

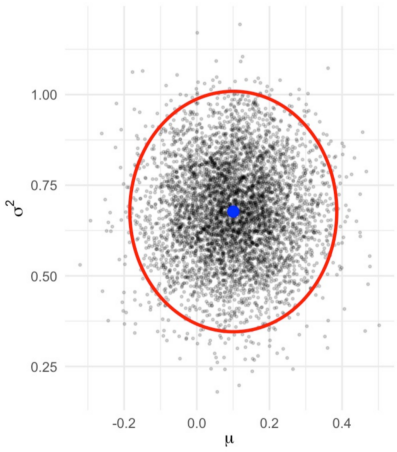


Figure 5.4: Asymptotic Gaussian posterior ($n = 50$) with $\chi^2_{2,0.95} \approx 5.99$

5.3.5 One- and two-side hypothesis testing

Bayesian hypothesis testing compares posterior probabilities of competing hypotheses. The Gaussian model allows analytic expressions for several relevant cases.

One-sided hypothesis test for the mean. Consider:

$$H_0 : \mu \leq \mu_*, \qquad H_1 : \mu > \mu_*$$

The posterior probability of H_1 is:

$$P(H_1 \mid D) = 1 - F_t\left(\frac{\mu_* - \mu_n}{\sqrt{\beta_n/(\alpha_n \kappa_n)}}\right)$$

where F_t denotes the CDF of a Student-t distribution with $2\alpha_n$ degrees of freedom.

A simple decision rule is the direct rejection criterion: reject H_0 if:

$$P(H_1 \mid D) > 1 - \alpha$$

Equivalently, H_0 is rejected if posterior mass aboveμ_* lies outside the lower bound of the corresponding credible interval.

Two-sided hypothesis test for the mean. Consider:

$$H_0 : \mu = \mu_*, \qquad H_1 : \mu \neq \mu_*$$

A common Bayesian procedure is the credible-interval test: reject H_0 if μ_* lies outside the $(1-\alpha)\%$ credible interval for μ.

Alternatively, one may compare models via the Bayes factor:

$$\mathrm{BF}_{10} = \frac{p(D \mid H_1)}{p(D \mid H_0)} = \frac{\int_{\mu^*}^{\infty} p(D \mid \mu) p(\mu)\, d\mu}{\int_{-\infty}^{\mu^*} p(D \mid \mu) p(\mu)\, d\mu}$$

where H_0 corresponds to the restricted model $\mu = \mu_*$ and H_1 to the full model with the conjugate prior. The interpretation is:

- $\mathrm{BF}_{10} > 1$: evidence favors H_1 (mean above μ_*)
- $\mathrm{BF}_{10} < 1$: evidence favors H_0
- Magnitude quantifies strength of evidence.

Also, it coincides with the odds ratio because there are no different priors on the hypotheses H_1 and H_2 (priors are on the parameters).

The marginal likelihoods can be computed analytically in the conjugate Gaussian setting.

Testing the variance. For hypotheses on the variance such as:

$$H_0 : \sigma^2 \leq \sigma_*^2, \qquad H_1 : \sigma^2 > \sigma_*^2$$

the posterior probability of H_1 is :

$$P(H_1 \mid D) = \int_{\sigma_*^2}^{\infty} \text{Inv-Gamma}(\sigma^2; \alpha_n, \beta_n)\, d\sigma^2$$

In the large-sample limit, this reproduces the classical χ^2-type behavior obtained from asymptotic normality.

5.4 Fitting as a special case of parametric inference

Classical model fitting can be understood as a special case of *parametric Bayesian inference*, where a likelihood is assumed for the data and a prior is optionally specified. The choice of a loss function then determines the specific point estimate. Suppose we observe data $D = \{ (x_i, y_i) \}_{i=1}^n$ and have a parametric model $f(x; \theta)$ with unknown parameters $\theta \in \Theta$. To formalize fitting in a Bayesian framework, assume the observations are noisy realizations of the model:

$$y_i = f(x_i; \theta) + \varepsilon_i, \qquad \varepsilon_i \sim i.i.d.\ \mathcal{N}\left(0, \sigma^2\right)$$

Then, the likelihood of the data is:

$$p(D \mid \theta) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - f(x_i; \theta))^2}{2\sigma^2} \right] \tag{5.5}$$

A prior $p(\theta)$ encodes any previous belief about the parameters. In classical fitting, one usually assumes a uniform (improper) prior, corresponding to no preference, $p(\theta) \propto 1$. Bayes' theorem as usual gives the posterior distribution: combining the information from the data (likelihood) and prior.

If we are only interested in estimating a numerical value for the parameters, given a loss function $L(\hat{\theta}, \theta)$, the Bayesian point estimate minimizes the *posterior expected loss*, as seen before. Using the 0-1 loss function leading to posterior mode with the likelihood (5.5) turns the MAP estimator into the *least-squares fit*:

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid D) = \arg\max_{\theta} p(D \mid \theta) = \arg\min_{\theta} \sum_{i=1}^{n} (y_i - f(x_i; \theta))^2$$

where we used $p(\theta \mid D) \propto p(D \mid \theta)$ for uniform priors. We have then show that classical least-squares fitting corresponds to a Gaussian likelihood, uniform prior, and 0-1 loss leading to the posterior mode (MAP estimate). Weighted least squares arises if the noise variance σ_i^2 depends on i, while other likelihoods (e.g., Poisson, binomial) lead to generalized fitting procedures.

The uncertainty of the fit is given by the Hessian:

$$H_{ab} = \frac{\partial^2}{\partial\theta_a \partial\theta_b} \sum_i \frac{(y_i - f(x_i; \theta))^2}{2\sigma^2}$$

Approximation at optimum:

$$H \approx \frac{1}{\sigma^2} J^T J, \qquad J_{ai} = \partial_{\theta_a} f(x_i; \hat{\theta}_{\text{MAP}})$$

The covariance is:

$$\text{Cov}(\theta) \approx \sigma^2 (J^T J)^{-1}$$

This is related to the propagation of uncertainty formula:

- curvature $\leftrightarrow$ uncertainty
- flat directions $\rightarrow$ large errors.

Chapter 6

Monte Carlo Methods

Monte Carlo methods are a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results. Bayesian inference defines probability distributions and Monte Carlo methods provide the computational machinery to explore them.

6.1 Why Monte Carlo?

In Bayesian inference, we want posterior expectations of the form:

$$\mathbb{E}\left[f(\theta)\right] = \int f(\theta)p(\theta \mid D)\, d\theta$$

from which we derive evidence, credible intervals, point estimates (MAP, mean, moments).

In most realistic problems we end up with posterior distributions that have no analytical solution and then we have to solve high-dimensional integrals. However, in this cases, classical numerical integration fails in high dimensions. Monte Carlo methods help estimating these integrals by **random sampling**, which basically consists in sampling from a distribution and then compute Equation 6.1.

Basic idea. Basically, the Monte Carlo idea is to replace integrals with averages over random samples. The key principle is:

$$\mathbb{E}_p[f(X)] \approx \frac{1}{N}\sum_{i=1}^{N} f(x_i), \quad x_i \sim p \tag{6.1}$$

Deterministic integration: curse of dimensionality. Another reason why we should switch to these stochastic methods is the so-called curse of dimensionality problem. Consider a curve such as the one in Figure 6.1 and we want to know the area under the curve (or the volume in d-dimension). In d-dimension, a grid-based methods with n points per dimension requires $N = n^d$ total points. The computational cost of this deterministic integration grows exponentially with d.

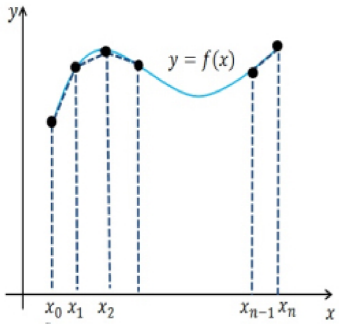


Figure 6.1

Example 6.1.1

For example, let's calculate the hyper-volume. Consider the integral over the unit hyper-cube:

$$I_d = \int_{[0,1]^d} \mathbb{1}_{\|x\|<1/2}\, dx$$

which is the volume fraction of $[0,1]^d$ inside a ball of radius $1/2$. For $d \gg 1$, most of the volume is on the surface. In fact, in d dimensions, the volume of any object scales with the size of the object to the power of d because all the points inside the volume are basically on the hypersurface. Therefore, we can write:

$$V_d \sim R^d,\ S_d \sim R^{d-1} \quad \Rightarrow \quad V_d \sim S_d \text{ for } d \gg 1$$

For this integral, this means that $I_d \to 0$ as $d \gg 1$ because we are looking at the fraction of the volume of the cube that stays inside it, which is in the center, and the larger the dimension, the more fraction of volume is going to be on in the surface.
Grid-based methods waste almost all n^d points outside the ball and the number of points becomes intractable.
The Monte Carlo idea is quite simple. Rather than starting with the previous discretization and summation, to estimate I_d, MC idea is to draw samples uniformly in $[0,1]^d$ and summing the points that satisfy the condition we want:

$$I_d \approx \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}_{\|x_i\|<1/2}$$

Basically, $I_d = P(\|X\| < 0.5)$ given $X_i \sim \mathcal{U}([0,1]^d)$ and the basic algorithm is something such as:

```
d <- 3
N <- 1000000
x <- matrix(runif(N * d), ncol = d)
# Compute squared modulus X
inside <- rowSums(x^2) < 0.25
mean(inside)
# [1] 0.065478
```

Let's consider a snippet code of Monte Carlo estimation in R and see what happens as d becomes larger:

```
N <- 1000000
    for (d in c(2, 5, 10, 20)) {
    start <- proc.time()
    x <- matrix(runif(N * d), ncol = d)
    inside <- rowSums(x^2) < 0.25
    val <- mean(inside)
    elapsed <- (proc.time() - start)['elapsed']
    cat('d =',d, -> ',val,' | time = ',elapsed,' 'sec') }
# d = 2 -> 0.195299 | time = 0.121 sec
# d = 5 -> 0.005191 | time = 0.151 sec
# d = 10 -> 2e-06 | time = 0.322 sec
# d = 20 -> 0 | time = 0.775 sec
```

As we can see, the larger d becomes, the smaller the are is. Moreover, observe that with this method, time scales linearly with d and the operation we're doing are not cursed by the dimensionality of the system. However, increasing the dimensionality, the estimation of the area inside the volume tends to zero. This means that the dimensionality does not impact the algorithm anymore, but the statistics. This happens because of the fact we mentioned earlier: most of the points lie on the surface when the dimension becomes larger and finding the point inside the sphere is a *rare event*.

6.2 Monte Carlo estimator

Consider $x_1, \dots, x_N \sim p(x)$ i.i.d. samples. Define the **estimator** for a generic integral $I = \mathbb{E}_p[f(X)]$:

$$\hat{I}_N = \frac{1}{N} \sum_{i=1}^{N} f(x_i) \tag{6.2}$$

This is a random object since it depends on the random variables x_i. It has two important properties which we already have seen when we talked about estimators. For large values of N, it converges to the true expectation value (it's unbiased) and the variance scales as $1/N$ (due to the fact that $\hat{I}_N$ is given by the sum of i.i.d. variables):

$$\mathbb{E}[\hat{I}_N] = \mathbb{E}_p[f] \text{ (unbiased)}, \qquad \mathrm{Var}(\hat{I}_N) = \frac{1}{N} \mathrm{Var}_p(f) \tag{6.3}$$

Therefore, the error scales as $\hat{I}_N - \mathbb{E}_p[f] = \sqrt{\mathrm{Var}_p(f)/N} \sim \mathcal{O}\big(N^{-1/2}\big)$. A very crucial feature is the fact that there is no explicit dependence on the dimension of the problem because the only thing that appear is how many variables there are. A point to be careful is that $\mathrm{Var}_p(f)$ can still be a problem in large d because it has to do with the statistics, the shape of the distribution in d dimensions. Basically, the variance can still be a function of d.

Example 6.2.1

Let's consider again the previous example. The volume fraction I_d shrinks exponentially with d. The function f that we average in the MC scheme is the characteristic function:

$$f(x) = \mathbb{1}_{\|x\|<0.5}$$

The variance is $\mathrm{Var}(f) = I_d(1 - I_d) \approx I_d$ for $d \gg 1$ However, also the expectation value of $I_d \to 0$ as $d \gg 1$. Therefore, we have to compare the relative error which grows exponentially in d:

$$\frac{\sqrt{\mathrm{Var}(\hat{I}_N)}}{I_d} \xrightarrow{d\gg 1} \frac{1}{\sqrt{NI_d}}$$

Therefore, to keep the fluctuations under control, we need $N \sim 1/I_d$ samples to keep the relative error $\mathcal{O}(1)$. This is not due to numerical reasons but to deal with the fluctuations of I_d. So, the method on how we sample introduce this kind of error.
In the following code we are asking for the volume of a hypersphere of radius 0.5 in d dimension that lies in the positive orthant. The true value is given, analytically, by:

$$I_d = \frac{\pi^{d/2}}{\Gamma(\frac{d}{2}+1)} 0.5^d \times 0.5^d$$

```
true_val <- function(d){pi^(d/2) / gamma(d/2 + 1) * 2^(-2*d)
N <- 1e6
for (d in c(2, 5, 10, 15, 20)) {
    start <- proc.time()
    x <- matrix(runif(N * d), ncol = d)
    inside <- rowSums(x^2) < 0.25
    k <- sum(inside)
    val <- k / N
    p_true <- true_val(d)
    rel_err <- if (k > 0) abs(val - p_true) / p_true else
       Inf
    elapsed <- (proc.time() - start)['elapsed']
    cat('d =', d,
        '| hits =', k,
        '| est =', signif(val, 3),
        '| rel.err =', signif(rel_err, 3) }
#d = 2 | hits = 195569 | est = 0.196 | rel.err = 0.00398
#d = 5 | hits = 5066 | est = 0.00507 | rel.err = 0.0145
#d = 10 | hits = 1 | est = 1e-06 |rel.err = 0.589
#d = 15 | hits = 0 | est = 0 | rel.err = Inf
#d = 20 | hits = 0 | est = 0 | rel.err = Inf
```

Here, *hits* indicate the number of points that are inside the ball, which decreases dramatically as d grows, as we already mentioned. Basically, in d dimensions most of the points are located in the hypersurface of the sphere and the estimation tends to zero because we miss the points in the center (which are few). Therefore, the relative error tends to infinite since we have an estimation of the integral of zero.

6.3 Overview

Basically, Monte Carlo methods reduce integration to sampling, but the core problem is: *how do we generate samples* $x_i \sim p(x)$*?* The rest of this course tries to tackle this fundamental question by introducing different methods that allow us to build increasingly general sampling methods.

The first method we will see is the *inverse transform* (CDF inversion) which is analytical (exact) but works only for simple distribution. Therefore, we will switch to approximated but smarter method: *importance sampling*, *rejection sampling*, *Markov chains*.

6.4 Inversion of the cumulative distribution function

The inversion of the CDF is one of the most fundamental and exact methods for sampling from a probability distribution. Recalling that the CDF is the object that tells us what's the probability that a random variable X takes values smaller or equal than x. Let X be a random variable with PDF $p(x)$. The CDF is then:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} p(x')\, dx' \tag{6.4}$$

Its properties are:

- $F(x)$ is monotonic increasing;
- $F(x) \in [0, 1]$
- $\lim_{x\to-\infty} F(x) = 0$, $\lim_{x\to\infty} F(x) = 1$

Therefore, it can be computed and inverted.

The idea of the CDF inversion is to sample uniformly in probability space, $U \sim \mathcal{U}([0, 1])$. Defining

$$X = F^{-1}(U)$$

with F cumulant of p. Then, the claim is that $X \sim p(x)$. Here, the inverse of F is $F^{-1}(U) = \inf\{x : F(x) \geq u\}$.

Proof. The result is based on the following equalities[1]:

$$P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) = \int_0^{F(x)} du = F(x)$$

where the last passage comes from the fact that $p_U(u) = 1$ in the interval $[0, 1]$. Therefore,

$$P(X \leq x) = F(x)$$

Hence, X has density $p(x)$. □

6.4.1 CDF inversion algorithm

The algorithm to sample $X \sim p(x)$ is the following:

1. Compute the CDF, $F(x)$
2. Invert it: $F^{-1}(u)$. This is the quantile function
3. Sample $u \sim \mathcal{U}([0, 1])$
4. Set $x = F^{-1}(u)$

[1] Remember that F is a monotonic function.

In Bayesian inference, this method has some advantages, since it allows for exact and independent sampling from simple posteriors, direct computation of expectations and efficient inference in conjugate Bayesian models. However, it is clearly limited to distributions with analytically invertible CDFs.

Example 6.4.1

Let's consider the example of a simple distribution such as the exponential distribution. The PDF is:

$$p(x) = \lambda e^{-\lambda x}, \qquad x \geq 0$$

and the CDF is:

$$F(x) = \lambda \int_0^x dx' \, e^{-\lambda x'} = 1 - e^{-\lambda x}$$

To find the inverse, we set $F(x) = u$ and solve:

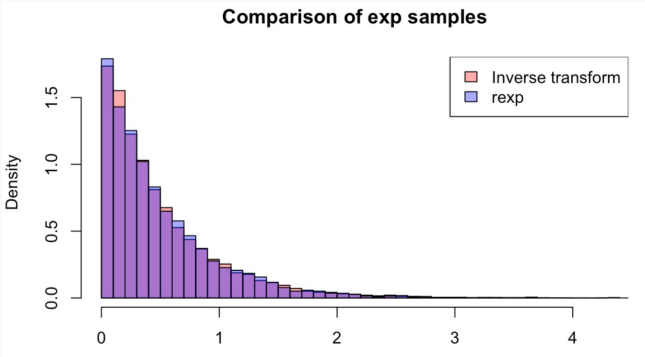
$$u = 1 - e^{-\lambda x} \quad \Rightarrow \quad x = -\frac{1}{\lambda} \log(1 - u)$$

Since $1 - u \sim \mathcal{U}([0, 1])$:

$$x = -\frac{1}{\lambda} \log u$$

A snippet code to compare the inverse transform sampling with the built-in exp generation in `R` is the following:

```
N <- 10000; u <- runif(N); lambda <- 2
x <- -log(u) / lambda
x_builtin <- rexp(N, rate = lambda)
hist(x, probability = TRUE, col = rgb(1,0,0,0.4), breaks =
    50, main = 'Comparison of exp samples', xlab = 'x')
hist(x_builtin, probability = TRUE, col = rgb(0,0,1,0.4),
    breaks = 50, add = TRUE)
legend('topright', legend = c('Inverse transform', 'rexp'),
    fill = c(rgb(1,0,0,0.4), rgb(0,0,1,0.4)))
```



As we can see, the two histograms are practically overlapping exactly.

Example 6.4.2

Let's consider another example. Imagine we want to sample from a distribution with very long tail. This could be the Pareto distribution, which typically describes a lot of things in population, such as large portion of wealth which is held by a small fraction of population (tail). Its PDF:

$$p(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}, \qquad x \geq x_m$$

which has two parameters, x_m which is a minimum parameter and α which represents the shape, modifying how fast the tail decreases.
The CDF is:

$$F(x) = 1 - \left(\frac{x_m}{x}\right)^\alpha, \qquad x \geq x_m$$

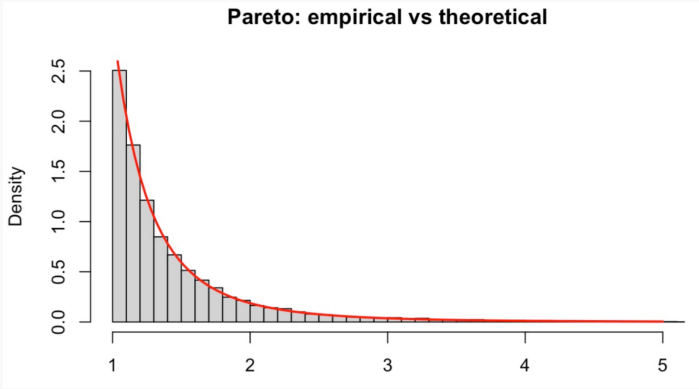
Therefore, the inversion leads to:

$$u = 1 - \left(\frac{x_m}{x}\right)^\alpha \quad \Rightarrow \quad x = x_m(1-u)^{-1/\alpha}$$

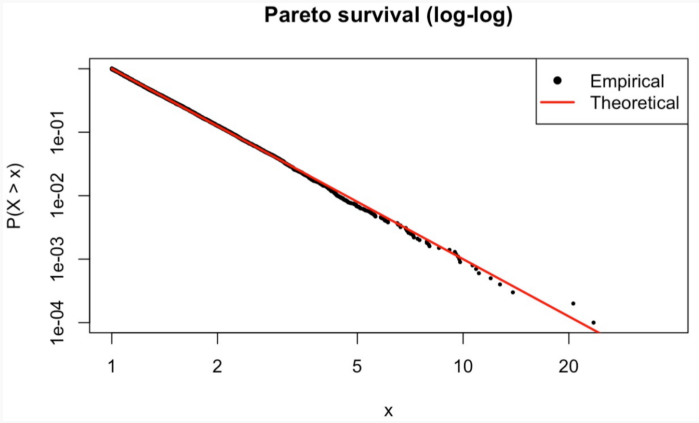
In R, we compare here the theoretical vs the empirical Pareto distribution.

```
N <- 10000; u <- runif(N); alpha <- 3; xm <- 1; x
    <- xm * (1 - u)^(-1/alpha)
# empirical survival
x_sorted <- sort(x)
n <- length(x_sorted)
surv_emp <- 1 - (1:n)/n
# histogram
hist(x, probability = TRUE, breaks = 100,
    main = 'Pareto: empirical vs theoretical',
    xlab = 'x', xlim = c(1, 5))
# theoretical density
dpareto <- function(x, xm, alpha) {
    ifelse(x >= xm,
        alpha * xm^alpha / x^(alpha + 1), 0)}
curve(dpareto(x, xm, alpha),
    col = 'red', lwd = 2, add = TRUE)
# plot empirical survival (1-CDF)
plot(x_sorted, surv_emp, log = 'xy', pch = 16, cex = 0.5,
    xlab = 'x', ylab = 'P(X > x)',
    main = 'Pareto survival (log-log)')
# analytical survival: P(X > x) = (xm/x)^alpha
curve((xm/x)^alpha,
    from = xm, to = max(x_sorted),
    col = 'red', lwd = 2, add = TRUE)
legend('upper right',
    legend = c('Empirical', 'Theoretical'),
    col = c('black', 'red'), pch = c(16, NA), lwd = c(NA, 2)
```

We can see that there is consistency between the two distributions. As we said, this distribution follows a power law. It has very large mass at large x.



To better see the evidence of a power law distribution, it is convenient to plot a log-log scale, which shows a straight line with a slope that correspond to the power exponent. In particular, the *survival function*, which corresponds to 1 − CDF. We can observe that for larger values of x, there are some fluctuations due to the fact that in that region the points are scarce (rare events).



These two examples show the simplicity of the inverse CDF method, which is exact but efficient only in one dimension. This is rarely available in complex models, and impossible in high dimensions. This leads to next question: *how to sample when F^{-1} is not available?*

6.5 Rejection sampling

Before introducing Monte Carlo Markov Chain methods, it is useful to discuss a conceptually simpler strategy for sampling from a target distribution: **rejection sampling**. Unlike MCMC, rejection sampling generates independent samples, at the price of potentially low efficiency in high dimensions.

Suppose we wish to sample from a target density $\pi(x)$ (defined on a space $\mathcal{X}$), known up to normalization:

$$\pi(x) = \frac{\tilde{\pi}(x)}{Z}$$

where $Z = \int_{\mathcal{X}} \tilde{\pi}(x)\, dx$ is unknown or difficult to compute normalization parameter. The idea is to introduce a proposal density $q(x)$ (with $q(x) > 0$ whenever $\tilde{\pi}(x) > 0$) which is easy to sample from and has the same support as π. This is defined such that there exists

a constant $M > 0$:

$$\frac{\tilde{\pi}(x)}{q(x)} \leq M, \quad \forall x \in \mathcal{X} \tag{6.5}$$

The interpretation is that the $Mq(x)$ provides an envelope bounding the unnormalized target density $\tilde{\pi}(x)$. The ideal choice for M is:

$$M^* \equiv \max_{x \in \mathcal{X}} \frac{\tilde{\pi}(x)}{q(x)}$$

as it maximizes the acceptance probability (see later). The main problem is that in high dimensions the max is often extremely hard to be determined.

6.5.1 Rejection sampling algorithm

The rejection sampling algorithm then proceeds as follows:

1. Sample $x \sim q(x)$ after having chosen a proposal density $q(x)$ that approximates the shape of $\pi(x)$ and is easy to sample from (e.g., Gaussian, uniform on a bounded domain, etc., but then must have the same support).

2. Sample $u \sim \mathcal{U}([0, 1])$.

3. Accept x if:

$$u \leq \frac{\tilde{\pi}(x)}{Mq(x)}$$

   with M chosen as close as possible to the estimated M^*

4. otherwise reject and return to step 1, repeating until the desired number of accepted samples is obtained. Notice that, by construction, the RHS of the inequality is always smaller than 1.

We can verify that the distribution of accepted samples is $\pi(x)$ and they are also independent. Why this works?

Let A denote the event that a proposed x is accepted. The joint density of proposing x and accepting it is:

$$p(x, A) = q(x)\frac{\tilde{\pi}(x)}{Mq(x)} = \frac{\tilde{\pi}(x)}{M}$$

The acceptance probability is therefore:

$$P(A) = \int_{\mathcal{X}} \frac{\tilde{\pi}(x)}{M}\, dx = \frac{1}{M}\int_{\mathcal{X}} \tilde{\pi}(x)\, dx = \frac{Z}{M}$$

which is less than or equal to 1 thanks to (6.5) and the fact that $\int_{\mathcal{X}} q(s) = 1$. The density of x conditioned on acceptance is:

$$p(x \mid A) = \frac{p(x, A)}{P(A)} = \frac{\tilde{\pi}(x)/M}{Z/M} = \frac{\tilde{\pi}(x)}{Z} = \pi(x)$$

which is exactly the target distribution $\pi(x)$. Thus, the accepted samples are independent and distributed according to the desired target density. For efficiency, one therefore wishes to choose $q(x)$ such that M is as small as possible, ideally close to M^*. Basically, the smaller M is, the larger the probability of acceptance will be. Because otherwise we throw away a lot of random number by non accepting them and this is a waste of time.

6.5.2 Rare event problem revisited

Let's consider again the rare event problem shown in Example 6.1.1. We had a target integral which is the fraction of hypercube of edge 1 occupied by hypersphere of radius 1/2:

$$I_d = \int_{[0,1]^d} \mathbb{1}_{\|x\|<1/2}\, dx$$

The naive Monte Carlo yields:

$$I_d \approx \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}_{\|x_i\|<1/2}, \qquad x_i \sim \mathcal{U}([0,1]^d)$$

We know that, for large d, event $\|x\| < 1$ is extremely rare, because the volume scale as $V_d \sim \ell^d$, the surface as $S_d \sim \ell^{d-1}$ and then $V_d \sim S_d$. Then, all the points we draw lie on the surface. We saw that the variance is $\mathrm{Var}(f) \approx I_d$ and the relative error is:

$$\frac{\sqrt{\mathrm{Var}(\hat{I}_N)}}{I_d} \sim \frac{1}{\sqrt{NI_d}}$$

which becomes very large as $I_d \to 0$. This naive strategy fails brutally as we're not storing an exponential amount of data, since we need N numbers to tackle the fluctuations of the objects we are sampling.

A smarter strategy is to use rejection sampling, whose idea is to sample from a proposal concentrated on the origin. We choose then $q(x)$ uniform on the hypersphere:

$$q(x) = \begin{cases} 2^d & \text{if } x \in [0,1/2]^d \\ 0 & \text{otherwise} \end{cases}$$

We can then rewrite our target density as:

$$\tilde{\pi}(x) = \mathbb{1}_{\|x\|<1/2}, \qquad I_d = \int_{[0,1]^d} \tilde{\pi}(x)\, dx = Z$$

which can be interpreted as a normalization factor of our distribution that has to be computed. We then set the bound as:

$$\frac{\tilde{\pi}(x)}{q(x)} \leq 2^{-d} \quad \Rightarrow \quad M = 2^{-d}$$

Therefore, this defines the acceptance probability:

$$P(A) = \frac{I_d}{M} = 2^d I_d$$

which is the probability to accept a certain draw from $q(x)$. This can mitigate our problem of not hitting the sphere, because I_d becomes exponentially small with d, but here we have a correction factor which is exponentially large. How do we use this formula to estimate our integral? We can compute our integral:

$$I_d = MP(A)$$

estimated as:

$$\hat{I}_d = 2^{-d}\frac{N_{\text{acc}}}{N}$$

with N total samples and N_{acc} total accepted samples. The key improvement here is that sampling is concentrated around the origin. In this way, the rare-event problem is reduced.

In R:

```
true_val <- function(d) {pi^(d/2)/gamma(d/2 + 1)*2^(-2*d)}
rej_estimate <- function(d, N = 1e6) {
    x <- matrix(runif(N * d, 0, 0.5), ncol = d)
    inside <- rowSums(x^2) < 0.25
    k <- sum(inside); val <- 2^(-d) * k / N #M=2^(-d)
    p_true <- true_val(d)
    rel_err <- if (p_true > 0) abs(val - p_true) / p_true
        else Inf
    cat('d =', d, '| accepted =', k, '| est =', signif(val,3),
        '| rel.err =', signif(rel_err, 3), '\n') }
for (d in c(2,5,10,15,20)) {rej_estimate(d)}
#d = 2 |accepted = 785239 |est = 0.196 |rel.err = 0.000203
#d = 5 |accepted = 164736 |est = 0.00515 |rel.err = 0.00147
#d = 10 |accepted = 2448 |est = 2.39e-06 |rel.err = 0.017
#d = 15 |accepted = 17 |est = 5.19e-10 |rel.err = 0.46
#d = 20 |accepted = 0 |est = 0 |rel.err = Inf
```

Just observe that the improvement here is that we can compute the integral for dimensions quite larger than before.

Because each accepted sample is independent, no burn-in is required and no autocorrelation is present (see later for MCMC). However, the computational cost per accepted sample is proportional to $1/P(A) = M/Z$. If M is much larger than Z, the method becomes inefficient.

The main limitation of rejection sampling appears in high dimensions. If $\pi(x)$ is concentrated in tiny regions of the support of $q(x)$, then M must be large and the acceptance probability becomes exponentially small in the dimension:

$$P(A) \to 0$$

typically exponentially in d. This leads to inefficiency since we draw a lot of samples and most of them are lost. This is the so-called *curse of dimensionality*, which restricts the applicability of rejection sampling in complex Bayesian models. For instance:

$$P(A) = \frac{I_d}{M} \propto \frac{2^{-2d}}{2^{-d}} = e^{-d\ln 2}$$

For this reason, while rejection sampling is conceptually simple and mathematically appealing, it is typically replaced in high-dimensional inference problems by Markov Chain Monte Carlo methods, which trade independence of samples for dramatically improved efficiency.

Since this method has a lot of potential inefficiency because only a fraction of samples we draw are used, we now introduce a method in which we make use of every random variable we sample. In fact there could be a fraction of samples that is less important but still might be used. This is *importance sampling*.

6.6 Importance sampling

As we said many times, in Bayesian inference, we are interested in expectations with respect to a posterior distribution $\pi(\theta) \equiv p(\theta \mid D)$ such as:

$$\mathbb{E}_\pi[f(\theta)] = \int f(\theta)\pi(\theta)\, d\theta$$

When direct sampling from $\pi(\theta)$ is hard, but evaluation of the unnormalised posterior $\tilde{\pi}(\theta) = p(D \mid \theta)p(\theta)$ is possible, **importance sampling** provides an alternative Monte Carlo approximation method.

Let introduce a **proposal** density $q(\theta)$ such that:

$$\text{supp}(\pi) \subseteq \text{supp}(q), \qquad q(\theta) > 0 \text{ whenever } \pi(\theta) > 0$$

Then, we may rewrite:

$$\mathbb{E}_\pi[f(\theta)] = \int f(\theta)w(\theta)q(\theta)\, d\theta \tag{6.6}$$

with the **importance weight** defined as:

$$w(\theta) \equiv \frac{\pi(\theta)}{q(\theta)} \quad \Rightarrow \quad \mathbb{E}_\pi[f] = \mathbb{E}_q[fw]$$

in such a way that we turned an expectation over π of f into an expectation over q of fw. This is the key point. We're not rejecting, but re-weighting the integral. The idea is to use q that put probability mass where it's hard to get probability mass. In this way, we could make events that under π are rare, we could make typical under q.

If $\theta_i \sim q(\theta)$ independently for $i = 1, \ldots, N$, then:

$$\hat{I}_N = \frac{1}{N}\sum_{i=1}^{N} f(\theta_i)w(\theta_i)$$

is an unbiased estimator of $\mathbb{E}_\pi[f(\theta)]$ provided π is known in normalized form.

However, in Bayesian applications, the posterior is typically known only up to normalization:

$$\pi(\theta) = \frac{\tilde{\pi}(\theta)}{Z}, \qquad Z = \int \tilde{\pi}(\theta)\, d\theta$$

Since Z is unknown, we use *self-normalized importance sampling*. Defining **unnormalized weights**:

$$\tilde{w}_i = \frac{\tilde{\pi}(\theta_i)}{q(\theta_i)}, \qquad \theta_i \sim q$$

the expectation values is:

$$\mathbb{E}\,[f] = \frac{\int f(\theta)\tilde{w}(\theta)q(\theta)\, d\theta}{\int \tilde{w}(\theta)q(\theta)\, d\theta}$$

and the estimator becomes:

$$\hat{I}_N = \frac{\sum_{i=1}^{N} \tilde{w}_i f(\theta_i)}{\sum_{i=1}^{N} \tilde{w}_i}$$

This estimator is consistent but no longer unbiased, since it is a ratio of random variables (so its expectation does not coincide with the ratio of expectations), but by the law of large numbers:

$$\hat{I}_N \xrightarrow{N\to\infty} \mathbb{E}_\pi[f(\theta)]$$

The variance of the basic (non-self-normalized) estimator is:

$$\mathrm{Var}(\hat{I}_N) = \frac{1}{N}\,\mathrm{Var}_q(f(\theta)w(\theta))$$

Efficiency therefore depends on weights variability. In particular when the variability is small, the variance is small. The optimal proposal $q(\theta)$, i.e. that minimizes variance for estimating a fixed expectation, is:

$$q^*(\theta) \propto |f(\theta)|\,\pi(\theta)$$

This can be shown by minimizing the second moment:

$$\mathbb{E}_q\left[(f(\theta)w(\theta))^2\right] \propto \int \frac{f(\theta)^2\pi(\theta)^2}{q(\theta)}\,d\theta$$

Introducing the Lagrange multiplier λ (q must be the PDF):

$$\int \frac{f(\theta)^2\pi(\theta)^2}{q(\theta)}\,d\theta + \lambda\left(\int q(\theta)\,d\theta - 1\right)$$

We can find the minimum by functional derivative:

$$-\frac{f(\theta)^2\pi(\theta)^2}{q^*(\theta)^2} + \lambda = 0 \quad \Rightarrow \quad q^*(\theta)^2 \propto f(\theta)^2\pi(\theta)^2$$

However, this choice is typically unavailable in practice.

A useful diagnostic is the **effective sample size** (ESS):

$$\mathrm{ESS} = \frac{\left(\sum_{i=1}^N \tilde{w}_i\right)^2}{\sum_{i=1}^N \tilde{w}_i^2}$$

Rationale: if $\tilde{w}_i = w_*$ for all i:

$$\mathrm{ESS} = \frac{(Nw_*)^2}{Nw_*^2} = N$$

If $\mathrm{ESS} \approx N$, indicates good weights with small variability. If the weights are highly uneven, $\mathrm{ESS} \ll N$, few samples dominate, indicating severe weight degeneracy.

In practice, since optimal q^* is not usable (depends on π), for good performance, one requires that:

- q should approximate the posterior π, matching its shape;
- the tails of q must be at least as heavy as those of π, otherwise the variance of the estimator may be infinite;
- in high dimensions, q must match the geometry (correlations, scale, and support) of π.

If $\sup_\theta \frac{\pi(\theta)}{q(\theta)} = \infty$, the variance of the estimator may be infinite and the method unreliable.

Now, we recall the hypercube example via importance sampling. Recalling the integral to be evaluated, we want to write it by averaging over q and re-weighting it by multiplying the measure by the weight:

$$I_d = \int dx\,\mathbb{1}_{\|x\|<1/2} = \mathbb{E}_q\left[\mathbb{1}_{\|x\|<1/2}w(x)\right]$$

We must choose a proposal $q(x)$ concentrated in the ball by making typical events that are rare, then it should be very peaked in the origin. The weight is:

$$w(x) = \frac{1}{q(x)}$$

In this way, there are many samples inside $\|x\| < 1/2$ and rare-event problem is largely mitigated. Importance sampling strategy is the following. Consider the integral (that can be seen as normalization factor over the distribution in the sphere):

$$I_d = \int \mathbb{1}_{\|x\|<1/2}\, dx$$

and introduce the proposal $q(x)$:

$$I_d = \int \mathbb{1}_{\|x\|<1/2} \frac{1}{q(x)} q(x)\, dx$$

The estimator is then:

$$\hat{I}_d = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}_{\|x_i\|<1/2} \frac{1}{q(x_i)}$$

While with naive MC methods there are almost no points in the sphere, with importance sampling many points are near the origin, there is no rejection (with improvement in efficiency) and continuous weighting, rather than throwing away stuff. Moreover, if $q \approx$ optimal, then $fw \approx$ const.. This leads to strong variance reduction.

How do we do it in practice? We want more samples near the origin. Therefore, we choose independent components of the vector x:

$$x_j \sim \text{Beta}\,(a, 1)\,, \qquad a < 1$$

where j runs over the dimension of the space d. Therefore, the PDF and the CDF are:

$$q(x) = \prod_{j=1}^{d} a x_j^{a-1} \quad \Rightarrow \quad F(x_j) = x_j^a \tag{6.7}$$

Then, the inverse CDF sampling yields:

$$x_j = u_j^{1/a}, \qquad u_j \sim \mathcal{U}([0, 1])$$

and x_j is distributed as beta function.

Then, we can repopulate the sampler by re-weighting the function with the weight:

$$w_i = \frac{1}{q(x_i)}$$

In R:

```
true_val <- function(d) {pi^(d/2)/gamma(d/2 + 1)*2^(-2*d)}
is_estimate <- function(d, N = 1e6, a = 0.35) {
    u <- matrix(runif(N * d), ncol = d); x <- u^(1/a)
    inside <- rowSums(x^2) < 0.25
    log_q <- d*log(a) + (a-1)*rowSums(log(x)) # log q(x)
    contrib<-numeric(N);contrib[inside]<-exp(-log_q[inside])
```

```
    val <- mean(contrib); p_true <- true_val(d)
    rel_err <- if (p_true > 0) abs(val - p_true) / p_true
        else Inf
    cat('d =', d, '| inside =', sum(inside),
    '| est =', signif(val, 3),'| rel.err =', signif(rel_err,3))}
for (d in c(2,5,10,15,20)) {is_estimate(d)}
#d = 2 | inside = 590529 | est = 0.196 | rel.err=0.000437
#d = 5 | inside = 211358 | est = 0.00507 | rel.err=0.0135
#d = 10 | inside = 25408 | est = 2.53e-06 | rel.err=0.0396
#d = 15 | inside = 2197 | est = 3.49e-10 | rel.err=0.0176
#d = 20 | inside = 143 | est = 8.24e-14 | rel.err=2.51
```

we can see that for higher dimension we maintain a lower relative error. Furthermore, we have to take some numerical precautions to solve problems of numerical instability. In fact, in large d, the product (6.7) is extremely small. Therefore, $q(x)$ underflows to 0 and this implies that $1/q(x)$ becomes ∞ or `NaN`. A simple solution is to compute in log-space:

$$\log q(x) = d \log a + (a-1) \sum_i \log x_i \quad \Rightarrow \quad \frac{1}{q(x)} = e^{-\log q(x)}$$

The advantages are clear: the products of small numbers become sums, which are more stable. The exponentiation is made only at the end. In extreme cases one uses the so-called LogSumExp identity:

$$\log \left(\sum_{i=1}^{n} e^{x_i} \right) = m + \log \left(\sum_{i=1}^{n} e^{x_i - m} \right), \qquad m = \max_i x_i$$

Note that importance sampling can also estimate the evidence:

$$Z = \int \tilde{\pi}(\theta)\, d\theta = \int \frac{\tilde{\pi}(\theta)}{q(\theta)} q(\theta)\, d\theta$$

via the estimator:

$$\hat{Z} = \frac{1}{N} \sum_{i=1}^{N} \frac{\tilde{\pi}(\theta_i)}{q(\theta_i)}, \qquad \theta_i \sim q$$

Some key takeaways:

- Naive MC fails due to rare events, typically:

$$\text{error} \sim e^{\text{const.} \times d}$$

- Rejection sampling method improves the proposal region, but still wastes samples (rejection).
- Importance sampling, instead, uses all samples and corrects bias via weights. In this way, it achieves strong variance reduction if q is well chose.

The core idea is to sample where the mass is. In this way, rare events uner π become typical under q.

In moderate/large d, importance sampling often suffers from *weight degeneracy*: a small number of samples dominate the weight mass. This phenomenon arises because probability mass in high dimensions concentrates in thin regions, making it difficult for a fixed proposal

distribution to approximate the posterior well. As dimension increases, the variance of the weights typically grows exponentially unless q is extremely well tuned. This curse of dimensionality is one of the main motivations for Markov chain Monte Carlo methods, which explore the posterior locally rather than relying on global reweighting. In practice, they construct artificial dynamics that samples from π for long times. They have the potential ability to sample efficiently in large d. However, the trade-off is that they introduce finite-time correlations. The key points to understand are the choice of the dynamical rule and monitoring of decaying autocorrelation.

6.7 Markov chains

When direct sampling methods such as CDF inversion are not available, one typically resorts to **Markov chain Monte Carlo** methods (MCMC). These algorithms construct a stochastic dynamics (random walk) in the parameter space, which explores the regions of high probability density. More precisely, they construct a Markov chain whose stationary distribution is the target probability distribution $\pi(x)$ of interest. Samples generated along the chain can then be used to estimate expectations with respect to $\pi(x)$. In fact, if X_t denotes the state of the chain at algorithmic time t, then at equilibrium we can replace the integral over the distribution with a finite average computed on the samples at time t:

$$\mathbb{E}_\pi[f] = \int f(x)\pi(x)\,dx \approx \frac{1}{N}\sum_{i=1}^{N} f(X_t)$$

In Bayesian inference, this is crucial because:

$$p(\theta \mid D) \propto p(D \mid \theta)p(\theta)$$

and MCMC only needs the posterior *up to normalization*.

6.7.1 Stochastic processes

To understand how to build and control the dynamics, we briefly introduce the basis for the Markov chains which rely on stochastic processes. A **stochastic process** is an ordered set of random variables indexed by (algorithmic) time $\{ X_0, X_1, \ldots, X_t, \ldots \}$. For a generic stochastic process, the next state depends on the entire previous history:

$$P(X_{t+1} = x' \mid X_t = x, X_{t-1}, \ldots, X_0)$$

Stochastic processes are called **Markovian** (or they have **Markov property)** if they're *memoryless*, meaning that they satisfy the condition:

$$P(X_{t+1} = x' \mid X_t = x, X_{t-1}, \ldots, X_0) = P(X_{t+1} = x' \mid X_t = x) \tag{6.8}$$

This is also called memoryless property, meaning that no memory beyond current state is retained. Therefore, the future step of a Markov process depends only on the current state and not on the past history.

6.7.2 Markov chain

A discrete-time **Markov chain** is a stochastic process $\{ X_t \}_{t \geq 0}$ characterized by the conditional probability:

$$P(X_{t+1} = x' \mid X_t = x) \equiv T(x \to x') \tag{6.9}$$

where $T(x \to x')$ is the transition probability from x to $x' \neq x$.

Basically, MCMC are a *design problem*, meaning that the goal of MCMC is to design transition probabilities T such that the target distribution $\pi(x)$ is a stationary distribution of the chain and the chain converges to it irrespective of the initial condition. We can interpret t as an algorithmic time with unit timestep.

In Figure 6.2 is shown a 2D parameter space explored by successive local moves. Each arrow is a stochastic transition drawn from $T(x \to x')$. The density of visited points at long times approaches $\pi(x)$. Basically, if we draw an histogram of these points, that histogram is a very good approximation of $\pi(x)$. However, because of the dynamics, even if we construct the moves locally, the trajectory will have some correlations. For instance, starting from x_0, reaching x in two steps is really improbable and after many many steps, even if the distribution will be broader, it will be still localized on x_0. Therefore, the dynamics will carry some correlations that we would like to avoid because they reduce the effective number of samples we can use. *There is an error in the drawing, it should be $x' \to x$.*

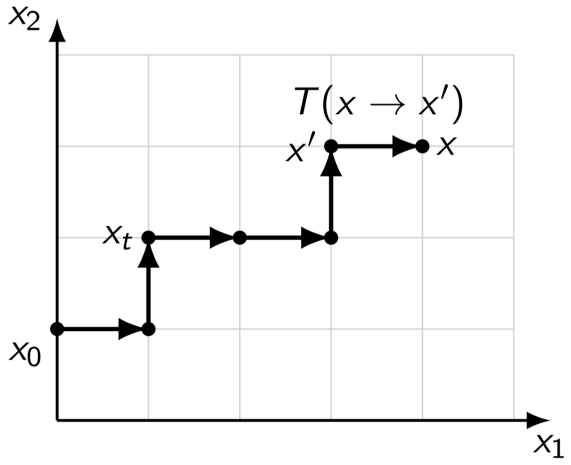


Figure 6.2

The evolution of the probability distribution $p_t(x) = P(X_t = x)$ associated with a Markov chain can be written in the form of a *master equation*. In discrete space, i.e. $x \in \mathbb{Z}$, if we introduce a probability vector $\boldsymbol{p}_t = (p_t(x_1), p_t(x_2), \ldots, p_t(x_N))$ the master equation reads:

$$\boldsymbol{p}_{t+1} = \boldsymbol{T}\boldsymbol{p}_t$$

where the transition matrix $\boldsymbol{T}$ has elements:

$$T_{xx'} = \begin{cases} T(x' \to x) & \text{if } x \neq x' \\ 1 - \sum_{y \neq x} T(x \to y) & \text{if } x = x' \end{cases}$$

where the diagonal entries are the *probabilities to stay still*, while the off-diagonal ones are the probabilities to move to another state nearby. Notice that the diagonal components are

chosen so as to ensure probability conservation and then having the probability normalized:

$$\mathbf{1} \cdot \boldsymbol{p}_t = \sum_i p_t(x_i) = 1, \qquad \forall t \geq 0$$

This follows from the column sums of $\boldsymbol{T}$:

$$\sum_x T_{xx'} = \sum_x T(x' \to x) + 1 - \sum_{y \neq x'} T(x' \to y) = 1$$

We want that the chain leads us to $\pi(x)$. If the distribution is already $\pi(x)$, if it evolves in time, it is still $\pi(x)$. Therefore, a distribution $\pi(x)$ is **stationary** if:

$$\boldsymbol{\pi} = \boldsymbol{T}\boldsymbol{\pi} \quad \iff \quad \pi(x) = \sum_{x'} T_{xx'}\pi(x')$$

Thus, under one update step, the distribution stays the same. In general, starting from an initial density $\boldsymbol{p}_0$, the solution for the evolution after n time steps is given

$$\boldsymbol{p}_t = \boldsymbol{T}^n \boldsymbol{p}_0 \tag{6.10}$$

Therefore, we want to construct $T_{xx'}$ in such a way that:

$$\boldsymbol{p}_t \xrightarrow{t\to\infty} \boldsymbol{\pi}$$

whatever $\boldsymbol{p}_0$ is.

To do that, it is useful to consider the continuous-time version of (6.10). Introducing a small time step dt such that $t = ndt$, we have the evolution:

$$\boldsymbol{p}_t = \boldsymbol{T}^{t/dt}\boldsymbol{p}_0$$

In order to make this independent of how small dt is, $\boldsymbol{T}$ must be of the form:

$$\boldsymbol{T} \approx \mathbf{1} + dt\, \boldsymbol{W} \quad \Rightarrow \quad \boldsymbol{p}_t \approx e^{t\boldsymbol{W}}\boldsymbol{p}_0 \tag{6.11}$$

where $\boldsymbol{W}$ is the **transition-rate matrix** (or generator) with entries:

$$W_{xx'} = \begin{cases} w(x' \to x), & x \neq x' \\ -\sum_{y \neq x} w(x \to y), & x = x' \end{cases} \tag{6.12}$$

where, for $x \neq x'$:

$$w(x' \to x) = \lim_{dt \to 0} \frac{T(x' \to x)}{dt}$$

Differentiating with respect to t leads to the **master equation**:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{p}_t = \boldsymbol{W}\boldsymbol{p}_t = \boldsymbol{W}e^{t\boldsymbol{W}}\boldsymbol{p}_0 \tag{6.13}$$

This is how the probability of a Markov chain evolves in continuous-time intervals. Since it is a linear equation, it's easier to understand the evolution of the chain. In fact, componentwise, the master equation (6.13) reads:

$$\frac{\mathrm{d}}{\mathrm{d}t}p_t(x) = \sum_{x'} \Big[\underbrace{p_t(x')w(x' \to x)}_{\text{in-flux term}} - \underbrace{p_t(x)w(x \to x')}_{\text{out-flux term}} \Big]$$

The *in-flux* term accounts for transitions into state x, while the *out-flux* term represents transitions out of x. By introducing the probability current from x to x':

$$J_t(x \to x') = p_t(x) w(x \to x')$$

Equation 6.13 takes the compact form:

$$\frac{\mathrm{d}}{\mathrm{d}t} p_t(x) = \sum_{x'} \left[J_t(x' \to x) - J_t(x \to x') \right]$$

This expression has the structure of a continuity equation in probability space.

We want the distribution $\pi(x)$ to be stationary in continuous time, i.e. :

$$\frac{\mathrm{d}}{\mathrm{d}t} \pi(x) = 0, \qquad \forall x$$

Therefore, at stationarity, the total incoming probability current must balance the total outgoing one at each point x:

$$\sum_{x'} \left[\pi(x') w(x' \to x) - \pi(x) w(x \to x') \right] = 0$$

This condition corresponds to *zero net probability currents* and is the defining property of a steady state (this is the first condition we have to enforce). What we want is that the Markov chain in the long time converges to the stationary distribution $\pi(x)$, i.e. :

$$\lim_{t \to \infty} p_t(x) = \pi(x) \tag{6.14}$$

for all initial distributions p_0. The second fundamental condition we must enforce is *irreducibility*. The Markov chain is **irreducible** if, for any pair of states x and x', there exists a finite time t such that:

$$P(X_t = x' \mid X_0 = x) > 0$$

This means that we must have transitions that are non-zero in any volume of the space, otherwise one risk to get stuck somewhere. In other words, the state space consists of a single communicating class, meaning that the all space is connected. Irreducibility prevents the dynamics from decomposing into disconnected regions, each with its own stationary distribution. When a Markov chain is irreducible, the stationary distribution is unique and the chain is said to be **ergodic**[2]. In this case, convergence (6.14) holds for all p_0.

Burn-in and spectral relaxation. The first question that might come up is: how long do we have to run the chain? When will it be close to stationarity? As we said, for some steps we will have some memory of the starting point, meaning that early samples still remember the starting distribution $\boldsymbol{p}_0$. This justify the use of a **burn-in** period, during which early samples – still influenced by the initial condition $\boldsymbol{p}_0$ – are thrown out. After

[2] A second necessary condition for discrete-time chains is *aperiodicity*. A state x is said to have period $k > 1$ if returns to x can only occur at times that are integer multiples of k. If such periodic behavior is present, the distribution pt may fail to converge pointwise, instead oscillating indefinitely. Aperiodicity ensures that the chain can return to a state at irregular times, allowing relaxation to stationary distribution. From a physical perspective, ergodicity ensures that the artificial dynamics underlying MCMC explores the entire configuration space, while aperiodicity guarantees effective mixing.

burn-in, time averages along the Markov chain provide consistent estimators of expectations with respect to the target posterior distribution.

Therefore, the question is how long should the burn-in period be? Longer than the longest relaxation time of the chain. The latter can be precisely established looking at the eigenvalues λ_k of the matrix $\boldsymbol{W}$ which set the inverse timescales of the relaxation.

Expanding (6.11) into the left and right eigenvectors of $\boldsymbol{W}$, $\boldsymbol{l}_k$ and $\boldsymbol{r}_k$[3], defined by the equations:

$$\boldsymbol{W}\boldsymbol{r}_k = \lambda_k \boldsymbol{r}_k, \qquad \boldsymbol{W}^T \boldsymbol{l}_k = \lambda_k \boldsymbol{l}_k$$

we obtain for the probability evolution:

$$\boldsymbol{p}_t = \sum_k c_k \boldsymbol{r}_k e^{t\lambda_k}$$

where $c_k = \boldsymbol{l}_k \cdot \boldsymbol{p}_0$ is the projection of the initial probability on the left eigenvector k. Given the specific form of the transition rate matrix (6.12), the Perron-Frobenius theorem states that:

Theorem 6.7.1 (Perron-Frobenius theorem)**.** *There exists an eigenvalue equal to zero, $\lambda_0 = 0$, while all other eigenvalues have smaller (thus, negative) real parts, i.e. :*

$$0 = \mathfrak{Re}(\lambda_0) \geq \mathfrak{Re}(\lambda_1) \geq \dots$$

The zero mode (right eigenvector) corresponding to λ_0 is thus the stationary distribution:

$$r_0(x) = \pi(x)$$

and the left eigenvector is flat

$$l_0(x) = const.$$

as a consequence of probability conservation.

This means that there other modes will decay in time with a rate that is given by $\mathfrak{Re}(\lambda_k)$ for $k > 0$.

Since in general $\boldsymbol{W}$ is not symmetric, there can be also an imaginary part. Thus, separating the stationary modes, the spectral decomposition (6.7.2) becomes:

$$\boldsymbol{p}_t = \boldsymbol{\pi} + \sum_{k>0} c_k \boldsymbol{r}_k e^{-t|\mathfrak{Re}(\lambda_k)|} e^{it|\mathfrak{Im}(\lambda_k)|}$$

For long times, the slowest non-zero mode dominates:

$$\|\boldsymbol{p}_t - \boldsymbol{\pi}\| \overset{t\to\infty}{\approx} e^{-t|\mathfrak{Re}(\lambda_1)|}$$

Therefore, the time one has to wait before discarding the Markov chain is the inverse of the argument of the exponential. The characteristic **relaxation time** is:

$$t_{\text{mix}} \approx \frac{1}{|\mathfrak{Re}(\lambda_1)|}$$

Thus, convergence is attained for times $t \gtrsim t_{\text{mix}}$.

[3]Unless detailed balance holds (see later), $\boldsymbol{W}$ is not symmetrizable, so its left and right eigenvectors differ and the eigenvalues are not real.

The quantity $\lambda_0 - |\mathfrak{Re}(\lambda_1)| = |\mathfrak{Re}(\lambda_1)|$ (because $\lambda_0 = 0$) is called **spectral gap**[4] of the generator and it should be as larger as possible. The timescale over which convergence occurs is called **mixing time**, here showed to be proportional to the inverse of the spectral gap of the transition rate matrix. Slow mixing, corresponding to a small spectral gap, leads to long autocorrelation times, then long memory, and inefficient sampling.

6.7.3 Detailed balance

We can introduce an important simplification by selecting a subclass of stationary dynamics. A particularly important special case of stationary state is that of **detailed balance**:

$$\pi(x)T(x \to x') = \pi(x')T(x' \to x), \qquad \forall x, x'$$

which requires that probability currents cancel *pairwise*:

$$J_t(x \to x') = J_t(x' \to x), \qquad \forall x, x'$$

However, detailed balance is a sufficient condition for stationarity, but not a necessary one: stationary distributions may exist with non-zero currents, analogous to non-equilibrium steady states in physical systems, where it implies that $\pi = \pi_{\text{eq}}$ is thermodynamic equilibrium (the stationary distributions are Gibbs-Boltzmann distributions).

When detailed balance holds, the Markov chain is said to be **reversible**. In this case, the stationary distribution can often be identified directly by inspection, since the condition above implies:

$$\frac{T(x \to x')}{T(x' \to x)} = \frac{\pi(x')}{\pi(x)} \tag{6.15}$$

This is technically convenient as it gives a direct condition to fix $T_{xx'}$. In fact we could design T using (6.15), then π is guaranteed to be stationary.

Most commonly used MCMC algorithm, such as the Metropolis-Hastings, are explicitly constructed to satisfy detailed balance with respect to the target distribution $\pi(x)$. It can be shown that Equation 6.15 allows to define a similarity transformation that symmetrizes $\boldsymbol{W}$. This means that under detailed balance conditions the eigenvalues λ_k are all real[5].

In Bayesian inference, enforcing detailed balance is conceptually appealing and technically convenient, as it guarantees convergence to the desired posterior distribution. At the same time, the analogy with equilibrium statistical mechanics highlights that MCMC with detailed balance samples configurations according to equilibrium weights, with the Monte Carlo time playing the role of an artificial dynamics.

Once the chain has reached stationarity, expectation values of observables $A(x)$ can be estimated as time averages along the chain:

$$\langle A \rangle = \int A(x)p(x)\,dx \approx \frac{1}{N}\sum_{i=1}^{N} A(x_t)$$

where $\{\, x_t \,\}$ are successive samples generated by the Markov chain after the burn-in period. Under the condition of ergodicity, this estimator converges to the correct expectation values as $N \to \infty$.

[4] In the discrete time case, one considers the eigenvalues of the transition matrix $\boldsymbol{T} = e^{\boldsymbol{W}}$ which are simply $\gamma_k = e^{\lambda_k}$. The spectral gap is still defined as the difference of the real part of the largest eigenvalues of $\boldsymbol{T}$, namely $\gamma_0 - |\gamma_1| = 1 - e^{\mathfrak{Re}(\lambda_1)}$.

[5] It can also be proved that among all choices of rates that ensure convergence to $\pi(x)$, those that satisfy detailed balance have the slowest convergence – yet, they are the easiest to devise.

In Bayesian inference, MCMC is particularly powerful because it allows sampling from posterior distribution of the form:

$$p(\theta \mid D) \propto p(D \mid \theta)p(\theta)$$

even when the normalization constant (the evidence) is unknown or intractable. Since MCMC only requires evaluation of the posterior up to a multiplicative constant, it scales naturally.

6.8 Monte Carlo implementation of discrete-time Markov chains

So far, we have discussed the evolution of the probability distribution $p_t(x)$ under a transition matrix $\boldsymbol{T}$. In practice, however, we do not evolve the full distribution $p_t(x)$. Instead, we generate a *single stochastic trajectory*:

$$x_0 \to x_1 \to x_2 \to \dots$$

whose statistics reproduce the master equation in expectation for long times. This is the essence of a Monte Carlo implementation of a discrete-time Markov chain.

Let the state space be either discrete or continuous. Given a current state x_t, the algorithm must generate a new state x_{t+1} according to the transition probabilities $T(x_t \to x')$. Formally, the Monte Carlo procedure consists of:

1. For the current state x_t, construct a probability distribution $T(x_t \to x')$ over possible next states;
2. Draw a uniform random number $u \sim \mathcal{U}([0, 1])$;
3. Choose x_{t+1} according to the CDF associated with $T(x_t \to x')$.

In discrete state spaces, step (3) amounts to selecting $x_{t+1} = x'$ with probability $T(x_t \to x')$. In continuous spaces, direct sampling from $T(x \to x')$ is difficult. Thus, one typically decomposes the transition into two steps: proposing a candidate x' from a proposal density $q(x' \mid x)$ and then accepting or rejecting it with a suitable probability. This leads to the Metropolis-Hastings construction.

6.8.1 Metropolis-Hastings algorithm

Suppose we wish to sample from a target density $\pi(x)$ up to normalization. The **Metropolis-Hastings** (MH) algorithm defines a transition probability of the form:

$$T(x \to x') = q(x' \mid x)\alpha(x \to x')$$

where $q(x' \mid x)$ is the proposal density and $\alpha(x \to x')$ is an acceptance probability to be determined. We impose detailed balance that ensures (more than!) stationarity with respect to $\pi(x)$:

$$\pi(x)q(x' \mid x)\alpha(x \to x') = \pi(x')q(x \mid x')\alpha(x' \to x)$$

A sufficient choice is:

$$\alpha(x \to x') = \min\left(1, \frac{\pi(x')q(x \mid x')}{\pi(x)q(x' \mid x)}\right)$$

Indeed, if the ratio inside the minimum is larger than one, the move is always accepted; otherwise, it is accepted with probability equal to that ration. Substituting this form into the detailed balance condition verifies that;

$$\pi(x)T(x \to x') = \pi(x')T(x' \to x)$$

so that $\pi(x)$ is stationary.

Proof. Define the ratio:

$$r(x, x') = \frac{\pi(x')q(x \mid x')}{\pi(x)q(x' \mid x)}$$

Then,

$$\alpha(x \to x') \min(1, r), \qquad \alpha(x' \to x) = \min\left(1, \frac{1}{r}\right)$$

Now consider the two cases: **Case 1:** $r \geq 1$, then:

$$\alpha(x \to x') = 1, \qquad \alpha(x' \to x) = \frac{1}{r}$$

so:

$$\pi(x)q(x' \mid x)\alpha(x \to x') = \pi(x)q(x' \mid x) = \pi(x')q(x \mid x')\frac{1}{r} = \pi(x')q(x \mid x')\alpha(x' \to x)$$

Case 2: $r < 1$, then:

$$\alpha(x \to x') = r, \qquad \alpha(x' \to x) = 1$$

and the same identity follows.

Therefore, MH satisfies detailed balance.

□

The practical implementation of one MH step from the current state x_t is:

1. Given x_t, draw a proposal $x' \sim q(x' \mid x_t)$. This is a distribution from which one can easily draw a candidate sample x' for the next point in the chain given x;
2. Compute the acceptance ratio:

$$r = \frac{\pi(x')q(x_t \mid x')}{\pi(x_t)q(x' \mid x_t)}$$

3. Draw $u \sim \mathcal{U}([0, 1])$
4. If $u < \min(1, r)$, accept and set $x_{t+1} = x'$, otherwise reject and set $x_{t+1} = x_t$.

The possibility of rejection is crucial: it modifies the diagonal elements of the transition matrix so as to guarantee probability conservation and detailed balance. A particularly common case is that of a *symmetric proposal* (a Gaussian for example):

$$q(x' \mid x) = q(x \mid x')$$

so that the acceptance probability simplifies to:

$$\alpha(x \to x') = \min\left(1, \frac{\pi(x')}{\pi(x)}\right)$$

This is the original Metropolis algorithm.

6.8.2 Estimating observables from MC

After burn-in, expectation values are estimated by time averages:

$$\mathbb{E}_\pi[f] = \int f(x)\pi(x)\,dx \approx \frac{1}{N}\sum_{t=1}^{N} f(x_t)$$

Under ergodicity, this estimator converges as $N \to \infty$ (ensemble average). However, successive samples are correlated: error is larger than for N independent draws. Then, convergence diagnostics and autocorrelation analysis are essential.

6.8.3 Convergence diagnostics and autocorrelation

Even when irreducibility and aperiodicity hold in principle, in practical computation one must verify that the chain has effectively reached stationarity and that samples are sufficiently decorrelated. One simple way is to run the algorithm many times from different initial points in the parameter space.

A more quantitative convergence diagnostic is the time taken by the chain to decorrelate. Along a stationary Markov chain we define the time autocorrelation function of an observable $f(x)$ as:

$$C_f(t) = \mathbb{E}\left[f(x_t)f(x_{t+\tau})\right] - \mathbb{E}\left[f^2\right]$$

where the average is taken over Monte Carlo time in the stationary regime. The normalized autocorrelation function is:

$$\rho_f(t) = \frac{C_f(t)}{C_f(0)}$$

If the chain is governed by a transition matrix with eigenvalues λ_k, the autocorrelation decays as a linear combination of exponentials with characteristic time $1/\lambda_k$. Asymptotically, the decay is dominated by the slowest mode with characteristic time given by the inverse spectral gap:

$$\rho_f(t) \sim e^{-t/t_{\text{exp}}} \overset{t\to\infty}{\sim} e^{-t|\mathfrak{Re}(\lambda_1)|}$$

Thus, slow decays of $\rho_f(t)$ means small spectral gap.

In numerical practice, given a finite trajectory $\{\, x_t \,\}_{t=1}^{N}$, defining the estimator:

$$\hat{I}_N = \frac{1}{N}\sum_{t=1}^{N} f(x_t)$$

a standard estimator of the normalized autocorrelation function $\rho_f(t)$ is:

$$\hat{\rho}_f(t) \simeq \frac{\frac{1}{N-t}\sum_{\tau=1}^{N-t}(f(x_\tau) - \hat{I}_N)(f(x_{t+\tau}) - \hat{I}_N)}{\frac{1}{N}\sum_{\tau=1}^{N}(f(x_\tau) - \hat{I}_N)^2}$$

A practical criterion for convergence is that $|\rho_f(t)|$ decays rapidly to zero compared to the total length of the chain, that is it should become $\ll 1$ on a timescale much shorter than N.

The **integrated autocorrelation time** is defined as:

$$t_{\text{int}} = 1 + 2\sum_{t=1}^{\infty} \rho_f(t)$$

which in practice the sum is truncated at the first t when $\rho_f(t) \ll 1$. The variance of the Monte Carlo estimator is the inflated by a factor $2t_{\text{int}}$:

$$\text{Var}\left(\frac{1}{N}\sum_{t=1}^{N} f(x_t)\right) \approx \frac{2t_{\text{int}}}{N}\,\text{Var}_\pi(f)$$

where the effective number of **effectively independent samples** is:

$$N_{\text{eff}} \approx \frac{N}{2t_{\text{int}}}$$

From the spectral perspective discussed earlier, slow decay of autocorrelations corresponds to a small spectral gap and hence to slow mixing (poor sampling efficiency). Monitoring the empirical autocorrelation function therefore provides a practical numerical proxy for the theoretical convergence properties of the transition matrix.

In the following lines, an MCMC MH algorithm is shown. Our target distribution is a 1D sum of two Gaussians. Even if the target distribution is normalized, in general, the normalization is not important because in algorithm there are just ratios of $\pi(x)$.

```
# Target distribution: bimodal
# pi(x) = 0.5 N(-3, 1) + 0.5 N(3, 1)
target_density <- function(x)
{ 0.5 * dnorm(x, mean = -3, sd = 1) + 0.5 * dnorm(x, mean = 3, sd
    = 1)}
log_target_density <- function(x){log(target_density(x))}
# Metropolis-Hastings algorithm
metropolis_hastings <- function(n_steps, x0, proposal_sd){
    x <- numeric(n_steps)
    x[1] <- x0; n_accept <- 0
    for (t in 2:n_steps){
        x_current <- x[t - 1]
        # Symmetric Gaussian proposal
        x_prop <- rnorm(1, mean = x_current, sd = proposal_sd)
        log_r <- log_target_density(x_prop) - log_target_density(
            x_current)
        if (log(runif(1)) < log_r){
            x[t] <- x_prop
            n_accept <- n_accept + 1 }
            else { x[t] <- x_current}}
    acceptance_rate <- n_accept / (n_steps - 1)
    return(list(samples = x, acceptance_rate = acceptance_rate))}
```

while for the running part:

```
#Run sampler
n_steps <- 50000; burn_in <- 5000
proposal_sd <- 2.0; x0 <- -8
mh <- metropolis_hastings(n_steps = n_steps, x0 = x0,
    proposal_sd = proposal_sd)
samples <- mh$samples
samples_before_burnin <- samples[1:burn_in]
samples_after_burnin <- samples[(burn_in + 1):n_steps]
cat('Acceptance rate:', mh$acceptance_rate, '\n')
# Normalized autocorrelation
lag_max <- 500
```

```
acf_result <- acf(samples_after_burnin, lag.max = lag_max, plot =
    FALSE)
lags <- as.numeric(acf_result$lag)
rho <- as.numeric(acf_result$acf)
pos_lags <- which(rho[-1] > 0)
if (length(pos_lags) == 0) { t_int <- 1} else {last_positive <-
   pos_lags[1]
    for (i in seq_along(pos_lags)) {
        if (pos_lags[i] == i) {last_positive <- pos_lags[i] }
           else { break} }
    t_int <- 1 + 2 * sum(rho[2:(last_positive + 1)])}
N <- length(samples_after_burnin); N_eff <- N / (2 * t_int)
cat('Integrated autocorrelation time t_int:', t_int, '\n')
cat('Effective sample size N_eff:', N_eff, '\n')
```

In Fig. 6.3, the two histograms at the top show the sampled distribution before and after burn-in. A burn-in of 5000 iterations was adopted, although this choice appears to be conservative. In fact, no evident transient phase is visible, suggesting that the chain reaches the stationary regime much earlier. The two histograms are almost identical, indicating that the initial samples have little impact on the final distribution. After burn-in removal, the sampled distribution is in good agreement with the target density.

The Markov chain trajectory is shown in the third panel. Starting from $x = -8$, the chain rapidly moves toward the high-probability region of the target distribution and then explores both modes, repeatedly transitioning between the two maxima. This indicates that the chain is able to sample the entire target distribution without becoming trapped in a single mode.

The bottom panel shows the normalized autocorrelation function. The autocorrelation decreases smoothly and becomes negligible after approximately 100 lags, while the remaining fluctuations are compatible with statistical noise. The integrated autocorrelation time is $t_{\text{int}} = 42.76$ meaning that approximately 43 correlated samples contain the same amount of information as one independent sample. Therefore, a reasonable burn-in could be chosen as a few multiples of the integrated autocorrelation time, corresponding to a few hundred iterations. The effective number of independent samples is approximately $N_{\text{eff}} \simeq 526$.

Then, summarizing:

- MCMC: dynamics whose stationary distribution is the target $\pi(x)$.
- Irreducibility ensures convergence to a unique stationary state.
- Detailed balance is a simple sufficient condition for stationarity.
- Metropolis-Hastings implements this by proposal plus accept/reject.
- Correlations matter: the relevant sample size is $N_{\text{eff}} < N$.
- The spectrum controls relaxation (burn-in) and correlation (ESS).

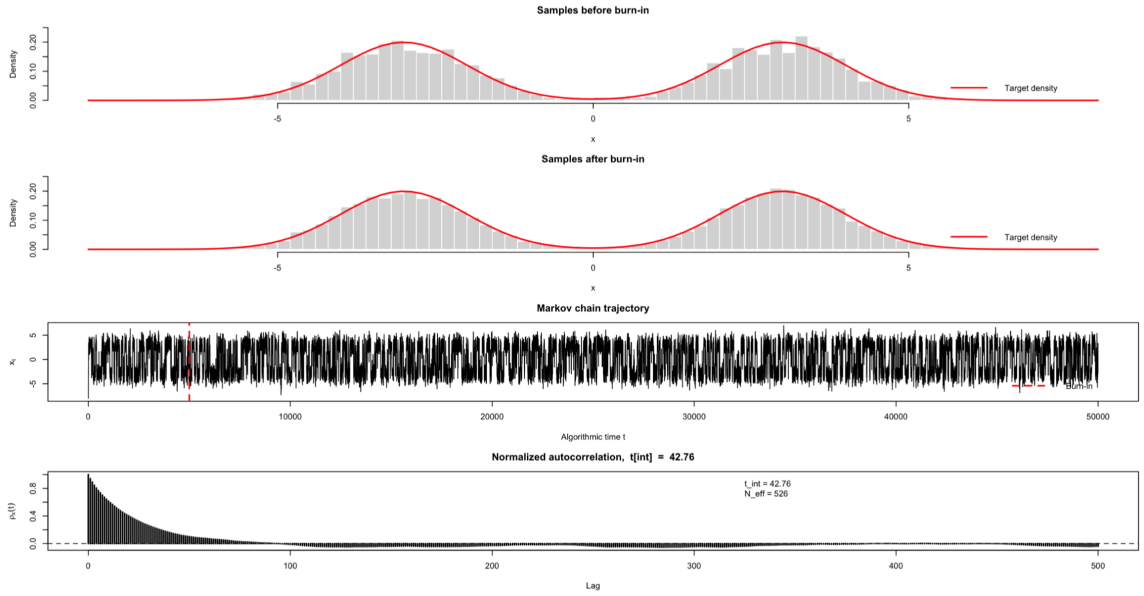


Figure 6.3

6.8.4 The independence sampler

In the random-walk Metropolis algorithm, the proposal is local, that is, $q(x' \mid x)$ depends on the current state x. This means that the proposal allows us to move locally in the parameter space by taking account on where we are.

A particularly important choice of the Hastings construction is obtained when the proposal distribution is chosen independently of the current state. Therefore, the proposal is global:

$$q(x' \mid x) = q(x')$$

In this case, the proposal value is a fresh draw x' from a fixed density q, regardless of the present configuration x_t. Therefore is a *global move*.

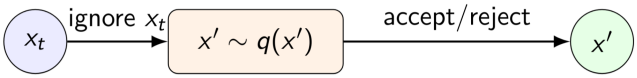


Figure 6.4

While for the generic Metropolis-Hastings step the acceptance probability is:

$$\alpha(x \to x') = \min\left(1, \frac{\pi(x')q(x \mid x')}{\pi(x)q(x' \mid x)}\right)$$

for the independence sampler we have:

$$q(x' \mid x) = q(x'), \qquad q(x \mid x') = q(x)$$

Therefore, the acceptance probability becomes simply:

$$\alpha(x \to x') = \min\left(1, \frac{\pi(x')q(x)}{\pi(x)q(x')}\right) = \min\left(1, \frac{\pi(x')/q(x')}{\pi(x)/q(x)}\right)$$

This algorithm is known as the **independent Metropolis-Hastings** algorithm (or **independence sampler**). Although proposals are independent draws from q, successive accepted states are still correlated because acceptance depends on the current state through the factor $q(x)$ in the ratio above.

Let's interpret the independence sampler through important weights. We define the importance weight:

$$w(x) = \frac{\pi(x)}{q(x)}$$

Then, the acceptance function is:

$$\alpha(x \to x') = \min\left(1, \frac{w(x')}{w(x)}\right)$$

If $w(x') > w(x)$, x' is accepted, while if $w(x') < w(x)$, x' is accepted with probability $w(x')/w(x)$. If $q \approx \pi$, then $w(x)$ is almost constant and acceptance is high. If q is poor, $w(x)$ varies strongly and the chain may get stuck at high-weight states.

The independence sampler is closely related to rejection sampling. In rejection sampling, one proposes $x' \sim q(x')$ and, given a target density π, it accepts with probability:

$$\frac{\tilde{\pi}(x')}{Mq(x')}$$

where M satisfies $\tilde{\pi}(x) \leq Mq(x)$ for all x.

In contrast, the independence sampler replaces the global bound M with a comparison to the current state x:

$$\alpha(x \to x') = \min\left(1, \underbrace{\frac{\pi(x')/q(x')}{\pi(x)/q(x)}}_{M}\right)$$

One may therefore interpret the independence sampler as a rejection scheme in which the effective bound is updated dynamically along the Markov chain. Unlike rejection sampling, it does not require prior knowledge of a global envelope constant M, but it produces correlated samples, which is the price to pay. However, the performance of the independence sampler depends crucially on how well $q(x)$ approximates $\pi(x)$. If $q(x) \approx \pi(x)$ so that the ratio $\pi(x)/q(x)$ is nearly constant, then acceptance probabilities are high, and autocorrelation is low, and the chain mixes rapidly, approaching the efficiency of independent sampling. On the other hand, if $q(x)$ poorly approximates $\pi(X)$ – for instance, underestimating its tails or missing modes – then the ratio $\pi(x)/q(x)$ can vary strongly across the state space. In this case, acceptance probabilities become small and the chain may remain trapped for long periods, leading to slow mixing and large autocorrelation times. This problem is particularly severe in high-dimensional spaces. A useful theoretical criterion is the following. If there exists a finite constant $C > 0$ such that:

$$\pi(x) \leq Cq(x), \qquad \forall x$$

then the independence sampler is *uniformly ergodic*, and the spectral gap is bounded away from zero. This condition is formally analogous to the envelope condition required in rejection sampling and again highlights the close relation between the two methods.

In the following lines our target is the same bimodal distribution given by the sum of the two Gaussian. We sample it using a t-Student proposal because it has long tails, which allow

us to sample from small to large values, covering the two modes of the target distribution. The diagnostic is in shown in Figure 6.5. Notice that the acceptance probability we get is not so impressive since the proposal distribution is not so good (it's no bimodal). However, it has the same support, and it allows to sample everywhere on the real axis. From the autocorrelation plot, the autocorrelation time is extremely small compared with the previous example with the random-walk MH algorithm. This means that the burn-in period we choose is way larger than it should be.

```
# Target: bimodal density
# pi(x) = 0.5 N(-3, 1) + 0.5 N(3, 1)
target_density <- function(x) {
    0.5 * dnorm(x, mean = -3, sd = 1) +
    0.5 * dnorm(x, mean = 3, sd = 1)}
log_target <- function(x) log(target_density(x))
# Independence proposal: heavy-tailed Student-t
# x = proposal_scale * T_df
proposal_scale <- 4
proposal_df <- 4
rproposal <- function(n) {
    proposal_scale * rt(n, df = proposal_df)}
log_proposal <- function(x) {
    dt(x / proposal_scale, df = proposal_df, log = TRUE) - log(
       proposal_scale)}
# IS function
independence_mh <- function(n_steps, x0) {
    x <- numeric(n_steps)
    x[1] <- x0
    n_accept <- 0
    for (t in 2:n_steps) {
        x_current <- x[t - 1]
        x_proposed <- rproposal(1)
        log_r <- log_target(x_proposed) + log_proposal(x_current)
        - log_target(x_current) - log_proposal(x_proposed)
        if (log(runif(1)) < log_r) {
            x[t] <- x_proposed
            n_accept <- n_accept + 1}
            else { x[t] <- x_current } }
    list(samples = x, acceptance = n_accept / (n_steps - 1)) }
fit_ind <- independence_mh(n_steps = 50000, x0 = 0)
fit_ind$acceptance
#[1] 0.4201284
burn_in <- 5000
samples_ind <- fit_ind$samples[(burn_in + 1):50000]
```

Some key observations:

- The chain can jump directly from one mode to the other. While in the previous example, the path of the sampler was spending some time over one move and some time over the other, reproducing the bimodal distribution. Here, the path is very different.
- Samples still correlated because rejection depends on previous state, but they die fast.
- A heavy-tailed proposal is safer than a too-narrow proposal because even if we do not cover the shape of the target distribution, it still spreads probability mass everywhere

– probably lowering the acceptance rate – but avoiding to get stuck in some part of the parameter space.

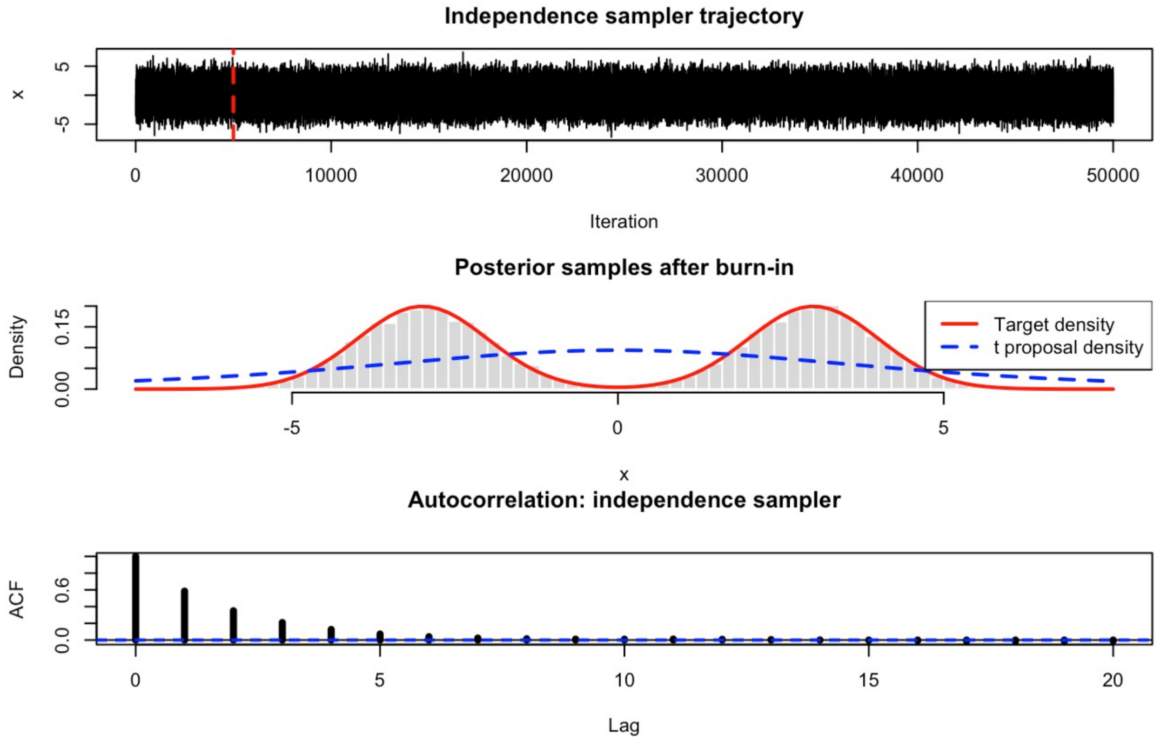


Figure 6.5

In Figure 6.6 are shown the results of a narrower proposal. In this case, same everything but rescaled proposal, the main difference is that the acceptance ratio dropped a lot, because most of the time it samples nearby where most of the mass is (near zero). This reduces a lot the number of samples and, on top of that, we have that the effective number of samples is much reduced. There is a longer correlation time and up to 20 iteration steps the samples are still correlated.

The independence sampler is effective when a good global approximation to the posterior is available, for example from a Laplace approximation or a variational approximation. In such cases, one may use the approximation as proposal density $q(x)$. However, when the posterior distribution is highly concentrated, strongly correlated, or high-dimensional, constructing a proposal $q(x)$ that globally approximates $\pi(x)$ is often difficult. In those situations, local proposal mechanisms (such as Metropolis) typically yield better performance.

As we saw many times, in high dimension, probability is highly concentrated. In this case, independence sampler fails. To understand why, recall the acceptance probability:

$$\alpha(x \to x') = \min\left(1, \frac{w(x')}{w(x)}\right), \qquad w(x) = \frac{\pi(x)}{q(x)}$$

In $d \gg 1$, the log-weight typically decomposes as a sum:

$$\log w(x) = \log \pi(x) - \log q(x) \approx \sum_{i=1}^{d} \Delta_i(x) = \mathcal{O}(d)$$

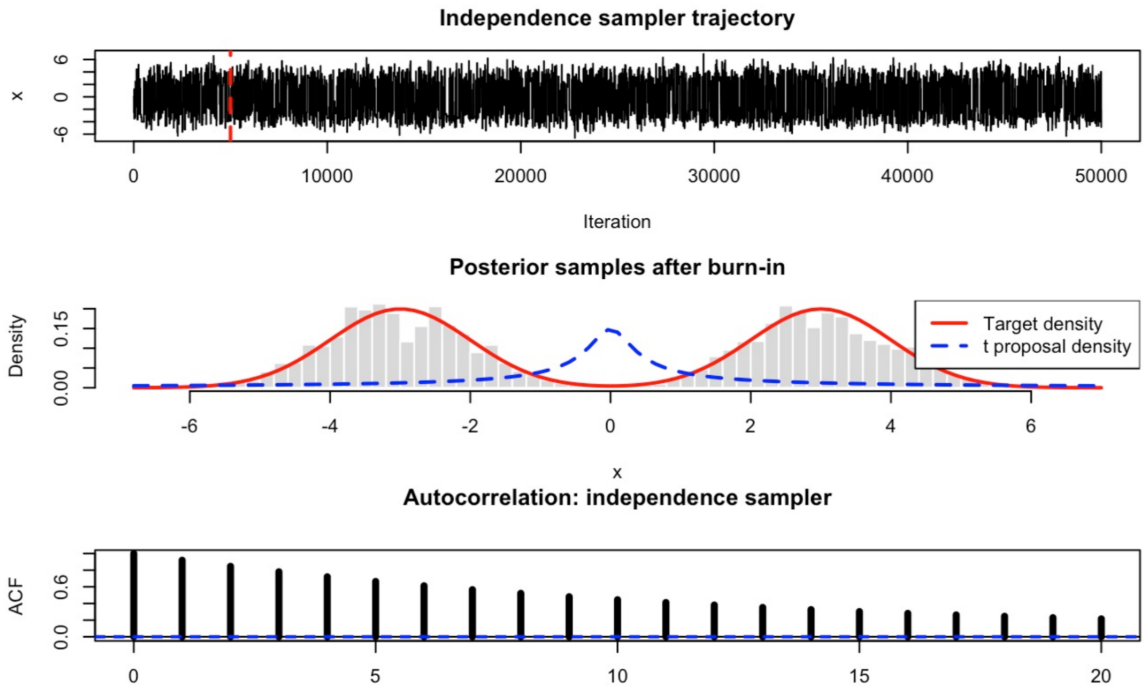


Figure 6.6

Even if each mismatch Δ_i is small:

$$\log w(x) = \mathcal{O}(d)\,, \qquad w(x) = \exp \mathcal{O}(d)$$

weights vary exponentially across the space. Therefore, acceptance probability tends to zero exponentially fast with d unless $q \approx \pi$ with exponentially small error. This means that the larger is the dimension of the space, the most accurate must be the guess of $q(x)$.

Geometrically, high-dimensional distributions concentrate on thin regions. For instance, take $\pi = \mathcal{N}(0, I_d)$, where I_d is the identity matrix (uncorrelated variables). In this case, $\|x\|^2 \sim \chi^2_d$, so:

$$\mathbb{E}\left[\|x\|^2\right] = d, \qquad \mathrm{Var}(\|x\|^2) = 2d$$

and $\|x\| \approx \sqrt{d}$ with high probability. Imagine q is even slightly mispecified (wrong scale, mean, etc). Almost all proposals land in *irrelevant regions*, which is the same failure of rejection/importance sampling in high dimension.

Therefore, independence sampler requires a **global approximation** $q \approx \pi$. In d dimensions, small per-coordinate errors accumulate:

$$\text{small error} \times d \quad \Rightarrow \quad \text{large global mismatch}$$

Typically, the acceptance rate decays like $P(A) \sim \exp -cd$ and it's another manifestation of the *curse of dimensionality*. This is a motivation to use:

- local proposals (random-walk Metropolis);
- gradient-based methods (e.g., HMC);
- Gibbs sampling (coordinate-wise updates).

6.9 Gibbs sampling

In Bayesian inference, one is often interested in expectations with respect to a high-dimensional posterior distribution:

$$p(\boldsymbol{\theta} \mid D), \qquad \boldsymbol{\theta} = (\theta_1, \ldots, \theta_d)$$

such as:

$$\mathbb{E}\left[f(\boldsymbol{\theta}) \mid D\right] = \int f(\boldsymbol{\theta}) p(\boldsymbol{\theta} \mid D)\, d\boldsymbol{\theta}$$

Direct integration (it would require storing a huge amount of data) or direct sampling from the joint posterior is typically infeasible when d is large or the posterior has a complicated structure.

However, sometimes the conditional distributions are easy and then we can draw a single value from it. **Gibbs sampling** is a MCMC method that generates samples from the joint posterior using only the *full conditional distributions* of each parameter. The construction of the Markov chain requires that for each component θ_i, the full conditional distribution is:

$$p(\theta_i \mid \theta_{-i}, D), \qquad \theta_{-i} \equiv (\theta_1, \ldots, \theta_{i-1}, \theta_{i+1}, \ldots, \theta_d)$$

is known and can be sampled from directly. The Gibbs sampler constructs a Markov chain by sequentially updating each coordinate from its full conditional distribution while keeping the other coordinates fixed. The algorithm proceeds as follows:

1. Initialize by choosing an initial state:

$$\boldsymbol{\theta}^{(0)} = (\theta_1^{(0)}, \ldots, \theta_d^{(0)})$$

2. For $t = 1, 2, \ldots$, update each component sequentially drawing from the full conditional given all the other parameters:

$$\begin{aligned}
\theta_1^{(t)} &\sim p(\theta_1 \mid \theta_2^{(t-1)}, \ldots, \theta_d^{(t-1)}, D) \\
\theta_2^{(t)} &\sim p(\theta_2 \mid \theta_1^{(t)}, \theta_3^{(t-1)}, \ldots, \theta_d^{(t-1)}, D) \\
&\vdots \\
\theta_d^{(t)} &\sim p(\theta_d \mid \theta_1^{(t)}, \ldots, \theta_{d-1}^{(t)}, D)
\end{aligned}$$

Notice that Gibbs sampling updates one parameter at a time. At each step, a parameter is sampled from its full conditional distribution while all the remaining parameters are kept fixed. The update of coordinate i can be written as:

$$T_i(\boldsymbol{\theta} \to \boldsymbol{\theta}') = p(\theta_i' \mid \theta_{-i}, D)\delta\big(\theta_{-i}' - \theta_{-i}\big) \tag{6.16}$$

where the new value θ_i' is drawn from the exact full conditional, while all the remaining coordinates remain unchanged.

Notice that the updates are performed sequentially. Therefore, each update uses the most recently available values of the other parameters. For example, when updating θ_2, the algorithm already uses the newly sampled value $\theta_1^{(t)}$.

A complete iteration of the algorithm, usually called a sweep, consists of updating all the coordinates once. Starting from $\boldsymbol{\theta}^{(t-1)}$, the sequence of updates produces the new state

$\boldsymbol{\theta}^{(t)}$. The corresponding transition kernel is obtained by composing the single-coordiante updates:

$$T \equiv T_d \circ T_{d-1} \circ \ldots \circ T_1$$

where the composition must be read from right to left and T_i denotes the update acting only on coordinate i. Since each T_i leaves the target posterior distribution $p(\boldsymbol{\theta} \mid D)$ invariant, the full sweep also preserves the same distribution.

Geometrically, each single-coordinate update changes only one parameter, while all the others remain fixed. As a consequence, the trajectory moves parallel to one of the coordinate axes in parameter space. This is shown Figure 6.7, where a complete sweep corresponds to a sequence of axis-aligned moves, represented by the red arrow followed by the blue one.

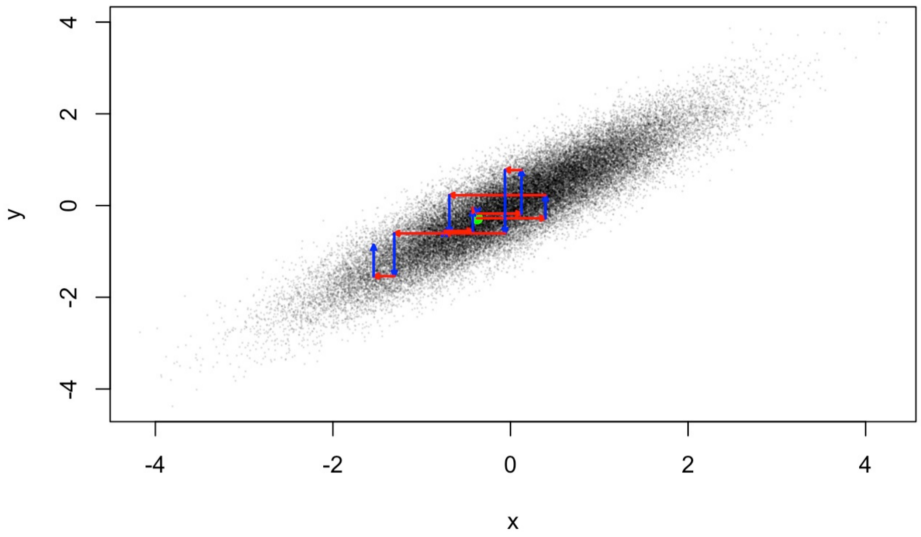


Figure 6.7: Gibbs sampling steps in 2D parameter space.

Proof. To show that the joint posterior $p(\boldsymbol{\theta} \mid D)$ is invariant under Gibbs updates, consider a single-coordinate update T_i. Gibbs sampling can be viewed as an MH algorithm where the proposal is the full conditional

$$q(\theta_i' \mid \boldsymbol{\theta}) = p(\theta_i' \mid \theta_{-i}, D)\delta\big(\theta_{-i}' - \theta_{-i}\big)$$

where δ enforces that all the other components remain unchanged. The acceptance probability is $\alpha(\boldsymbol{\theta} \to \boldsymbol{\theta}') = 1$. Notice an important point: Gibbs proposals are always accepted, samples are of course correlated. If the posterior has strong correlations, coordinate-wise updates may move slowly through parameter space.

We verify detailed balance (6.17) with respect to the joint posterior, starting from:

$$p(\boldsymbol{\theta} \mid D)T_i(\boldsymbol{\theta} \to \boldsymbol{\theta}') = p(\theta_i, \theta_{-i} \mid D)p(\theta_i' \mid \theta_{-i}, D)\delta\big(\theta_{-i}' - \theta_{-i}\big)$$

Using the factorization:

$$p(\theta_i, \theta_{-i} \mid D) = p(\theta_i \mid \theta_{-i}, D)p(\theta_{-i} \mid D)$$

we obtain:

$$\begin{aligned} p(\boldsymbol{\theta} \mid D) T_i(\boldsymbol{\theta} \to \boldsymbol{\theta}') &= p(\theta_i \mid \theta_{-i}, D) p(\theta_{-i} \mid D) p(\theta_i' \mid \theta_{-i}, D) \delta\left(\theta_{-i}' - \theta_{-i}\right) = \\ &= p(\theta_i \mid \theta_{-i}', D) \underbrace{p(\theta_{-i}' \mid D) p(\theta_i' \mid \theta_{-i}', D)}_{p(\theta_i', \theta_{-i}' \mid D)} \delta\left(\theta_{-i}' - \theta_{-i}\right) = \\ &= p(\theta_i', \theta_{-i}' \mid D) \underbrace{p(\theta_i \mid \theta_{-i}', D) \delta\left(\theta_{-i}' - \theta_{-i}\right)}_{T_i(\boldsymbol{\theta}' \to \boldsymbol{\theta})} = p(\boldsymbol{\theta}' \mid D) T_i(\boldsymbol{\theta}' \to \boldsymbol{\theta}) \end{aligned}$$

which is the detailed balance condition:

$$p(\boldsymbol{\theta} \mid D) T_i(\boldsymbol{\theta} \to \boldsymbol{\theta}') = p(\boldsymbol{\theta}' \mid D) T_i(\boldsymbol{\theta}' \to \boldsymbol{\theta}) \tag{6.17}$$

□

Thus, each coordinate update satisfies detailed balance, and since a composition of reversible kernels with the same invariant distribution remains invariant, the full Gibbs transition kernel leaves $p(\boldsymbol{\theta} \mid D)$ invariant. Under the usual assumptions of irreducibility and aperiodicity, the chain is ergodic, and therefore:

$$\boldsymbol{\theta}^{(t)} \sim p(\boldsymbol{\theta} \mid D), \qquad \text{for } t \to \infty$$

Example 6.9.1

Let's consider an example target *bivariate normal distribution*. Consider:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right]$$

It's easy to show the full conditionals are:

$$X \mid Y = y \sim \mathcal{N}\left(\rho y, 1 - \rho^2\right), \qquad Y \mid X = x \sim \mathcal{N}\left(\rho x, 1 - \rho^2\right)$$

This gives an exact Gibbs sampler and it is a simple example where we know the true target distribution. When ρ is close to 1, the correlation increases and Gibbs moves slowly along a narrow diagonal ridge, giving a more correlated chain.
Let's derive the full conditionals. Recall that $p(x \mid y) = \frac{p(x,y)}{p(y)} \propto p(x, y)$ up to constants (at fixed y), the joint density can be written, up to normalization, as:

$$\log p(x, y) = -\frac{1}{2(1 - \rho^2)}(x^2 - 2\rho xy + y^2)$$

Fixing y, collect terms in x:

$$\log p(x \mid y) \propto -\frac{1}{2(1 - \rho^2)}(x^2 - 2\rho yx)$$

Completing the square in x gives:

$$X \mid Y = y \sim \mathcal{N}\left(\rho y, 1 - \rho^2\right)$$

```
gibbs_bivar_norm <- function(n_steps, rho, start = c(0, 0)){
x <- numeric(n_steps)
```

```r
y <- numeric(n_steps)
x[1] <- start[1]
y[1] <- start[2]
conditional_sd <- sqrt(1 - rho^2)
for (t in 2:n_steps) {
    # Update x given previous y
    x[t] <- rnorm(1, mean = rho * y[t - 1], sd = conditional_sd)
    # Update y given updated x
    y[t] <- rnorm(1, mean = rho * x[t], sd = conditional_sd)
    }
cbind(x = x, y = y) }
rho <- 0.9
chain_gibbs <- gibbs_bivar_norm(n_steps = 50000, rho = rho,
    start = c(-4, 4))
```

The results are shown in Figure 6.8. The chain samples from the correct two-dimensional Gaussian. The updates are coordinate-wise: first $x \mid y$, then $y \mid x$. Every proposal is accepted because it is drawn from the exact conditional. For large ρ, the posterior has a narrow diagonal structure. Coordinate-wise moves then produces strong autocorrelation. In Figure 6.9, the GS path at different steps is shown.

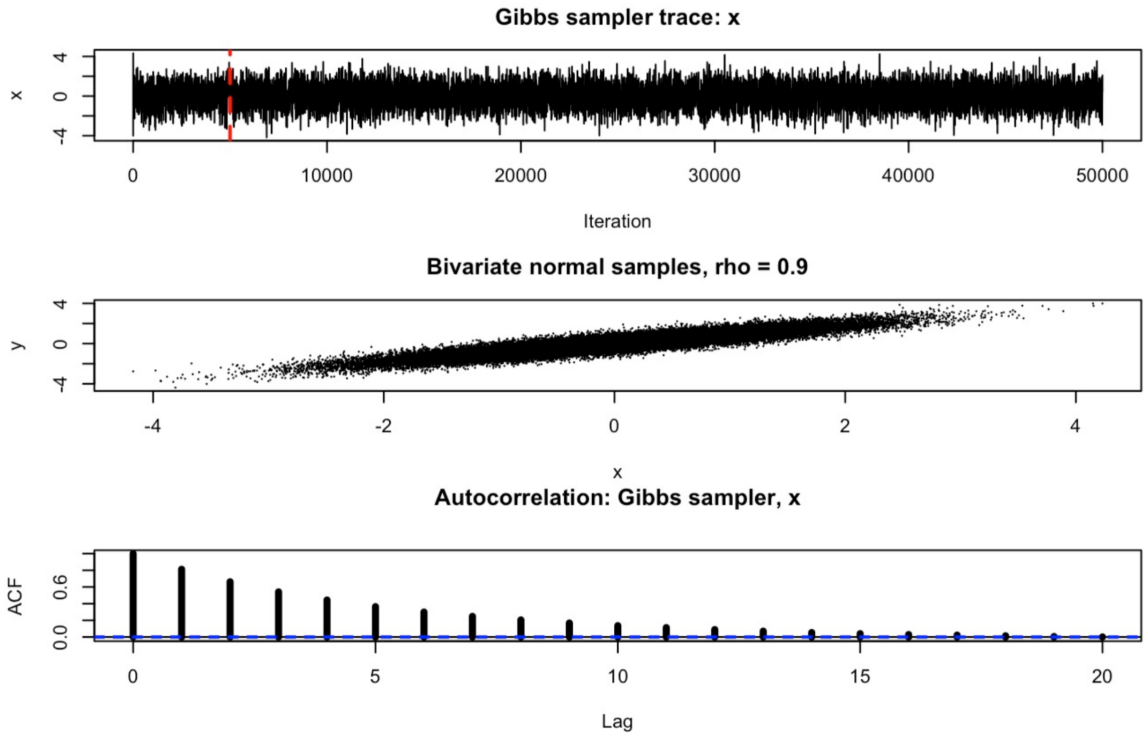


Figure 6.8

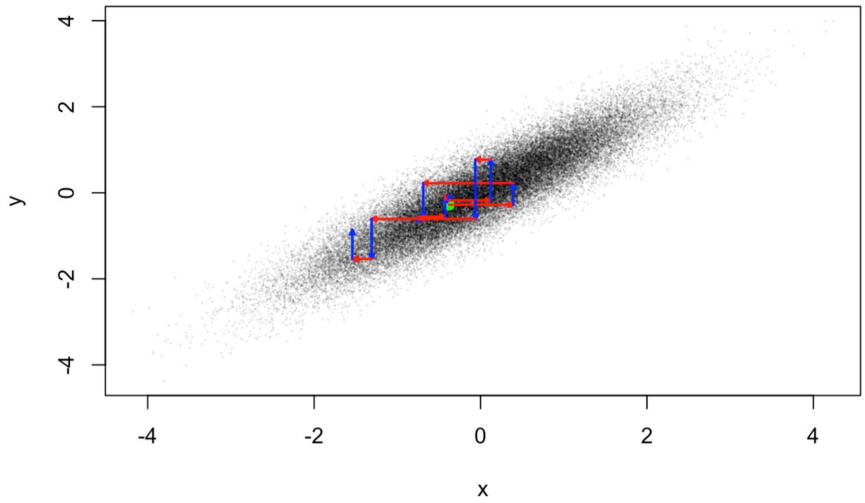


Figure 6.9

Gibbs sampling can be viewed as a special case of the Metropolis-Hastings algorithm with $q(x' \mid x) = q(x')$. Indeed, updating coordinate i corresponds to choosing the proposal:

$$q(\theta_i' \mid \boldsymbol{\theta}) = p(\theta_i' \mid \theta_{-i}, D)\delta\big(\theta'_{-i} - \theta_{-i}\big)$$

that is, the proposal is accepted. Gibbs sampling is therefore an MH algorithm with acceptance probability equal to one. When the full conditional distributions are not available in closed form, one may embed Metropolis steps inside the coordinate updates. This strategy is known as **Metropolis-within-Gibbs**. Its acceptance probability compares weights $\pi(x)/q(x)$. It can make long-distance jumps, but only works well if q approximates π globally.

Although Gibbs updates are always accepted, it doesn't mean independen, successive samples remain correlated and autocorrelation can still be large. In particular, if components of $\boldsymbol{\theta}$ are strongly correlated under the posterior, updating them one at a time may lead to slow exploration of the joint space. Autocorrelation functions of observables and integrated autocorrelation times provide practical diagnostics of convergence, exactly as in the general MCMC case. Both methods show the central MCMC trade-off: easy proposals versus efficient exploration.

Gibbs sampling is thus conceptually simple (no proposal tuning parameters are required) and easy to implement (acceptance probability is identically one), particularly effective in hierarchical and conjugate Bayesian models, where conditional distributions take standard forms. As a major drawback, it requires that all full conditional distributions are known and directly samplable, mixing is slow when parameters are strongly correlated. Additionally, performance deteriorates in high-dimensional or multimodal problems.

Finally, let's make a comparison between the independence and the Gibbs sampler. The independence sampler is useful when we have a good global approximation to the posterior (Laplace, variational approximation), while Gibbs sampling is useful when the modal has simple full conditionals (conjugate Bayesian models, hierarchical Gaussian models). In real applications, samplers often combine ideas: Metropolis-within-Gibbs, Block Gibbs sampling, where rather than updating sequentially at each direction in the parameter space, one update a certain number of parameter globally, adaptive or gradient-based proposals.

	Independence sampler	Gibbs sampler
Proposal	$q(x')$	$p(\theta_i \mid \theta_{-i}, D)$
Move type	global	coordinate-wise
Acceptance	MH accept/reject	always accepted
Main requirement	good global q	samplable conditionals
Main weakness	poor q causes sticking	correlations cause slow mixing
Typical use	approximate posterior proposal	conjugate/hierarchical models

Table 6.1

6.10 Hamiltonian Monte Carlo

The Markov chain construction discussed so far (MH and GS) typically rely on *local* updates of the state. As a consequence, successive samples are strongly correlated, especially in high-dimensional problems or when the target distribution exhibits strong correlations among its components. This leads to slow exploration of the parameter space, long autocorrelation times and small effective sample size. Global proposals, such as those proposed by the independence sampler, are also very inefficient in high-dimensional spaces as acceptance probabilities drop dramatically.

Hamiltonian Monte Carlo (HMC) is a MCMC method designed to overcome these limitations by proposing distant *global* moves informed by the geometry of the target density. The key idea is to introduce an auxiliary dynamical system whose trajectories efficiently explore regions of high probability. This is done by considering an augmented state space equipped with Hamiltonian dynamics which basically turns random walk of MH into a ballistic motion. Instead of moving slowly around the space, one goes straight along the dynamics, that is more like a deterministic ballistic motion.

Let $x \in \mathbb{R}^d$ denote the parameter of interest, with target density $\pi(x)$. HMC introduces an auxiliary variable $p \in \mathbb{R}^d$, called the **momentum**, and defines a joint distribution:

$$\pi(x, p) = \pi(x) f(p \mid \mathbf{M})$$

whose marginal distribution over x is the target $\pi(x)$. Here, $\mathbf{M}$ is a positive definite mass matrix and $f(p \mid M)$ is the normal PDF with zero mean and covariance $\mathbf{M}$. Since $\int \pi(x, p)\, dp = \pi(x)$, sampling from $\pi(x, p)$ and discarding p gives samples from $\pi(x)$.

It is then convenient to introduce the Hamiltonian:

$$H(x, p) = U(x) + K(p)$$

where the physics analogy suggest us to refer to U and K as potential and kinetic energy, respectively:

$$U(x) = -\log \pi(x), \qquad K(p) = \frac{1}{2} p^T \mathbf{M}^{-1} p$$

and $\mathbf{M}$ is the mass matrix which tunes which direction we put more energy. The joint density can then be written as:

$$\pi(x, p) \propto e^{-H(x,p)}$$

Hamiltonian dynamics defines deterministic evolution in the augmented space (x, p) via:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \nabla_p H(x,p) = \mathbf{M}^{-1}p$$
$$\frac{\mathrm{d}p}{\mathrm{d}t} = -\nabla_x H(x,p) = -\nabla_x U(x)$$

Hence, the position x evolves according to the momentum p. The momentum evolves according to the gradient of the log-target density:

$$-\nabla_x U(x) = \nabla_x \log \pi(x)$$

HMC proposals therefore use gradient information to move toward regions of high probability.

These dynamics have two crucial properties:

- *Energy conservation*: $H(x,p)$ is conserved along trajectories.

  $$\frac{\mathrm{d}}{\mathrm{d}t}H(x(t),p(t)) = \dot{x}\cdot\nabla_x H + \dot{p}\cdot\nabla_p H = \nabla_p H\cdot\nabla_x H - \nabla_x H\cdot\nabla_p H = 0$$

  Therefore, $H(x(t),p(t)) = H(x(0),p(0))$ and exact Hamiltonian trajectories move along level sets of constant joint density $\pi(x,p) \propto e^{-H(X,p)}$. This allows us to explore regions with the same ridge (it has not to be misunderstood with exploring same levels of every probable state for the parameters).

- *Volume preservation*: the transformation in phase space preserves volume (Liouville's theorem). The phase-space velocity is $(\dot{x},\dot{p}) = (\nabla_p H - \nabla_x H)$ and the volume preservation means that the divergence of the phase-space velocity is zero:

  $$\nabla_x\cdot\dot{x} + \nabla_p\cdot\dot{p} = \nabla_x\cdot\nabla_p H - \nabla_p\cdot\nabla_x H = 0$$

  This means that it is interpretable as an incompressible fluid

These properties have a few consequences: they imply that the Hamiltonian flow preserves the joint distribution $\pi(x,p)$. If only Hamiltonian dynamics is run, only the portion of parameter space determined by initial condition (x_0,p_0) is explored. Then, to complement the dynamics, we need to jump from one level set of H to another (jumping from one energy shell to another) by refreshing the momentum p regularly. Basically, in this way we sample from our target distribution with a dynamics that is very fast, ballistic (it makes a jump) and then repeat, so on and so forth.

In practice, Hamiltonian dynamics cannot be simulated exactly and must be discretized. The resulting proposal is then corrected using a Metropolis step. One iteration of the HMC algorithm proceeds as follows:

1. Given the current state x_t, draw a fresh momentum:

   $$p_t \sim \mathcal{N}(0,\mathbf{M})$$

2. Starting from (x_t,p_t), simulate Hamiltonian dynamics for a fictitious time T using a numerical integrator obtaining a proposal (x',p').

3. Accept or reject the proposal according to the Metropolis rule:

   $$\alpha((x,p)\to(x',p')) = \min\left(1,\exp\left[-H(x',p') + H(x,p)\right]\right)$$

4. If accepted, set $x_{t+1} = x'$; otherwise $x_{t+1} = x_t$.

Why the Metropolis step? If Hamiltonian dynamics could be simulated exactly, then $H(x', p') = H(x, p)$, and every proposal would be accepted. In practice, numerical integration introduces discretization error

$$\Delta H = H(x', p') - H(x, p) \neq 0$$

and the Hamiltonian is not exactly conserved along the simulated trajectory. The Metropolis correction accepts with probability

$$\alpha = \min\left(1, e^{-\Delta H}\right)$$

which restores exact invariance of the target π. Notice that the momentum drawn is Gibbs sampling.

For a well-tuned integrator, ΔH is small, leading to high acceptance rates. If the step size is too large, energy errors accumulate and proposals are frequently rejected, degrading performance.

The momentum variable is discarded after each iteration. The acceptance step ensures that the Markov chain satisfies detailed balance with respect to $\pi(x, p)$, and therefore leaves $\pi(x)$ invariant.

So, as we said, Hamiltonian dynamics must be discretized. A crucial component of HMC is the numerical integration of Hamiltonian dynamics. The integrator must preserve volume and be approximately energy conserving in order to ensure π to be the stationary sampled distribution and maintain high acceptance rates. The most commonly used scheme is the **leapfrog integrator**. Given a step size ε, one leapfrog step updates (x, p) as follows:

$$\begin{aligned} p_{t+\frac{\varepsilon}{2}} &= p_t - \frac{\varepsilon}{2}\nabla U(x_t) \\ x_{t+\varepsilon} &= x_t + \varepsilon M^{-1} p_{t+\frac{\varepsilon}{2}} \\ p_{t+\varepsilon} &= p_{t+\frac{\varepsilon}{2}} - \frac{\varepsilon}{2}\nabla U(x_{t+\varepsilon}) \end{aligned}$$

Repeating this update L times produces a trajectory of total length $T = L\varepsilon$. Why the leapfrog? The leapfrog integrator has three key properties:

- it is *time-reversible*;
- it preserves phase-space volume;
- it approximately conserves the energy, with errors of order $\mathcal{O}\left(\varepsilon^2\right)$ per step.

These properties ensure that the Metropolis acceptance probability remains close to one when ε is sufficiently small and $\Delta H \approx 0$, Therefore, $\alpha = \min\left(1, e^{-\Delta H}\right)$ is usually close to one. If ε is too large, energy errors accumulate and the rejection rate increases.

The efficiency of HMC depends on several parameters to be tuned:

- **Step size ε:** a small step size improves energy conservation and acceptance probability but increases computational cost. A large step size reduces computation per proposal but leads to poor acceptance. In practice, ε is chosen to balance these effects.
- **Number of steps L:** the trajectory length $T = L\varepsilon$ determines how far proposals move in the state space. If L is too small, the method reduces to a local random walk. If L is too large, trajectories may retrace and double back the path, wasting computation.

- **Mass matrix M:** the choice of $\mathbf{M}$ controls the geometry of the momentum distribution. A good choice for efficient sampling requires matching the scale and the correlation structure of the target.

Compared to Metropolis-Hastings, HMC generates proposals approximately following the geometry of the target distribution $\pi(x)$. Recall that standard random-walk Metropolis proposes local moves according to:

$$x' = x + \eta, \qquad \eta \sim q(\eta)$$

so the proposed state is obtained by adding a random perturbation to the current position. As a consequence, the algorithm often explores the parameter space slowly, especially in high-dimensional problems.

In contrast, HMC uses gradient information to propose distant moves that approximately follow trajectories of constant probability density. This typically results in larger moves, higher acceptance rates due to approximate energy conservation, lower autocorrelation between successive samples (losing memory faster), better performance in high dimensions.

The price to pay are that gradients of $\log \pi(x)$ are required, numerical integration of Hamilton's equations is required, and the tuning is more complex.

Modern implementations of HMC often use No-U-Turn Sampler (NUTS), which automatically adapts ε and L during a warm-up phase. Basically, it stops trajectories when they begin to double back, avoiding unnecessary long simulations. This greatly reduces the burden of manual tuning and is one of the main reasons why HMC can be used effectively in general-purpose Bayesian software such as `Stan`.

Example 6.10.1

Consider a correlated bivariate normal target π such that:

$$X \sim \mathcal{N}\left[\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1 & \rho\\ \rho & 1\end{pmatrix}\right]$$

```
rho <- 0.95
Sigma <- matrix(c(1, rho, rho, 1), 2, 2)
Sigma_inv <- solve(Sigma)
log_pi <- function(x) {-0.5 * as.numeric(t(x) %*% Sigma_inv
    %*% x)}
grad_U <- function(x) {Sigma_inv %*% x}
```

Let's compare random-walk Metropolis with HMC. The random-walk Metropolis implementation is:

```
rw_mh <- function(n_iter, step = 0.3) {
    x <- c(0, 0)
    samples <- matrix(NA, n_iter, 2)
    acc <- 0
    for (i in 1:n_iter) {
        x_prop <- x + rnorm(2, 0, step)
        log_alpha <- log_pi(x_prop) - log_pi(x)
        if (log(runif(1)) < log_alpha) {
        x <- x_prop
        acc <- acc + 1
```

```
    }
    samples[i, ] <- x
    }
    list(samples = samples, acc_rate = acc / n_iter)
}
```

and the leapfrog integrator is:

```
leapfrog <- function(x, p, epsilon, L) {
    p <- p - 0.5 * epsilon * grad_U(x)
    for (i in 1:L) {
        x <- x + epsilon * p
            if (i != L) {
        p <- p - epsilon * grad_U(x)
        }
    }
    p <- p - 0.5 * epsilon * grad_U(x)
    list(x = x, p = -p)
}
```

Here, the final momentum flip makes the proposal exactly reversible:

$$q(\theta \mid \theta') = q(\theta' \mid \theta)$$

We used $\mathbf{M} = \mathbf{I}$ for simplicity.
The HMC in R is:

```
U <- function(x) {-log_pi(x)}
K <- function(p) {0.5 * sum(p^2)}
hmc <- function(n_iter, epsilon = 0.15, L = 20) {
    x <- c(0, 0)
    samples <- matrix(NA, n_iter, 2)
    acc <- 0
    for (i in 1:n_iter) {
        p <- rnorm(2)
        current_H <- U(x) + K(p)
        proposal <- leapfrog(x, p, epsilon, L)
        x_prop <- proposal$x
        p_prop <- proposal$p
        proposed_H <- U(x_prop) + K(p_prop)
        log_alpha <- -proposed_H + current_H
        if (log(runif(1)) < log_alpha) {
            x <- x_prop
            acc <- acc + 1
        }
        samples[i, ] <- x
    }
    list(samples = samples, acc_rate = acc / n_iter)
}
```

Running the comparison:

```
n_iter <- 5000
fit_mh <- rw_mh(n_iter, step = 0.3)
fit_hmc <- hmc(n_iter, epsilon = 0.15, L = 20)
fit_mh$acc_rate
```

```
#[1] 0.61
fit_hmc$acc_rate
#[1] 0.9642
```

The acceptance rate for the HMC is way bigger with respect to the MH. HMC uses gradient information and can move much farther while maintaining high acceptance with not-too-small ε. In a strongly correlated target, random-walk MH moves slowly along the ridge. The improvement is visible by comparing trace plots and autocorrelation function (Figure 6.10 and 6.11). The empirical autocorrelation function estimates how quickly the chain forget its past. HMC shows much faster decay for autocorrelation, leading to larger ESS. The autocorrelation plot of HMC is a typical signature of HMC dynamics

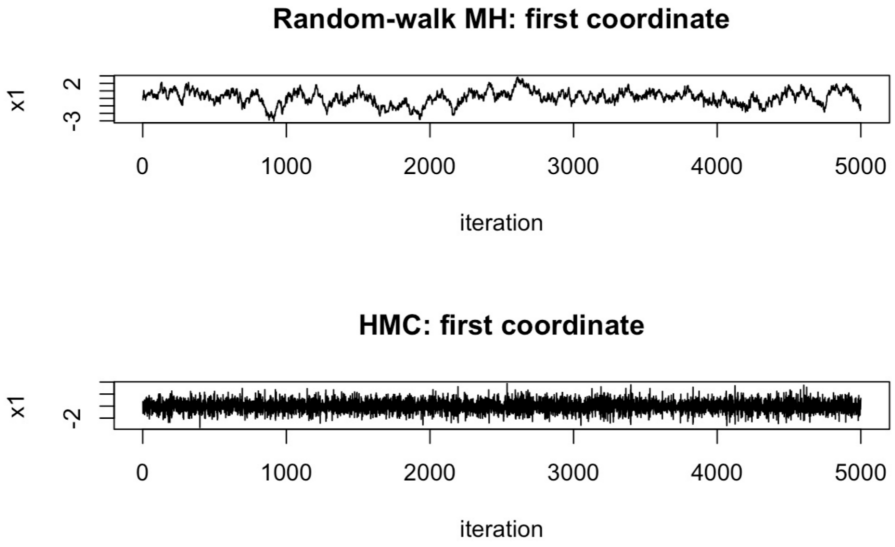


Figure 6.10

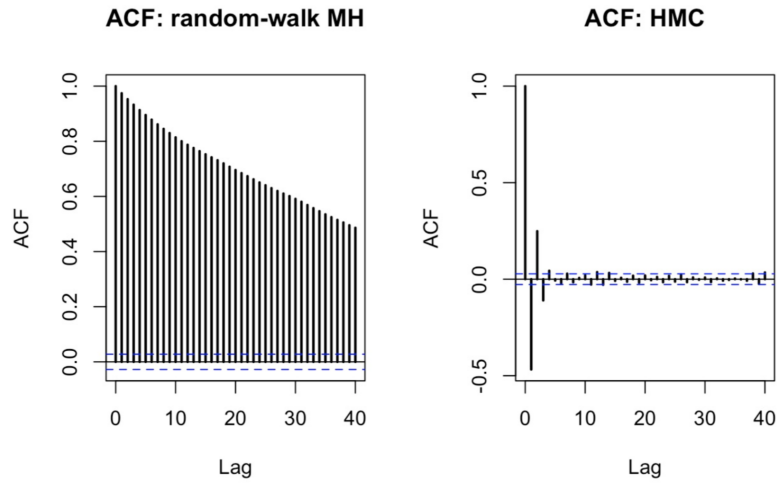


Figure 6.11

Wrapping everything up:

- HMC augments the target space with momenta:

$$\pi(x, p) = \pi(x) f(p \mid \mathbf{M})$$

- It uses Hamiltonian dynamics to propose distant moves.
- Exact dynamics preserves Hamiltonian energy, phase space volume and the joint density $\pi(x, p)$.
- Leapfrog integration gives practical approximate trajectories.
- A Metropolis correction removes discretization bias.
- Good tuning of ε, L and $\mathbf{M}$ is essential.
- HMC is especially useful for high-dimensional, strongly correlated Bayesian posteriors.

6.10.1 Multimodal targets problem

Consider a simple bimodal target $\pi(x)$, mixture of 2 Gaussian:

$$X \sim \frac{1}{2}\mathcal{N}(\mu_1, \Sigma) + \frac{1}{2}\mathcal{N}(\mu_2, \Sigma)$$

where X is a 2D vector. Consider the two modes well separated in space. This creates a *low-probability region* between them. We ran both MH and HMC algorithm in `R` and see what happens.

```
mu1 <- c(-5, 0)
mu2 <- c(5, 0)
Sigma <- diag(2)
Sigma_inv <- solve(Sigma)
log_pi <- function(x) {
        log(0.5 * exp(-0.5 * t(x - mu1) %*% Sigma_inv %*% (x -
           mu1)) +
            0.5 * exp(-0.5 * t(x - mu2) %*% Sigma_inv %*% (x -
               mu2)))}
grad_U <- function(x) {
    # numerical gradient for simplicity
    eps <- 1e-5
    g <- numeric(2)
    for (i in 1:2) {
        e <- rep(0,2); e[i] <- eps
        g[i] <- ( -log_pi(x + e) + log_pi(x - e) ) / (2*eps)
    }
g
}
n_iter <- 5000
fit_mh <- rw_mh(n_iter, step = 0.5)
fit_hmc <- hmc(n_iter, epsilon = 0.1, L = 30)
```

Both chains typically explore *only one mode*. Transitions between modes are extremely rare, leading to getting stuck in the points where we initialized. HMC is slightly better, in the sense that it is able to jump from one mode to another but one is still more present

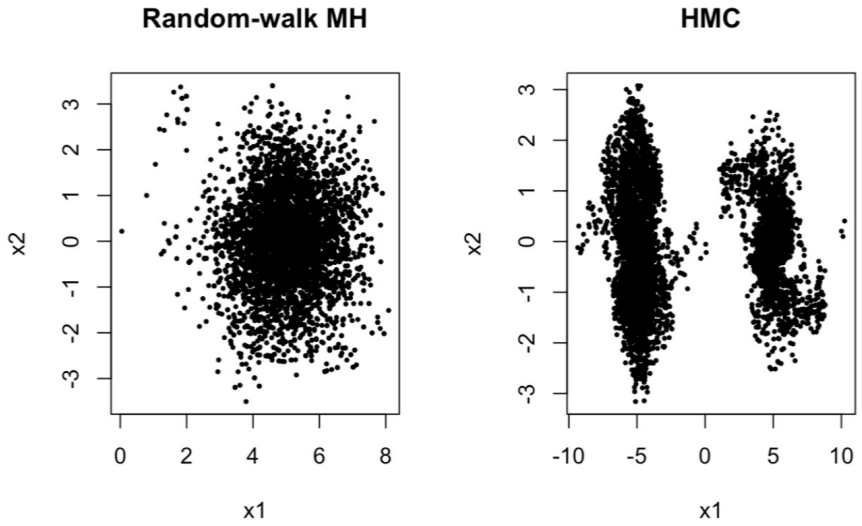


Figure 6.12

than the other. Instead, random-walk MH gets stuck on the right hand side mode of the distribution.

Why both methods fail? Since the two modes are separated by a region of very low probability:

- Random-walk MH proposes local moves and crossing the low-density region is very unlikely;
- HMC follows approximate level sets of the Hamiltonian and tends to stay within one mode. Transitions require *unusually large* kinetic energies.

In both cases, there is poor mixing and biased estimates if only one mode is explored.

Essentially, the problem is the presence of *energy barriers* between modes. To fix this, the idea is to modify the target distribution to facilitate exploration. Introducing a family of distributions:

$$\pi_\beta(x) \propto e^{-\beta U(x)}, \qquad \beta \geq 0$$

where β is a inverse-temperature-like parameter. For small β (large temperatures), the distribution is flatter and modes become easier to traverse. This leads to method such as *simulated annealing* (optimization) or *parallel tempering* (sampling).

6.11 Multimodality and energy barriers: tempering and annealing

Recalling that our goal in Bayesian inference is sampling $\pi(\theta) = p(\theta \mid D)$. We saw that standard MH and HMC algorithms can struggle when π has several well-separated modes. The difficulty is not only $d \gg 1$, but also *energy barriers*. By writing the posterior in exponential form $\pi(\theta) = e^{-E(\theta)}$, modes of the posterior correspond to minima of $E(\theta)$. Regions of small $\pi(\theta)$ correspond to high-energy barriers.

Consider a local minimum θ^*_α of $E = -\ln \pi$ divided from a second minimum θ^*_β by a saddle $\theta^\ddagger$. A random walk takes time τ to jump from θ^*_α over the barrier:

$$\Delta E = E(\theta^\ddagger) - E(\theta^*_\alpha)$$

If $\Delta E \gg 1$ escaping from the minimum is a *rare event*, which can be characterized by statistical physics through **Arrhenius formula**, which gives the escape time:

$$\tau \approx e^{\Delta E} \tag{6.20}$$

The idea to tackle this problem is to draw a parallel with statistical physics introducing a method that leverage on a temperature parameter $T > 0$.

Let's use $\pi(\theta) = p(\theta \mid D)$ to denote the posterior distribution we're interested to compute.

We introduce a *temperature* parameter $T > 0$ and define the tempered distribution:

$$\pi_T(\theta) = \frac{\pi(\theta)^{1/T}}{Z_T}, \qquad Z_T = \int \pi(\theta)^{1/T}\, d\theta \tag{6.21}$$

Writing the posterior in exponential form, thus exploiting the analogy with equilibrium statistical mechanics:

$$\pi(\theta) = \exp\left(-E(\theta)\right)$$

the tempered distribution becomes:

$$\pi_T(\theta) = \frac{\exp\left(-E(\theta)/T\right)}{Z_T}$$

Equivalently, using the inverse temperature $\beta = 1/T$, we get:

$$\pi_\beta(\theta) = \frac{\pi(\theta)^\beta}{Z_\beta}$$

with $\pi_\beta(\theta) \propto \exp\left(-\beta E(\theta)\right)$. Thus, $E(\theta)$ plays the role of an energy function and T acts as temperature in the Boltzmann distribution. Basically, tuning β, we change the relative importance of minima and maxima. Larger T means smaller β and the target is flatter. Therefore, large T reduces the contrast between modes and valleys (Figure 6.13) and barrier crossing becomes easier. Vice versa, small T concentrates probability.

Regarding the algorithms, using this trick, we can obtain random walks that are so energetic that flies over the barrier, reaching the other modes, and then sample back from the target distribution with $T = 1$.

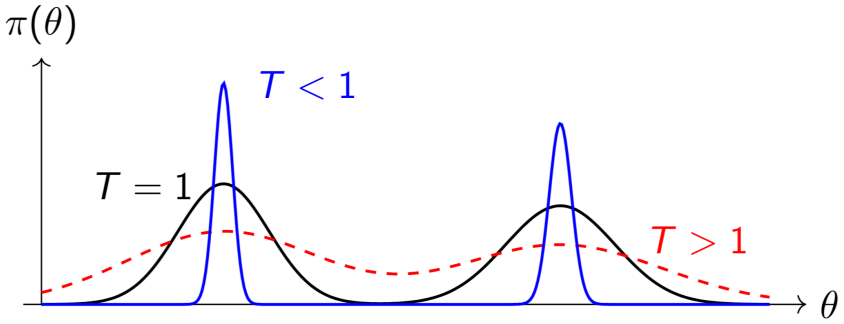


Figure 6.13

There are two of these tempering strategies that allow us to construct two kind of algorithms: *simulated annealing* and *parallel tempering* use this idea differently. Basically, simulated annealing uses one Markov chain at different temperatures and cools toward a

degenerate distribution. It's original goal was not sampling, but optimization. Basically, this method allows to explore the distribution by starting with higher temperatures and cooling down, we reach the target distribution, allowing to find local maxima in very complex distributions in multi-dimensions. Instead, parallel tempering is slightly more complex, since it runs many chains at different temperature, allowing the system to jump from one to another. Then, it keeps the cold chain at $T = 1$ and uses hot chains as helpers to fly over barriers. Therefore, it's good as sampling algorithm.

Method	Temperature use	Goal
Simulated annealing	one chain, changing T	optimization/MAP
Parallel tempering	many chains, fixed ladder	posterior sampling

Table 6.2

6.11.1 Simulated annealing

As we have seen, **simulated annealing** is a stochastic optimization method inspired by thermodynamic cooling. In Bayesian inference, it's usually used to approximate:

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} \pi(\theta) = \arg\min_{\theta} E(\theta)$$

The algorithm samples from a sequence of increasingly concentrated targets, i.e. colder distributions.

For temperature $T > 0$, recalling (6.21), if $\pi(\theta) = \exp(-E(\theta))$, then:

$$\pi_T(\theta) = \frac{\exp(-E(\theta)/T)}{Z_T}$$

which is our new target. Large T leads to flat distribution and frequent uphill moves. Small T concentrates the distribution and leads to almost greedy behavior.

In the low-temperature limit $T \to 0$, probability mass concentrates around minima of E. In the simplest case of a unique MAP estimator:

$$\pi_T(\theta) \xrightarrow{T\to 0} \delta\Big(\theta - \hat{\theta}_{\text{MAP}}\Big)$$

More generally, if the distribution is multimodal, the distribution concentrates on the multiple global minimizers of E, under mild regularity conditions. Therefore, simulated annealing transforms a sampling problem into a sequence of increasingly concentrated distributions whose limit identifies the MAP estimator.

In practice, we implement the algorithm by constructing a sequence of temperatures:

$$T_0 > T_1 > \ldots > T_k \to 0$$

At each fixed temperature T_k, one runs a Markov chain whose stationary distribution is π_{T_k} (invariant), typically using a Metropolis-Hastings transition with proposal $q(\theta' \mid \theta)$ and acceptance probability:

$$\alpha_T(\theta, \theta') = \min\left(1, \left(\frac{\pi(\theta')}{\pi(\theta)}\right)^{1/T} \frac{q(\theta \mid \theta')}{q(\theta' \mid \theta)}\right) = \min\left(1, \exp\left(-\frac{E(\theta') - E(\theta)}{T}\right)\frac{q(\theta \mid \theta')}{q(\theta' \mid \theta)}\right)$$

For large T, uphill moves ($E(\theta') > E(\theta)$) are frequently accepted, allowing the chain to escape local minima of $E(\theta)$ and exploring globally. As T decreases, the chain falls into an energy well and at low T the chain jiggles near a minimum: the algorithm becomes increasingly greedy.

Simulated annealing can be viewed as a non-homogeneous MCMC method: at each fixed temperature, the Markov kernel leaves π_T invariant, but the target distribution itself changes over time. Unlike standard MCMC, whose goal is to sample from a fixed posterior to approximate expectations, simulated annealing intentionally drives the chain toward a degenerate limit distribution concentrated at the MAP estimate.

Simulated annealing is typically used for:

- global optimization and MAP estimation,
- escaping local modes in highly multimodal posteriors,
- constructing good initial states for subsequent MCMC runs,
- approximating posterior modes in large-scale models.

6.11.2 Optimal cooling protocol

The behavior of simulated annealing depends critically on the *cooling schedule*. At low T classical convergence results require very slow cooling.

An optimal cooling protocol is to decrease T_k sufficiently slowly, for instance:

$$T_k \sim \frac{c}{\log k} \tag{6.22}$$

under suitable assumptions, the algorithm converges to the set of global optima with probability one. Such logarithmic cooling schedule arises from the exponential scaling of escape times from local minima, a phenomenon analogous to the Arrhenius law in statistical physics.

Why do we need optimal log cooling? Suppose the Markov chain is trapped in a local minimum of E that is separated from a deeper minimum by an energy barrier of height $\Delta E > 0$. At temperature T, a Metropolis proposal that increases the energy by ΔE is accepted with probability approximately:

$$p_T \approx \exp\left(-\frac{\Delta E}{T}\right)$$

Such a probability of crossing the barrier at one step is the same exponential factor that governs thermally activated escape processes in statistical mechanics. Then, from Arrhenius formula (6.20), the expected waiting time before escape is of order:

$$\tau(T) \approx \frac{1}{p_T} \approx \exp\left(\frac{\Delta E}{T}\right)$$

showing that escape times grow exponentially as temperature decreases.

Now consider a decreasing temperature schedule T_k. To guarantee convergence to the global minimum, the chain must retain the ability to escape local minima infinitely often as $k \to \infty$. Assuming the logarithmic schedule (6.22), the barrier-crossing probability at iteration k behaves as:

$$p_{T_k} \approx \exp\left(-\frac{\Delta E}{T_k}\right) = \exp\left(-\frac{\Delta E}{c}\log k\right) = k^{-\Delta E/c}$$

For the chain to escape a local minimum with probability one, the total number of escape opportunities must be infinite[6]. This requires:

$$\infty = \sum_{k=1}^{\infty} p_{T_k} \approx \sum_{k=1}^{\infty} k^{-\Delta E/c}$$

The second series diverges if and only if:

$$\frac{\Delta E}{c} \leq 1, \qquad \text{that is, } c \geq \Delta E$$

Therefore, if:

$$T_k \geq \frac{c}{\log k} \leq 1, \qquad \text{with, } c > \Delta^*$$

where Δ^* is the maximal depth of local minima, the algorithm converges to the global optimum with probability one under suitable regularity assumptions. Logarithmic cooling is therefore the slowest schedule that compensates for the exponential growth of metastable escape times. However, such logarithmic cooling is typically too slow for practical use. In applications, geometric schedules are common:

$$T_{k+1} = \gamma T_k, \qquad 0 < \gamma < 1$$

Faster schedules such as the geometric cooling, typically cause the chain to freeze in suboptimal local minima, while logarithmic cooling preserves the possibility of eventual barrier crossing. There is therefore a trade-off between theoretical guarantees and computational feasibility (Table 6.4).

Simulated annealing algorithm:

1. Choose initial state θ_0 and initial temperature T_0.
2. For $k = 0, 1, \ldots, K$:
   - (i) propose θ' from $q(\theta' \mid \theta_k)$;
   - (ii) accept using the tempered MH probability α_{T_k};
   - (iii) update the temperature to $T_{k+1} < T_k$.
3. Return θ with the smallest energy, or the final low-temperature state.

Such an algorithm may start near the wrong local mode, but high temperature helps it escape. Cooling drives it toward the dominant mode.

6.12 Parallel tempering

Parallel tempering, also called **replica exchange MCMC**, is a sampling method designed for posteriors distributions with multiple well-separated modes by also using tempered distributions.. Its purpose is different from simulated annealing, which lowers the temperature in order to concentrate the chain near the MAP estimator. Instead, parallel tempering runs several Markov chains at different temperatures at the same time, and

[6]The second Borel-Cantelli lemma states that if random events are independent and the sum of their probabilities is infinite, then the probability that infinitely many of them occur is 1.

uses the high-temperature chains to help the low-temperature chain explore the posterior distribution.

Let now $\pi(\theta) = p(\theta \mid D)$ denote the posterior distribution. As in simulated annealing, introduce a temperature parameter $T > 0$ and define:

$$\pi_T(\theta) = \frac{\pi(\theta)^{1/T}}{Z_T}, \qquad Z_T = \int \pi(\theta)^{1/T}\, d\theta$$

Equivalently, using the inverse temperature $\beta = 1/T$, we write:

$$\pi_\beta(|theta) = \frac{\pi(\theta)^\beta}{Z_\beta}, \qquad 0 < \beta < 1$$

The original posterior corresponds to $T = 1$, or $\beta = 1$. Values $T > 1$ correspond to $0 < \beta < 1$ and produce a flatter distribution. If the posterior is written in Boltzmann form:

$$\pi(\theta) = \exp\left(-E(\theta)\right)$$

Then,

$$\pi_\beta(\theta) = \frac{\exp\left(-\beta E(\theta)\right)}{Z_\beta}$$

Thus, increasing T reduces the energetic cost of crossing regions of low posterior probability. Energy barriers that are nearly impossible to cross at $T = 1$ become much easier to cross at high temperature.

Parallel tempering constructs a ladder of inverse temperatures:

$$1 = \beta_0 > \beta_1 > \ldots > \beta_L > 0 \quad \text{or} \quad 1 = T_0 < T_1 < \ldots < T_L$$

For each inverse temperature β_ℓ, one runs a Markov chain with stationary condition distribution π_{β_ℓ}. The joint state of the algorithm is $\Theta = (\theta_0, \theta_1, \ldots, \theta_L)$, where θ_ℓ is the current state of the replica at inverse temperature β_ℓ. The target distribution of the enlarged product space is:

$$\Pi(\Theta) = \prod_{\ell=0}^{L} \pi_{\beta_\ell}(\theta_\ell) \propto \prod_{\ell=0}^{L} \pi(\theta_\ell)^{\beta_\ell}$$

The marginal distribution of θ_0, corresponding to $\beta_0 = 1$, is exactly the posterior distribution of interest $\pi(\theta)$. Therefore, samples from the cold chain θ_0 are samples from the original posterior and hot chains are just *auxiliary variables* used to improve exploration.

The algorithm alternates between two types of moves:

1. First, each replica is updated at fixed temperature. For example, one may apply a Metropolis-Hastings transition that leaves π_{β_ℓ} invariant. If the proposal density for replica ℓ is $q_\ell(\theta'_\ell \mid \theta_\ell)$, the acceptance probability is:

$$\begin{aligned}\alpha_\ell(\theta_\ell, \theta'_\ell) &= \min\left(1, \left(\frac{\pi(\theta'_\ell)}{\pi(\theta_\ell)}\right)^{\beta_\ell} \frac{q_\ell(\theta_\ell \mid \theta'_\ell)}{q_\ell(\theta'_\ell \mid \theta_\ell)}\right) = \\ &= \min\left(1, \exp\left\{-\beta_\ell[E(\theta'_\ell) - E(\theta_\ell)]\right\} \frac{q_\ell(\theta_\ell \mid \theta'_\ell)}{q_\ell(\theta'_\ell \mid \theta_\ell)}\right)\end{aligned}$$

   For hot replicas, β_ℓ is small and uphill moves in energy are more likely to be accepted, so the hot chains move more freely between modes.

2. Second, the algorithm proposes swaps between replicas at different temperatures. Usually, one proposes to exchange the states of two adjacent temperatures, say ℓ and $\ell + 1$:

$$(\theta_\ell, \theta_{\ell+1}) \to (\theta_{\ell+1}, \theta_\ell)$$

The Metropolis acceptance probability is obtained from the product target $\Pi(\Theta)$. Since the normalizing constants cancel, the swap is accepted with probability:

$$\alpha_{\text{swap}} = \min\left(1, \frac{\pi(\theta_{\ell+1})^{\beta_\ell}\pi(\theta_\ell)^{\beta_{\ell+1}}}{\pi(\theta_\ell)^{\beta_\ell}\pi(\theta_{\ell+1})^{\beta_{\ell+1}}}\right)$$

Taking the log gives:

$$\begin{aligned}\log \alpha_{\text{swap}} &= \min\left(0, \exp\left\{(\beta_\ell - \beta_{\ell+1})[\log \pi(\theta_{\ell+1}) - \log \pi(\theta_\ell)]\right\}\right) = \\ &= \min\left(0, \exp\left\{(\beta_\ell - \beta_{\ell+1})[E(\theta_\ell) - E(\theta_{\ell+1})]\right\}\right)\end{aligned}$$

where we used $\log \pi(\theta) = -E(\theta)$. This acceptance probability satisfies detailed balance with respect to the product target $\Pi(\Theta)$.

The mechanism is the following. Hot chains, with small β, can cross energy barriers and move between modes relatively easily. When a hot chain discovers a new mode, a sequence of accepted swaps can transport that configuration down the temperature ladder toward the cold chain. The cold chain therefore explores the true posterior more efficiently than a single-temperature chain, while still having exact marginal target π.

Parallel tempering is therefore a homogeneous MCMC method on an enlarged state space. Unlike simulated annealing, the temperature ladder is fixed in time and the joint target distribution is fixed. The purpose is not to converge to a degenerate MAP distribution, but to improve sampling from the posterior itself.

In Bayesian inference, parallel tempering is used for:

- sampling from highly multimodal posteriors,
- moving between well-separated posterior modes,
- reducing sensitivity to the initial condition,
- diagnosing multimodality by comparing the behavior of cold and hot chains,
- improving exploration before or together with more local MCMC methods.

6.12.1 Temperature ladder and swap efficiency

The performance of parallel tempering depends strongly on the choice of temperature ladder. If adjacent temperatures are too far apart, the swap acceptance probability is very small. If they are too close, swaps are accepted often, but many replicas are needed to connect the posterior temperature $\beta = 1$ to a sufficiently hot chain.

The swap probability between adjacent inverse temperature β_ℓ and $\beta_{\ell+1}$ depends on the difference:

$$\Delta\beta_\ell = \beta_\ell + \beta_{\ell+1}$$

and on the energy difference between the two replicas. In energy notation,

$$\log \alpha_{\text{swap}} = \min\left\{0, (\beta_\ell - \beta_{\ell+1})[E(\theta_\ell) - E(\theta_{\ell+1})]\right\}$$

Thus, large gaps in inverse temperature lead to poor overlap between adjacent tempered distributions, and therefore low swap rates.

A common practical choice is a geometric ladder in temperature:

$$T_\ell = T_0 r^\ell, \qquad r > 1$$

or equivalently

$$\beta_\ell = \beta_0 r^{-\ell}$$

The largest temperature T_L should be high enough that the corresponding chain can move between modes. The spacing should be fine enough that swaps can propagate states between adjacent replicas.

A useful diagnostic is the empirical swap acceptance rate between adjacent replicas. Very small rates indicate poor communication along the ladder. Very high rates may indicate that the ladder is unnecessarily dense. In practice, one often adapts or tunes the number of temperatures and their spacing in pilot runs.

6.12.2 Comparison with simulated annealing

Both simulated annealing and parallel tempering use the family

$$\pi_\beta(\theta) \propto \pi(\theta)^\beta$$

However, they use this family for different purposes. In simulated annealing, the temperature changes with time. The algorithm starts at high temperature and slowly cools toward $T = 0$, or equivalently toward $\beta = \infty$ if the target is written as $\exp[-E/T]$. The limiting distribution concentrates on the global minimizers of E, so the goal is optimization and MAP estimation.

In parallel tempering, all temperatures are present simultaneously. The coldest replica is fixed at $T = 1$, so its marginal distribution is the original posterior. The hotter replicas are auxiliary chains used to improve exploration. The goal is sampling, not optimization.

Thus, simulated annealing is best viewed as a temperature-driven optimization method, while parallel tempering is an exact MCMC method on an enlarged state space whose cold marginal is the posterior of interest.

In summary, tempering introduces $\pi_T(\theta) \propto \pi(\theta)^{1/T}$. High temperature flattens the posterior and makes energy barriers easier to cross. Simulated annealing cools a single chain and targets the MAP estimator. Parallel tempering runs several chains at fixed temperatures and swaps their states. The cold chain of parallel tempering has the exact posterior as its marginal target. The central practical issue in both methods is the temperature schedule or ladder. Both methods exploit temperature, but they answer different statistical questions.

We briefly summarize in Table 6.4 all the methods we have seen in this chapter with their goal and typical use.

	Simulated annealing	Parallel tempering
Target	changes with time	fixed product target
Temperature	decreases to zero	fixed ladder
Output	MAP/optimizer	posterior samples
MCMC type	non-homogeneous	homogeneous on enlarged space
Cold chain	not fixed at $T = 1$	fixed at $T = 1$
Use of hot states	early exploration	continuous mode discovery

Table 6.3

Method	Goal	Bayesian use
CDF inversion	Exact sampling	Simple posteriors
Rejection sampling	Exact sampling	Low dimensions
Importance Sampling	Expectations	Evidence, re-weighting
MCMC	Posterior averages	General inference
Simulated annealing	MAP estimate	Optimization
Parallel tempering	Posterior sampling	Multimodal posteriors

Table 6.4

Chapter 7

Bayesian regression

7.1 Where are we in the course

In the previous chapters, MCMC gave us a general way to sample from complicated posterior distributions:

$$p(\theta \mid D) \propto p(D \mid \theta)p(\theta)$$

and computing expectation values under $p(\theta \mid D)$.

We now apply the same Bayesian logic to *regression*: learning a relation between inputs x_i and outputs y_i.

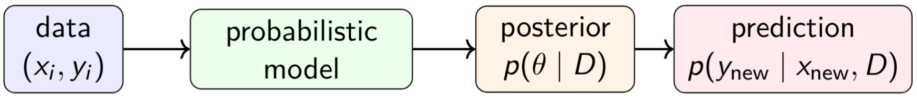


Figure 7.1

In the next sections, we will look at regression analysis extended to Bayesian framework:

1. We define a likelihood plus a prior
2. Distinguish linearity in parameters from linearity in x
3. Write linear and polynomial regression with a design matrix
4. Derive the Gaussian posterior for Bayesian linear regression
5. Interpret posterior predictive uncertainty
6. See why generalized linear (and non-linear) models usually require MCMC or approximations; worked examples of GLMs

7.2 Introduction to regression analysis

In many inference problems, we observe input-output pairs $\mathcal{D} = \{ (x_i, y_i) \}_{i=1}^{N}$ and we want to infer a relationship between the input – or **regressor** – variables x_i and the output, response – or **regressand** – variables y_i. The input variables x_i are independent and explanatory, while the dependent y_i variables are predicted.

In the Bayesian framework, this relationship is represented by a probabilistic generative model. The basic form is:

$$y_i = f(x_i, \theta) + \varepsilon_i \tag{7.1}$$

where $f(x_i; \theta)$ is the generative model or noise-free mean predictor – which is non-linear in general – and ε_i describes measurement scatter or intrinsic variability of the data.

For Gaussian noise, one assumes:

$$\varepsilon_i \sim \mathcal{N}\left(0, \sigma^2\right), \qquad y_i \mid x_i, \theta, \sigma \sim \mathcal{N}\left(f(x_i; \theta), \sigma^2\right)$$

If data points are conditionally independent given the parameters, the likelihood factorizes:

$$p(\mathcal{D} \mid \theta) = \prod_{i=1}^{N} p(y_i \mid x_i, \theta)$$

For the Gaussian model, this gives the log-likelihood:

$$\log p(\mathcal{D} \mid \theta, \sigma) = -N \log \sigma - \frac{N}{2} \log (2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^{N} [y_i - f(x_i; \theta)]^2 \tag{7.2}$$

Bayesian inference then proceeds by assigning a prior $p(\theta)$, or $p(\theta, \sigma)$ if the noise scale is unknown, and computing:

$$p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta) p(\theta)$$

The posterior distribution is the Bayesian object that replaces a single best-fit curve. It contains both the parameter values favored by the data and they uncertainty.

Throughout this section, we assume that the x_i values are fixed and known without uncertainty. If the inputs themselves are noisy, one should introduce latent true inputs x_i^{true} and infer them jointly with the regression parameters.

7.2.1 Linear versus nonlinear regression

The adjective "linear" in linear regression refers to linearity in the unknown parameters, not necessarily in the input variable. A model is **linear** if it is linear in its parameters $\boldsymbol{w}$:

$$y_i = \boldsymbol{\phi}(x_i)^T \boldsymbol{w} + \varepsilon_i$$

where $\boldsymbol{\phi}(x)$ is a vector of basis functions, or features, and $\boldsymbol{w}$ is a vector of coefficients. The feature map $\boldsymbol{\phi}(x)$ may be non-linear in x. In Table 7.1 are shown some examples of feature maps.

Feature map $\boldsymbol{\phi}(x)$	Model	Linear in parameters
$(1, x)$	$w_0 + w_1 x$	yes
$(1, x, x^2)$	$w_0 + w_1 x + w_2 x^2$	yes
$(1, \sin x, \cos x)$	$w_0 + w_1 \sin x + w_2 \cos x$	yes
$(1, e^{-\lambda x})$	$w_0 + w_1 e^{-\lambda x}$	no, if λ unknown

Table 7.1

Therefore, polynomial regression is still linear regression if the unknown coefficients enter linearly. For non-linear models, such as the exponential one, the posterior is generally non-Gaussian and cannot be computed in closed form.

For N data points and M basis functions, the **design matrix** is:

$$\boldsymbol{\Phi} = \begin{pmatrix} \phi_1(x_1) & \phi_2(x_1) & \dots & \phi_M(x_1) \\ \phi_1(x_2) & \phi_2(x_2) & \dots & \phi_M(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(x_N) & \phi_2(x_N) & \dots & \phi_M(x_N) \end{pmatrix}$$

This is also called **model** or **regressor matrix**. The linear Gaussian regression model can then be written compactly as:

$$\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{w} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}\left(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_N\right) \tag{7.3}$$

Therefore, the likelihood reads:

$$p(\boldsymbol{y} \mid \boldsymbol{\Phi}, \boldsymbol{w}, \sigma^2) = \mathcal{N}\left(\boldsymbol{y} \mid \boldsymbol{\Phi}\boldsymbol{w}, \sigma^2 \boldsymbol{I}_N\right) \tag{7.4}$$

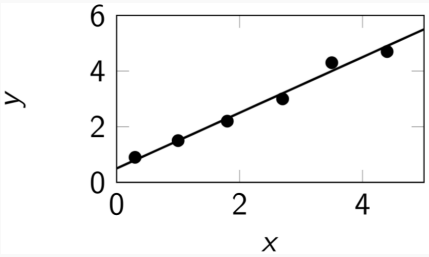
The important key-point to keep in mind is that, with respect to ordinary regression, here we provide a distribution for the parameters, which is the thing that we want to estimate.

Example 7.2.1

Two simple examples of Bayesian linear regression are:

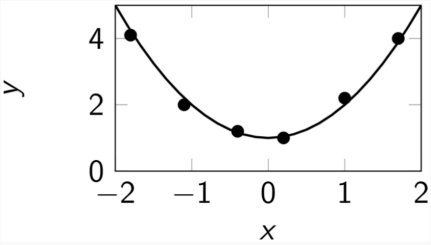
- A *straight line*:

$$\phi(x) = (1, x), \qquad f(x) = w_0 + w_1 x$$



- A *quadratic curve*:

$$\phi(x) = (1, x, x^2), \qquad f(x) = w_0 + w_1 x + w_2 x^2$$



Both of them are linear in $\boldsymbol{w}$.

7.2.2 Bayesian linear regression: prior and likelihood

We now assume that the noise variance σ^2 is known (not a parameter). Choose a Gaussian prior on the weights:

$$p(\boldsymbol{w}) = \mathcal{N}\left(\boldsymbol{w} \mid \boldsymbol{m}_0, \boldsymbol{S}_0\right)$$

Together with the Gaussian likelihood:

$$p(\boldsymbol{y} \mid \boldsymbol{\Phi}, \boldsymbol{w}) = \mathcal{N}\left(\boldsymbol{y} \mid \boldsymbol{\Phi}\boldsymbol{w}, \sigma^2 \boldsymbol{I}_N\right)$$

Bayes' theorem gives:

$$p(\boldsymbol{w} \mid \mathcal{D}) \propto \underbrace{p(\boldsymbol{y} \mid \boldsymbol{\Phi}, \boldsymbol{w})}_{\text{data fit}} \times \underbrace{p(\boldsymbol{w})}_{\substack{\text{reg./} \\ \text{prior}}}$$

where $p(\boldsymbol{w})$ acts as prior info or regularization. Because both the likelihood and the prior are Gaussian, the posterior is Gaussian (conjugacy):

$$p(\boldsymbol{w} \mid \mathcal{D}) = \mathcal{N}\left(\boldsymbol{w} \mid \boldsymbol{m}_N, \boldsymbol{S}_N\right)$$

with variance and mean, respectively:

$$\boldsymbol{S}_N^{-1} = \boldsymbol{S}_0^{-1} + \frac{1}{\sigma^2}\boldsymbol{\Phi}^T\boldsymbol{\Phi}, \qquad \boldsymbol{m}_N = \boldsymbol{S}_N\left(\boldsymbol{S}_0^{-1}\boldsymbol{m}_0 + \frac{1}{\sigma^2}\boldsymbol{\Phi}^T\boldsymbol{y}\right)$$

The posterior precision is the sum of a prior precision, $\boldsymbol{S}_0^{-1}$, and a data precision, $\sigma^{-2}\boldsymbol{\Phi}^T\boldsymbol{\Phi}$, making explicit the balance between prior information and info provided by the data.

A common special case is the zero-mean isotropic prior:

$$\boldsymbol{m}_0 = \boldsymbol{0}, \qquad \boldsymbol{S}_0 = \sigma_0^2 \boldsymbol{I}_M$$

Then, the posterior simplifies:

$$\boldsymbol{S}_N^{-1} = \frac{1}{\sigma_0^2}\boldsymbol{I}_M + \frac{1}{\sigma^2}\boldsymbol{\Phi}^T\boldsymbol{\Phi}, \qquad \boldsymbol{m}_N = \frac{1}{\sigma^2}\boldsymbol{S}_N\boldsymbol{\Phi}^T\boldsymbol{y}$$

The ordinary least-squares estimator is recovered when the prior becomes very broad, i.e. $\sigma_0 \to \infty$.

With a finite Gaussian prior, the MAP estimator is instead equivalent to **ridge regression**. Indeed, consider the linear model:

$$\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{w} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}\left(\boldsymbol{0}, \sigma^2 \boldsymbol{I}\right)$$

Ordinary least squares minimizes the residual sum of squares:

$$\mathcal{L}_{\text{OLS}} = \|\boldsymbol{y} - \boldsymbol{\Phi}\boldsymbol{w}\|^2$$

Ridge regression adds an L^2 penalty on the parameters:

$$\mathcal{L}_{\text{Ridge}} = \|\boldsymbol{y} - \boldsymbol{\Phi}\boldsymbol{w}\|^2 + \lambda \|\boldsymbol{w}\|^2$$

which penalizes large values of parameters.

The corresponding estimator is:

$$\hat{\boldsymbol{w}}_{\text{Ridge}} = \left(\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \lambda \boldsymbol{I}\right)^{-1}\boldsymbol{\Phi}^T\boldsymbol{y}$$

To understand the connection with Bayesian inference, assume a Gaussian likelihood and prior. Maximizing the posterior probability is equivalent to minimizing:

$$-\log p(\boldsymbol{w} \mid \boldsymbol{y}) \propto \|\boldsymbol{y} - \boldsymbol{\Phi}\boldsymbol{w}\|^2 + \frac{\sigma^2}{\sigma_0^2}\|\boldsymbol{w}\|^2$$

Thus, ridge regression is MAP estimation with a Gaussian prior, with regularization strength given by the ratio between the variance of the noise and the variance of the computed variables:

$$\lambda = \frac{\sigma^2}{\sigma_0^2}$$

Here, λ basically decides the maximum size that the weight might have.

Why use ridge regression, aka regularization? OLS may become unstable when

- predictor are highly correlated,
- the number of parameter is large,
- the design matrix is nearly singular,
- the model overfits noisy data.

In these cases, the estimated coefficients become large:

$$\hat{\boldsymbol{w}}_{\text{OLS}} = (\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\boldsymbol{y}$$

Ridge regression stabilizes the inversion by adding a positive term to the diagonal:

$$\hat{\boldsymbol{w}}_{\text{Ridge}} = (\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \lambda\boldsymbol{I})^{-1}\boldsymbol{\Phi}^T\boldsymbol{y}$$

This regularization is useful because it penalizes very large coefficients, by shrinking them. Then, it reduced variance, stabilizing fits when the columns of $\boldsymbol{\Phi}$ are correlated, and reducing overfitting, especially when using flexible basis expansions such as high-order polynomials. However, it increases bias. Overall, it improves numerical stability, improving generalization performance.

Why the prior helps? Because it acts as a regularizer. Without it, the model is too flexible. A high-degree polynomial may fit the noise in the observed data (Figure 7.2). Indeed, prior penalizes implausible coefficients and propagates remaining uncertainty to predictions. Bayesian model comparison can compare polynomial orders through the evidence $p(\mathcal{D})$.

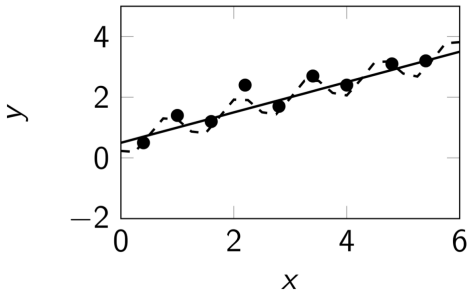


Figure 7.2

7.2.3 Prediction: integrate over parameter uncertainty

For a new input x_* with feature vector $\boldsymbol{\phi}_* = \boldsymbol{\phi}(x_*)$, the posterior predictive distribution is obtained by marginalizing over parameter uncertainty:

$$p(y_* \mid x_*, \mathcal{D}) = \int p(y_* \mid x_*, \boldsymbol{w}) p(\boldsymbol{w} \mid \mathcal{D})\, d\boldsymbol{w}$$

In the Gaussian linear case with known σ^2, this integral is analytic:

$$y_* \mid x_*, \mathcal{D} \sim \mathcal{N}\left(y_* \mid \boldsymbol{\phi}_*^T \boldsymbol{m}_N, \sigma^2 + \boldsymbol{\phi}_*^T \boldsymbol{S}_N \boldsymbol{\phi}_*\right)$$

There are two sources of predictive uncertainty. The first term in the variance, σ^2, is the irreducible observation noise. The second term, $\boldsymbol{\phi}_*^T \boldsymbol{S}_N \boldsymbol{\phi}_*$, is uncertainty in the fitted parameters w.

It is important to distinguish two types of uncertainty bands.

Credible band for the regression curve. This describes uncertainty in the noise-free mean function $f(x;\theta)$. At a point x_*,

$$p(f(x_*) \mid \mathcal{D}) = \int \delta(f(x_*) - f(x_*;\theta))\, p(\theta \mid \mathcal{D})\, d\theta$$

In practice, if $\theta^{(s)}$ are posterior samples (e.g., obtained via MCMC), one computes $f^{(s)}(x_*) = f(x_*;\theta^{(s)})$ and takes posterior quantiles, for example the 2.5% and 97.5% quantiles for a pointwise 95% credible interval.

Posterior predictive band. This describes uncertainty in a future observed value y_*. It includes both uncertainty in the mean function and the observational noise:

$$p(y_* \mid x_*, \mathcal{D}) = \int p(y_* \mid x_*, \theta) p(\theta \mid \mathcal{D})\, d\theta$$

For Gaussian noise, posterior predictive draws are obtained by first drawing $\theta^{(s)}$ from the posterior and then drawing:

$$y_*^{(s)} \sim \mathcal{N}\left(f(x_*;\theta^{(s)}), [\sigma^{(s)}]^2\right)$$

Predictive bands are therefore wider than credible bands for the mean curve. Curve bands computed pointwise should also not be confused with simultaneous bands that cover the whole curve for all x at once.

7.2.4 Unknown noise level and parameter transformations

So far, σ^2 has been assumed known. In practice, the noise scale σ is often unknown and it is inferred with the regression parameters $\theta = (\boldsymbol{w}, \sigma)$. The posterior is then:

$$p(\boldsymbol{w}, \sigma \mid \mathcal{D}) \propto p(\boldsymbol{y} \mid \boldsymbol{\Phi}, \boldsymbol{w}, \sigma) p(\boldsymbol{w}) p(\sigma)$$

Since $\sigma > 0$, it is convenient to work with an unconstrained parameter:

$$\eta = \log \sigma, \qquad \sigma = e^{\eta}$$

For a scale parameter, a weakly informative prior is often used. A classical non-informative choice in η is the Jeffreys prior $p(\sigma) \propto 1/\sigma$, equivalently uniform in $\log \sigma$.

This is the bridge back to the MCMC workflow used in the previous part of the course. Similar care is needed for other parametrizations.

7.2.5 MCMC view of regression fitting

Let's consider a simple example of Bayesian regression in 1D. Consider a straight line with unknown noise:

$$f(x) = b_0 + bx, \qquad \theta = (b_0, b_1, \sigma)$$

The log-posterior is, up to constants:

$$\log p(\theta \mid \mathcal{D}) = \sum_{i=1}^{N} \log \mathcal{N}\left(y_i \mid b_0 + b_1 x_i, \sigma^2\right) + \log p(b_0, b_1, \sigma)$$

The MCMC target is the unnormalized $\pi(\theta)$. Sample (b_0, b_1, σ) or transformed parameters such as $(b_0, b_1, \log \sigma)$. Then, as usually, plot traces, marginal densities, and parameter correlations. Finally, we can use samples to compute posterior predictive distributions.

MH with a change of variable. Since $\sigma > 0$, it is convenient to sample:

$$\eta = \log \sigma, \qquad \sigma = e^{\eta}$$

The posterior in the transformed variables becomes:

$$p(b_0, b_1, \eta \mid \mathcal{D}) = p(b_0, b_1, \sigma = e^{\eta} \mid \mathcal{D}) \left| \frac{\mathrm{d}\sigma}{\mathrm{d}\eta} \right|$$

Since $\frac{\mathrm{d}\sigma}{\mathrm{d}\eta} = e^{\eta} = \sigma$, the Jacobian contributes $\log |J| = \eta$. If the prior is Jeffreys $p(\sigma) \propto 1/\sigma$, then:

$$p(\eta) = p(\sigma) \left| \frac{\mathrm{d}\sigma}{\mathrm{d}\eta} \right| \propto \frac{1}{\sigma} \sigma = \text{const.}$$

The prior is flat in $\log \sigma$.

When the posterior is not analytically tractable, the same Bayesian target remains:

$$\pi(\theta) = p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta) p(\theta)$$

For example, a quadratic model with unknown noise can be written as:

$$y_i \mid \theta \sim \mathcal{N}\left(b_0 + \tan(\alpha) x_i + b_2 x_i^2, e^{2\zeta}\right), \qquad \theta = (b_0, \alpha, b_2, \zeta)$$

A random-walk Metropolis algorithm proposes:

$$\theta' = \theta + \xi, \qquad \xi \sim \mathcal{N}(0, \Sigma_q)$$

and accepts or rejects the proposal according to the posterior ratio. The practical difficulty is often not writing down the posterior but choosing a good parametrization and proposal covariance Σ_q.

Polynomial models can mix slowly because coefficients compensate each other. In a quadratic model, changes satisfying approximately:

$$\delta b_0 + \delta b_1 x + \delta b_2 x^2 \simeq 0$$

create long, tilted posterior contours. A diagonal proposal then moves inefficiently across the posterior. A first remedy is to center and scale the input:

$$z_i = \frac{x_i - \bar{x}}{s_x}$$

and fit the polynomial in z:

$$f(z) = a_0 + a_1 z + a_2 z^2$$

The columns $1, z, z^2$ are usually less correlated than $1, x, x^2$, and the posterior becomes closer to an axis-aligned cloud.

The proposal scale also matters. If it is too small, the acceptance rate is high but the chain moves by tiny steps, producing strong autocorrelation and a low effective sample size. If it is too large, most proposals fall in low-posterior regions and are rejected. Pilot chains can be used to estimate a suitable covariance, after which a production chain is run with a fixed proposal. Adaptation must not be performed naively during the final sampling stage, because changing the transition rule can violate detailed balance unless an appropriate adaptive-MCMC scheme is used.

For high-dimensional or strongly correlated posteriors, Hamiltonian Monte Carlo is often more efficient than random-walk Metropolis because it uses gradient information to propose distant moves with high acceptance probability. Probabilistic programming systems such as `JAGS` and Stan automate much of the sampling. `JAGS` uses graphical-model notation and samplers such as Gibbs and Metropolis steps, while Stan uses gradients and Hamiltonian Monte Carlo. In all cases, however, the statistical object being sampled is the same posterior, and posterior prediction is always obtained by propagating posterior draws through the observation model.

In a Bayesian regression analysis one should report not only point estimates, but also posterior uncertainty, posterior correlations, convergence diagnostics, the assumed likelihood and priors, and posterior predictive checks.

7.3 Generalized linear models

The Gaussian noise assumption is not always appropriate. **Generalized linear models** (GMLs) extend regression to other data types by combining three ingredients:

1. a random component, that is, a likelihood $p(y_i \mid \mu_i)$ appropriate for the data
2. a linear predictor:
$$f = \phi(x)^T w$$
3. a link function g relating the mean parameter to the linear predictor:
$$g(\mu) = f, \qquad \mu = g^{-1}(f)$$

The inverse link maps the real-valued linear predictor into the range allowed by the model parameter. Gaussian linear regression is the special case with Gaussian likelihood and identity link, $g(\mu) = f$.

Common examples are summarized as follows:

GLMs are connected to an exponential-family form, defined as:

$$p(y \mid f) = \exp\{yf - A(f)\}h(y)$$

where f is the natural parameter and $A(f)$ is the log-partition function.

In the Bayesian setting, we place a prior on $\boldsymbol{w}$ and infer:

$$p(\boldsymbol{w} \mid \mathcal{D}) \propto p(\mathcal{D} \mid \boldsymbol{w})p(\boldsymbol{w})$$

Unlike the Gaussian linear model, the posterior is usually not available in closed form. For logistic and Poisson regression, Gaussian conjugacy is usually lost. We need approximation such as Laplace method, variational inference, or MCMC sampling.

Data type	Likelihood	Mean parameter	Common link
Continuous	Gaussian	μ	identity: $\mu = f$
Binary	Bernoulli	π	logit: $\log \frac{\pi}{1-\pi} = f$
Counts	Poisson	λ	log: $\log \lambda = f$

Table 7.2: $f = \phi(x)^T \boldsymbol{w}$

7.3.1 Logistic regression for binary data

For binary observations $y_i \in \{0, 1\}$, the Bernoulli likelihood is appropriate:

$$y_i \mid x_i, \boldsymbol{w} \sim \text{Bernoulli}(\pi_i), \qquad \pi_i = h(\phi(x_i)^T \boldsymbol{w})$$

where:

$$h(z) = \frac{1}{1 + e^{-z}}$$

The likelihood is:

$$p(\mathcal{D} \mid \boldsymbol{w}) = \prod_{i=1}^{N} \pi_i^{y_i} (1 - \pi_i)^{1-y_i}$$

With Gaussian prior:

$$p(\boldsymbol{w} \mid \mathcal{D}) \propto p(\mathcal{D} \mid \boldsymbol{w}) \mathcal{N}(\boldsymbol{w} \mid 0, \boldsymbol{S}_0)$$

Equivalently, up to an additive constant:

$$\log p(\boldsymbol{w} \mid \mathcal{D}) = \sum_{i=1}^{N} \left[y_i \log \sigma(\phi(x_i)^T \boldsymbol{w}) + (1 - y_i) \log(1 - \sigma(\phi(x_i)^T \boldsymbol{w}) \right] - \frac{1}{2} \boldsymbol{w}^T \boldsymbol{S}_0^{-1} \boldsymbol{w} + \text{const.}$$

This posterior is not Gaussian. A common approximation is the Laplace approximation: find the MAP estimate and approximate the log posterior to second order around it, producing a Gaussian approximation.

The logit link has a useful interpretation. The odds of success are $\pi_i/(1 - \pi_i)$, and a one-unit increase in a scalar predictor x multiplies the odds by e^{w_1}. The probability itself changes non-linearly with x.

Predictions are obtained by integrating over the posterior:

$$p(y_* = 1 \mid x_*, \mathcal{D}) = \int \sigma(\phi(x_*)^T \boldsymbol{w}) p(\boldsymbol{w} \mid \mathcal{D}) \, d\boldsymbol{w}$$

For posterior samples $\boldsymbol{w}^{(s)}$, one computes:

$$\pi_*^{(s)} = \sigma(\phi(x_*)^T \boldsymbol{w}^{(s)})$$

Quantiles of $\pi_*^{(s)}$ give uncertainty in the success probability. Posterior predictive binary outcomes are generated by:

$$y_*^{(s)} \sim \text{Bernoulli}(\pi_*^{(s)})$$

Binary data are noisy by construction: each observation is only 0 or 1, and:

$$\text{Var}(y_i \mid \pi_i) = \pi_i(1 - \pi_i)$$

which is largest near $\pi_i = 1/2$ and small near 0 or 1. Logistic regression estimates a detection or success probability, not a continuous measured response.

7.3.2 Poisson regression for count data

For count observations $y_i \in \mathbb{N}$:

$$y_i \mid x_i, \boldsymbol{w} \sim \text{Poisson}(\lambda_i), \qquad \lambda_i = \exp\left(\phi(x_i)^T \boldsymbol{w}\right)$$

a Poisson likelihood is appropriate. Poisson regression uses the log link which ensures $\lambda_i > 0$. The likelihood is:

$$p(\mathcal{D} \mid \boldsymbol{w}) = \prod_{i=1}^{N} \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!}$$

and the log-likelihood is

$$\log p(\mathcal{D} \mid \boldsymbol{w}) = \sum_{i=1}^{N} \left[y_i \eta_i - e^{\eta_i} - \log(y_i!)\right]$$

With a Gaussian prior, the posterior satisfies

$$\log p(\boldsymbol{w} \mid \mathcal{D}) = \sum_{i=1}^{N} \left[y_i \phi(x_i)^T \boldsymbol{w} - e^{\phi(x_i)^T \boldsymbol{w}}\right] - \frac{1}{2} \boldsymbol{w}^T \boldsymbol{S}_0^{-1} \boldsymbol{w} + \text{const.}$$

Again this posterior is not analytically tractable in general, so one uses Laplace approximations, variational methods, or MCMC.

The log link has a multiplicative interpretation. For a scalar predictor with

$$\log \lambda(x) = w_0 + w_1 x$$

a one-unit increase in x changes the expected count by

$$\frac{\lambda(x+1)}{\lambda(x)} = e^{w_1}$$

Thus $w_1 > 0$ means expected counts increase with x, while $w_1 < 0$ means expected counts decrease with x. Log-linear models are therefore natural for rates that change by multiplicative factors.

The posterior predictive distribution is:

$$p(y_* \mid x_*, \mathcal{D}) = \int \text{Poisson}(y_* \mid e^{\phi(x_*)^T \boldsymbol{w}}) p(\boldsymbol{w} \mid \mathcal{D})\, d\boldsymbol{w}$$

For posterior samples $\boldsymbol{w}^{(s)}$, compute

$$\lambda_*^{(s)} = \exp\left\{\phi(x_*)^T \boldsymbol{w}^{(s)}\right\}$$

Quantiles of $\lambda_*^{(s)}$ give the credible interval for the mean rate. Predictive samples are generated by:

$$y_*^{(s)} \sim \text{Poisson}(\lambda_*^{(s)})$$

The posterior mean prediction is:

$$\mathbb{E}[y_* \mid x_*, \mathcal{D}] = \mathbb{E}_{p(\boldsymbol{w}|\mathcal{D})}[e^{\phi(x_*)^T \boldsymbol{w}}]$$

If observations correspond to different exposure times, detector areas, bin widths, or populations at risk, the Poisson mean should include an exposure E_i:

$$y_i \sim \text{Poisson}(\lambda_i), \qquad \lambda_i = E_i r_i, \qquad \log r_i = \phi(x_i)^T \boldsymbol{w}$$

Equivalently,

$$\log \lambda_i = \log E_i + \phi(x_i)^T \boldsymbol{w},$$

where $\log E_i$ is an offset with known coefficient. Ignoring exposure can produce biased conclusions about rates.

7.4 Connection to supervised learning

Bayesian regression models are central examples of supervised learning algorithms. The dataset consists of labeled examples $\mathcal{D} = \{ (x_i, y_i) \}_{i=1}^N$, and the goal is to learn a mapping from inputs to outputs that generalizes to unseen data. In Bayesian language, learning means updating a prior distribution into a posterior distribution using Bayes' theorem. While classical regression often returns a single fitted curve, Bayesian regression returns a *distribution over possible curves* through Bayes' theorem.

Different choices of likelihood, link function, priors, and feature maps correspond to different learning algorithms. The Bayesian approach provides a principled way to quantify uncertainty, regularize flexible models through priors, compare models via the evidence $p(\mathcal{D})$, and make predictions by integrating over parameter uncertainty rather than conditioning only on a single fitted parameter value. Many modern machine-learning models, including neural networks, can be viewed as nonlinear regression models with very many parameters. Their Bayesian counterparts follow the same principles discussed here.

What we have seen up to know prepares to use MCMC to fit a straight line and a quadratic curve with unknown noise, diagnose chains and inspect posterior correlations, compute posterior predictive distributions from samples and finally move from hand-coded MH algorithms to `JAGS` and `Stan`. The conceptual bridge is: analytic Bayesian linear regression is the clean reference case. MCMC and probabilistic programming let us keep the same model-building logic when conjugacy disappears.

In summary, the take-home messages:

- Bayesian regression models $p(y \mid x, \theta)$ and infers $p(\theta \mid \mathcal{D})$.
- Linear regression means linearity in parameters, not necessarily in x.
- Gaussian prior plus Gaussian likelihood gives an analytic Gaussian posterior.
- Posterior prediction integrates over parameter uncertainty.
- Unknown noise, nonlinear models, logistic regression, and Poisson regression naturally lead back to MCMC.

7.5 Applications of Bayesian regression with MCMC: linear and quadratic fitting

In the previous sections, we introduced regression as a probabilistic model:

$$y_i = f(x_i; \theta) + \varepsilon_i$$

We will see in details two examples that we hinted before – linear and quadratic regression – using MCMC to infer the posterior distribution of the parameters.

$$\begin{aligned} \text{linear:} \quad & f(x) = b_0 + b_1 x \\ \text{quadratic:} \quad & f(x) = b_0 + b_1 x + b_2 x^2 \end{aligned}$$

In both cases, the noise amplitude σ is unknown and inferred from data.

7.5.1 Model ingredients

The data are $\mathcal{D} = \{ (x_j, y_j) \}_{j=1}^N$. We assume the x_j are fixed and measured without uncertainty (if not, latent variables x_j^{true} are needed).

The response variable is generated as:

$$y_j = f(x_j; \theta) + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}\left(0, \sigma^2\right)$$

This accounts for measurement error in y, intrinsic scatter, or unmodelled variation:

$$p(y_j \mid x_j, \theta) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y_j - f(x_j;\theta))^2}{2\sigma^2} \right]$$

Assuming independent data points:

$$p(\mathcal{D} \mid \theta) = \prod_{j=1}^{N} p(y_j \mid x_j, \theta)$$

The log-likelihood is:

$$\begin{aligned} \log p(\mathcal{D} \mid \theta) &= \sum_{j=1}^{N} \log p(y_j \mid x_j, \theta) = \\ &= -N \log \sigma - \frac{N}{2} \log 2\pi - \frac{1}{2\sigma^2} \sum_{j=1}^{N} [y_j - f(x_j;\theta)]^2 \end{aligned}$$

We use the unnormalized posterior in MCMC:

$$p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta) p(\theta \mid I)$$

Why transform parameters? There are may ways to parametrize the model and some choices are more meaningful than others. Some parameters are constrained. For example, $\sigma > 0$. A convenient choice is to sample:

$$\eta = \ln \sigma, \qquad \sigma = e^{\eta}$$

This automatically prevents negative values of σ.

For a line slope, one can sample the angle:

$$b_1 = \tan \alpha$$

Then, the sampled parameters for the line are:

$$\theta = (b_0, \alpha, \ln \sigma)$$

respectively, the intercept, angle and logarithm of variance.

7.5.2 Linear model

For the linear, the generative model is:

$$f(x; b_0, b_1) = b_0 + b_1 x$$

The sampled parameters are:

$$\theta = (b_0, \alpha, \eta), \qquad b_1 = \tan\alpha, \qquad \sigma = e^{\eta}$$

If we have no prior information on the models, the least informative choice that we can make is:

$$b_0 \sim \mathcal{N}\left(0, \delta^2\right), \qquad \alpha \sim \text{Uniform}, \qquad \eta = \ln\sigma \sim \text{Uniform}$$

Jeffreys prior for scale parameter. Flat prior in $b_1 \neq$ flat prior in α.

The two priors are related by the usual change of variables rule:

$$p(b_1)\, db_1 = p(\alpha)\, d\alpha$$

Since $b_1 = \tan\alpha$, we have:

$$\frac{\mathrm{d}b_1}{\mathrm{d}\alpha} = 1 + \tan^2\alpha = 1 + b_1^2 = \sec^2\alpha$$

Therefore,

$$d\alpha = \frac{db_1}{1 + b_1^2}$$

Case 1: flat prior on the angle. Suppose all orientations of the line are equally plausible before seeing the data. Then, we choose a flat prior on the line orientation:

$$p(\alpha) = \frac{1}{\pi}, \qquad \text{with } \alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

Then,

$$p(b_1)\, db_1 = p(\alpha)\, d\alpha = \frac{1}{\pi}\frac{db_1}{1 + b_1^2}$$

A flat prior in angle corresponds to a Cauchy prior on the slope. It disfavors extremely large slopes. This happens because a very large interval of slopes corresponds to a very small interval of angles near $\alpha = \pm\pi/2$.

Case 2: flat prior on the slope. Suppose instead we choose a flat prior directly on the slope:

$$p(b_1) = \text{const.}$$

Then, the induced prior on the angle is:

$$p(\alpha) = p(b_1)\left|\frac{\mathrm{d}b_1}{\mathrm{d}\alpha}\right| \propto \sec^2\alpha = 1 + \tan^2\alpha$$

A flat prior on the slope strongly favors orientations close to vertical lines $\alpha \to \pm\pi/2$.

Consider a straight-line data-generating model with true parameters: $b_0 = 0$ and $b_1 = \sigma = 1$ (Figure 7.3). We generate a small number of data, i.e. $N = 10$. We then run a generic

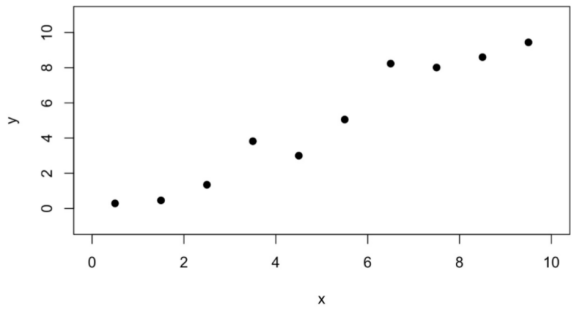


Figure 7.3

MH sampler with starting point $b_0 = 2$, $\alpha = \pi/8$ and $\ln \sigma = \ln 3$ with proposal covariance diagonal multivariate Gaussian.

The trace plots and marginal posteriors are shown in Figure 7.4. On the RHS we can observe that the marginal posteriors are very far from the true values since the amount of data is very small and we are far from the CLT. Looking at joint posteriors (Figure 7.5), we can observe b_0-α anticorrelation. One has to ask itself wether is due to data or the parametrization, that is, it's a real feature of the data or it's due to the parameters? In this particular case it's due to choice of the parameters: $-\delta b_0 = \delta b_1 x > 0$. A standard solution is recentering: $x_j \mapsto x_j - \bar{x}$ (explain why).

Therefore, once we have a decent number of samples, the MAP is the sampled point with largest posterior value, where posterior means MC averages.

What do error bars and curve intervals mean? The vertical error bars on observed points usually represent the observational noise scale:

$$y_j \mid x_j, \theta \sim \mathcal{N}\left(f(x_j;\theta), \sigma^2\right)$$

If σ unknown, estimate it from the posterior, e.g., by $\hat{\sigma}_{\text{MAP}}$. Bands around the fitted curve can be:

- credible band for the mean curve: uncertainty on $f(x;\theta)$,
- posterior predictive band: uncertainty on a new observation y_*.

Credible interval for the regression curve. We want the uncertainty of the **mean function**:

$$p(f(x_*) \mid \mathcal{D}) = \int f(x_*) p(b_0, \alpha, \sigma \mid \mathcal{D})\, db_0\, d\alpha\, d\sigma$$

The 95% central credible interval is obtained from the posterior quantiles:

$$[q_{2.5\%}(f(x_*)), q_{97.5\%}(f(x_*))]$$

In practice, for each posterior sample $\theta^{(k)}$ of MCMC, evaluate the noise-free curve:

$$f_*^{(k)} = b_0^{(k)} + \tan \alpha^{(k)} x_*$$

Calculate lower/upper limit from 2.5/97.5 percentiles of $\{ f_*^{(k)} \}$

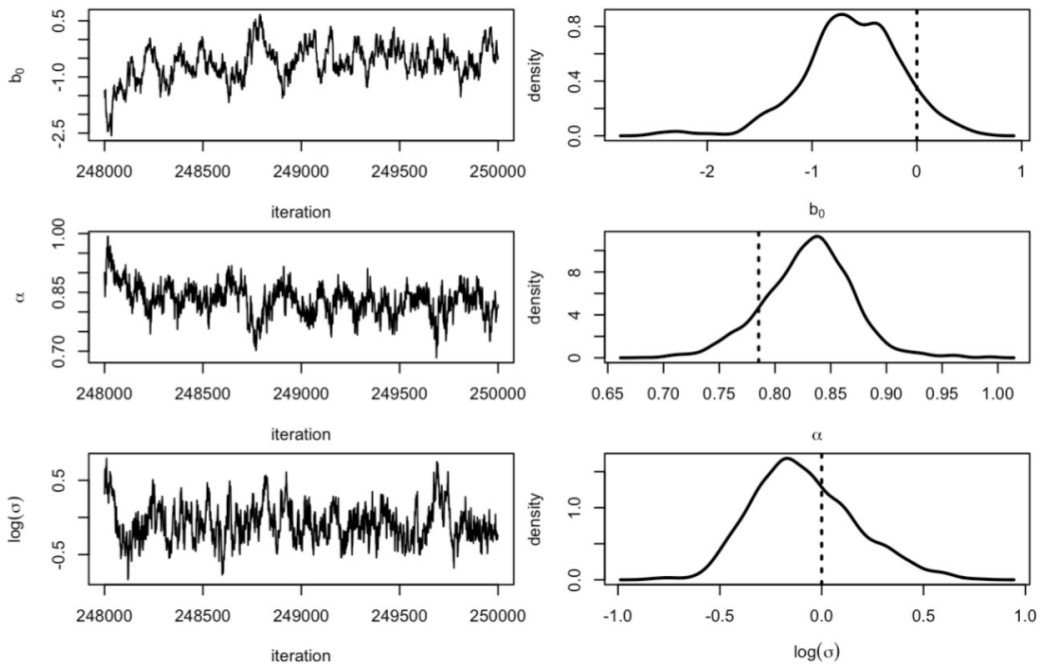


Figure 7.4

Posterior predictive interval. We want the uncertainty for a future observed point. The Posterior predictive distribution is:

$$p(y_* \mid \mathcal{D}) = \int \underbrace{p(y_* \mid x_*, b_0, \alpha, \sigma)}_{\mathcal{N}(f(x_*),\sigma^2)} p(b_0, \alpha, \sigma \mid \mathcal{D})\, db_0\, d\alpha\, d\sigma$$

In practice, for each posterior sample $\theta^{(k)}$ of MCMC, evaluate the noise-free curve $f^{(k)}(x_*)$, then generate a predictive draw:

$$y^{(k)} \sim \mathcal{N}\left(f^{(k)}(x_*), (\sigma^{(k)})^2\right)$$

Calculate lower/upper limit from 2.5/97.5% percentiles of $\{\, y_*^{(k)} \,\}$.

In Figure 7.6a is shown a linear fit with error bars and credible interval for regression curve. In Figure 7.6b is shown the linear posterior predictive interval.

Interpretation: The regression curve credible interval tells us what is the true regression line. While the posterior predictive interval tells us where should future observed data points fall.

7.5.3 Quadratic model

We now fit a quadratic generative model:

$$y_j = b_0 + b_1 x_j + b_2 x_j^2 + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}\left(0, \sigma^2\right)$$

The sampled parameters are:

$$\theta = (b_0, \alpha, b_2, \eta), \qquad b_1 = \tan\alpha, \qquad \sigma = e^{\eta}$$

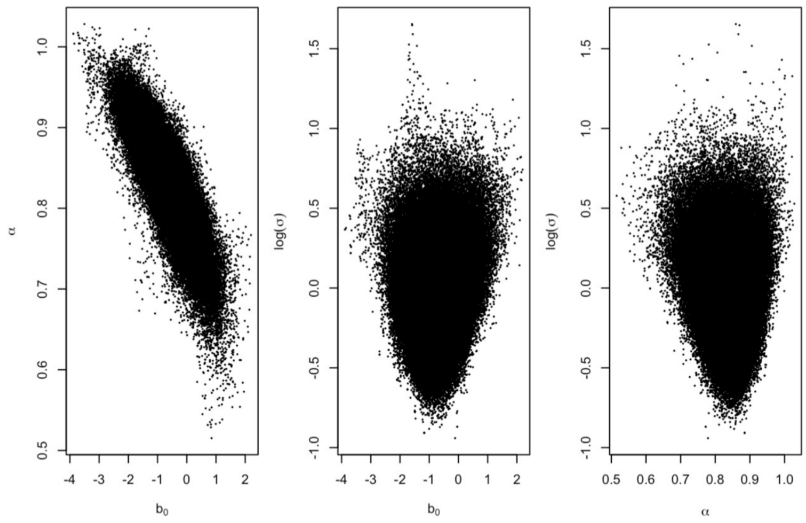


Figure 7.5

Compared with the straight-line model we have: one additional parameter b_2, stronger posterior correlations, more tuning of the proposal covaraince, typically longer chains.

We simulate data from:

$$f(x) = 25 - 10x + x^2, \qquad \sigma^2 = 2$$

We choose priors:

$$b_0 \sim \mathcal{N}\left(0, 10^2\right), \qquad \alpha \sim \text{Uniform}, \qquad b_2 \sim \mathcal{N}\left(0, 5^2\right), \qquad \ln \sigma \sim \text{Uniform}$$

The quadratic posterior can be more difficult to explore. A reasonable starting point can be obtained from a classical least-squares fit. Typically, longer burn-in and longer sampling are required with MCMC. The acceptance rate can be lower than in the straight line model. Also, too large correlation time and tiny ESS, and tuning of the proposal is needed.

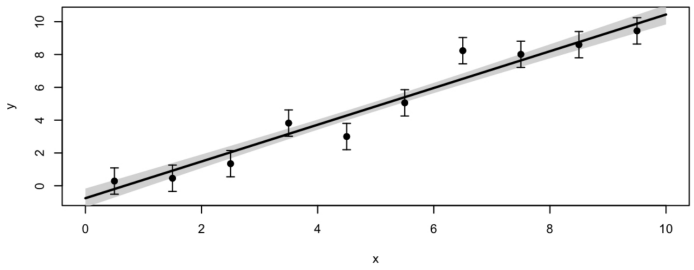
7.5.4 Linear vs quadratic: what changes?

	Linear model	Quadratic model
Mean function	$b_0 + b_1 x$	$b_0 + b_1 x + b_2 x^2$
Sampled parameters	(b_0, α, η)	(b_0, α, b_2, η)
Noise	inferred	inferred
Posterior	3-dimensional	4-dimensional
Correlations	present	usually stronger
MCMC cost	moderate	larger

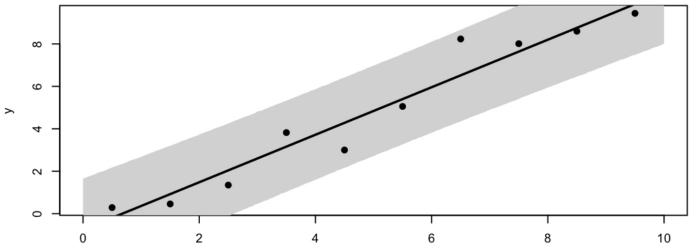
Table 7.3

More flexible models can fit more shapes, but they also introduce more posterior uncertainty and correlations. Tuning of MCM and advanced methods (e.g., HMC) are needed.

Summary of the workflow:



(a)



(b)

Figure 7.6

1. Specify the generative model $f(x;\theta)$ and the noise model.
2. Choose priors for the parameters.
3. Write functions for log-prior, log-likelihood, and log-posterior.
4. Run MCMC and tune proposal covariance.
5. Check trace plots, autocorrelation, effective sample size, and joint posteriors.
6. Summarize parameters with MAP, posterior means, covariance, and correlations.
7. Plot:
    - data with observational error bars,
    - fitted mean curve,
    - credible band for the mean curve,
    - posterior predictive band for new observations

7.6 Application

Appendices

Appendix A: change of variables

Given a random variable X, let $Y = g(X)$ be a new random variable obtained by applying a function g (defined on the support of X to X. The distribution of Y is determined by requiring that probabilities are preserved under the transformation.

Discrete case. If X is discrete with pmf $p_X(x)$, then the pmf of $Y = g(X)$ is obtained by summing the probabilities of all values of X that are mapped into the same value y:

$$p_Y(y) = P(Y = y) = P(g(X) = y) = \sum_{x:\, g(x)=y} p_X(x) \tag{7.5}$$

If g is one-to-one, this reduces simply to

$$p_Y(y) = p_X\left(g^{-1}(y)\right),$$

for all y in the range of g.

Continuous case. If X is continuous with pdf $f_X(x)$ and g is one-to-one and differentiable, then the pdf of $Y = g(X)$ is

$$f_Y(y) = f_X\left(g^{-1}(y)\right)\left|\frac{d}{dy}g^{-1}(y)\right|$$

Equivalently, writing $x = g^{-1}(y)$,

$$f_Y(y) = \frac{f_X(x)}{|g'(x)|}$$

The absolute value of the derivative is the *Jacobian* of the inverse transformation. It appears because a density is not a probability: probabilities of small intervals must be preserved,

$$f_Y(y)\,dy = f_X(x)\,dx$$

If g is not one-to-one, one has to sum over all preimages of y:

$$f_Y(y) = \sum_{x:\, g(x)=y} \frac{f_X(x)}{|g'(x)|}$$

where the sum runs over all solutions of $g(x) = y$.

Example 7.6.1

Let X have pdf f_X and define

$$Y = aX + b, \qquad a \neq 0$$

Then

$$x = \frac{y-b}{a}, \qquad \left|\frac{dx}{dy}\right| = \frac{1}{|a|}$$

and therefore

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

Thus, translations shift the density, while rescalings also change its height so that the total probability remains equal to one.

Appendix B: σ-Algebras

In probability theory, not every subset of the sample space Ω can be consistently assigned a probability. A *σ-algebra* specifies which subsets of Ω are regarded as admissible events, i.e. which questions about the outcome of an experiment are meaningful.

Definition 7.6.1. Let Ω be a non-empty set. A collection $\mathcal{F} \subseteq 2^\Omega$ is called a *σ-algebra* if it satisfies:

1. Containment of the empty set:
$$\varnothing \in \mathcal{F}$$

2. Closure under complement: if $A \in \mathcal{F}$, then:
$$\overline{A} = \Omega \setminus A \in \mathcal{F}$$

3. Closure under countable unions: if $\{ A_n \}_{n=1}^{\infty} \subset \mathcal{F}$, then:
$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$$

From these properties it follows, via De Morgan's laws, that $\mathcal{F}$ is also closed under countable intersections.

Interpretation. Elements of a σ-algebra are called *events*. The defining properties ensure that:

- if an event is meaningful, then its negation is meaningful;
- if countably many events are meaningful, then the event that at least one of them occurs is also meaningful;
- limits of sequences of events remain well-defined.

This logical stability is essential for defining probabilities and for taking limits.

Examples.

- The trivial σ-algebra $\{\varnothing, \Omega\}$, containing only impossible and certain events.
- If Ω is finite or countable, the power set 2^{Ω} is a σ-algebra.
- On $\Omega = \mathbb{R}$, the *Borel σ-algebra* is generated by open intervals and is the standard choice for defining probability densities.

Relation to probability measures. A *probability measure* is a function:

$$\mathbb{P} : \mathcal{F} \to [0, 1]$$

defined on a σ-algebra $\mathcal{F}$ and satisfying the Kolmogorov axioms. The σ-algebra determines which events can be assigned probabilities, while the probability measure assigns numerical values to them. In Bayesian inference, σ-algebras provide the rigorous mathematical foundation for defining random variables, conditioning, and posterior distributions.

Appendix C: Events as a Boolean algebra

The operations on events introduced in probability theory are expressed in the language of set theory, but they can equivalently be rewritten in the language of Boolean algebra. The underlying mathematics is the same, only the point of view changes.

Events form a Boolean algebra. Given a sample space Ω and a σ-algebra $\mathcal{F}$, the elements of $\mathcal{F}$ (events), together with the operations:

$$A \cap B \quad (\texttt{AND}), \qquad A \cup B \quad (\texttt{OR}) \qquad A^c \equiv \overline{A} \quad (\texttt{NOT})$$

form a Boolean algebra. Thus, passing from set-theoretic notation to logical notation does not alter the structure, but only the language used to describe it.

Dictionary between set theory and Boolean algebra. In Boolean algebra, events are treated as truth values, and the operations obey logical rules rather than geometric ones.

Set-theoretic language	Boolean-algebra language	Meaning
Event $A \subset \Omega$	Proposition A	Statement about the outcome
$A \cap B$	$A \wedge B$ or $A \cdot B$	Logical `AND`
$A \cup B$	$A \vee B$ or $A + B$	Logical `OR`
A^c or $\overline{A}$	$\neg A$	Logical `NOT`
$\varnothing$ (empty set)	False	Impossible event
Ω (full space)	True	Certain event

Table 7.4

Formal remark: why probability uses set theory. Probability theory is formulated within measure theory, where probability measures act on sets, integrals require measurable sets, limits of sequences of events require closure under countable operations. Unfortunately, Boolean algebra alone is too weak to support these constructions. A σ-algebra is precisely a Boolean algebra that is closed under countable unions and intersections.

The bridge: indicator functions. The connection between Boolean logic and set-theoretic probability is provided by the indicator function:

$$\mathbb{1}_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$$

Under this correspondence, Boolean operations become algebraic identities:

$$\mathbb{1}_{A\cap B} = \mathbb{1}_A \mathbb{1}_B, \qquad \mathbb{1}_{\overline{A}} = 1 - \mathbb{1}_A, \qquad \mathbb{1}_{A\cup B} = \mathbb{1}_A + \mathbb{1}_B - \mathbb{1}_A \mathbb{1}_B$$

This correspondence explains why logical notation and algebraic products sometimes appear in probability formulas.

Interpretaion in Bayesian inference. In Bayesian inference, events often represent logical statements such as hypotheses, model assumptions, or parameter constraints. Probabilities quantify degrees of belief in these statements, making the Boolean interpretation conceptually natural. Nevertheless, the set-theoretic formulation is essential to ensure mathematical consistency and to support conditioning, integration, and limits. Events in probability theory form a Boolean algebra, and the familiar set-theoretic operations are simply the concrete realization of logical operations within measure theory.